

Question ID: 5fe2bf83

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

A right triangle has legs with lengths of **11** centimeters and **9** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

Answer

- A. $\sqrt{40}$
- B. $\sqrt{202}$
- C. 20
- D. 202

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that a right triangle has legs with lengths of **11** centimeters and **9** centimeters. Substituting **11** for a and **9** for b in the formula $c^2 = a^2 + b^2$ yields $c^2 = 11^2 + 9^2$, which is equivalent to $c^2 = 121 + 81$, or $c^2 = 202$. Taking the square root of each side of this equation yields $c = \pm\sqrt{202}$. Since c represents a length, c must be positive. Therefore, the length of the triangle's hypotenuse, in centimeters, is $\sqrt{202}$.

Choice A is incorrect. This is the result of solving the equation $c^2 = 11 (2) + 9 (2)$, not $c^2 = 11^2 + 9^2$.

Choice C is incorrect. This is the result of solving the equation $c (2) = 11 (2) + 9 (2)$, not $c^2 = 11^2 + 9^2$.

Choice D is incorrect. This is the result of solving the equation $c = 11^2 + 9^2$, not $c^2 = 11^2 + 9^2$.

Question ID: 36eb78e2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of a side of square X is **9** centimeters. The area of rectangle Y is **32** square centimeters. What is the total area, in square centimeters, of square X and rectangle Y?

Answer

- A. **145**
- B. **113**
- C. **82**
- D. **81**

Correct Answer: B

Rationale

Choice B is correct. The area of a square is s^2 , where s is the side length of the square. It's given that the length of a side of square X is **9** centimeters. It follows that the area of square X is **9²**, or **81** square centimeters. It's also given that the area of rectangle Y is **32** square centimeters. Therefore, the total area, in square centimeters, of square X and rectangle Y is **81 + 32**, or **113**.

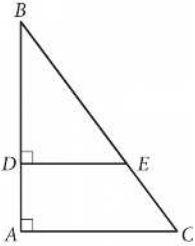
Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area of square X, not the total area of square X and rectangle Y, in square centimeters.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\tan B = \frac{3}{4}$. If $BC = 15$ and $DA = 4$, what is the length of $\overline{DE}$?

Rationale

The correct answer is 6. Since $\tan B = \frac{3}{4}$, $\triangle ABC$ and $\triangle DBE$ are both similar to 3-4-5 triangles. This means that they are both similar to the right triangle with sides of lengths 3, 4, and 5. Since $BC = 15$, which is 3 times as long as the hypotenuse of the 3-4-5 triangle, the similarity ratio of $\triangle ABC$ to the 3-4-5 triangle is 3:1. Therefore, the length of $\overline{AC}$ (the side opposite to $\angle B$) is $3 \times 3 = 9$, and the length of $\overline{AB}$ (the side adjacent to $\angle B$) is $4 \times 3 = 12$. It is also given that $DA = 4$. Since $AB = DA + DB$ and $AB = 12$, it follows that $DB = 8$, which means that the similarity ratio of $\triangle DBE$ to the 3-4-5 triangle is 2:1 ($\overline{DB}$ is the side adjacent to $\angle B$). Therefore, the length of $\overline{DE}$, which is the side opposite to $\angle B$, is $3 \times 2 = 6$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangles ABC and DEF , angles B and E each have measure 27° and angles C and F each have measure 41° . Which additional piece of information is sufficient to determine whether triangle ABC is congruent to triangle DEF ?

Answer

- A. The measure of angle A
- B. The length of side AB
- C. The lengths of sides BC and EF
- D. No additional information is necessary.

Correct Answer: C

Rationale

Choice C is correct. Since angles B and E each have the same measure and angles C and F each have the same measure, triangles ABC and DEF are similar, where side BC corresponds to side EF . To determine whether two similar triangles are congruent, it is sufficient to determine whether one pair of corresponding sides are congruent. Therefore, to determine whether triangles ABC and DEF are congruent, it is sufficient to determine whether sides BC and EF have equal length. Thus, the lengths of BC and EF are sufficient to determine whether triangle ABC is congruent to triangle DEF .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect. The given information is sufficient to determine that triangles ABC and DEF are similar, but not whether they are congruent.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of the acute angles is $(9x - 560)^\circ$. The sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. What is the value of w ?

Correct Answer: 1660

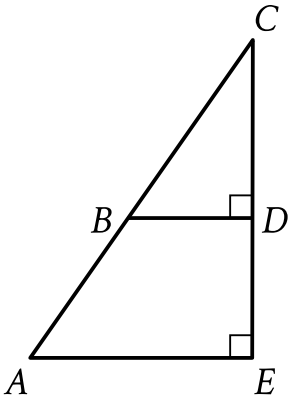
Rationale

The correct answer is **1,660**. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. When two parallel lines are intersected by a transversal line, the angles formed have the following properties: two adjacent angles are supplementary, and alternate interior angles are congruent. Therefore, each of the four acute angles have the same measure, and each of the four obtuse angles have the same measure. It's also given that the measure of one of the acute angles is $(9x - 560)^\circ$. If two angles are supplementary, then the sum of their measures is 180° . Therefore, the measure of the obtuse angle adjacent to any of the acute angles is $(180 - (9x - 560))^\circ$, or $(180 - 9x + 560)^\circ$, which is equivalent to $(-9x + 740)^\circ$. It's given that the sum of the measures of one of the acute angles and three of the obtuse angles is $(-18x + w)^\circ$. It follows that $(9x - 560) + 3(-9x + 740) = (-18x + w)$, which is equivalent to $9x - 560 - 27x + 2,220 = -18x + w$, or $-18x + 1,660 = -18x + w$. Adding $18x$ to both sides of this equation yields $1,660 = w$.

Question ID: 5bd482c2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, triangle CAE is similar to triangle CBD . The measure of angle CBD is 57° , and $AE = 26(BD)$. What is the measure of angle CAE ?

- Answer
- A. $(26 \cdot 57)^\circ$
 - B. $(26 + 57)^\circ$
 - C. 57°
 - D. 26°

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle CAE is similar to triangle CBD . Corresponding angles in similar triangles have equal measure. Angle BCD and angle ACE represent the same angle. It follows that angle BCD and angle ACE have equal measure and are corresponding angles. It's given in the figure that angle BDC and angle AEC are right angles and therefore have equal measure. It follows that angle BDC and angle AEC are corresponding angles. Therefore, angle CBD and angle CAE are corresponding angles and have equal measure. It's given that the measure of angle CBD is 57° , so the measure of angle CAE is 57° .

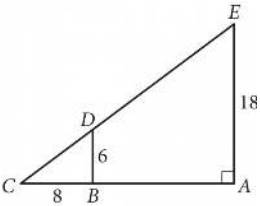
Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the figure above, $\overline{BD}$ is parallel to $\overline{AE}$. What is the length of $\overline{CE}$?

Rationale

The correct answer is 30. In the figure given, since $\overline{BD}$ is parallel to $\overline{AE}$ and both segments are intersected by $\overline{CE}$, then angle BDC and angle AEC are corresponding angles and therefore congruent. Angle BCD and angle ACE are also congruent because they are the same angle. Triangle BCD and triangle ACE are similar because if two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. Since triangle BCD and triangle ACE are similar, their corresponding sides are proportional. So in triangle BCD and triangle ACE, $\overline{BD}$ corresponds to $\overline{AE}$ and $\overline{CD}$ corresponds to $\overline{CE}$. Therefore, $\frac{BD}{CD} = \frac{AE}{CE}$. Since triangle BCD is a right triangle, the Pythagorean theorem can be used to give the value of CD: $6^2 + 8^2 = CD^2$. Taking the square root of each side gives $CD = 10$. Substituting the values in the proportion $\frac{BD}{CD} = \frac{AE}{CE}$ yields $\frac{6}{10} = \frac{18}{CE}$. Multiplying each side by CE, and then multiplying by $\frac{10}{6}$ yields $CE = 30$. Therefore, the length of $\overline{CE}$ is 30.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Circle A has a radius of $3n$ and circle B has a radius of $129n$, where n is a positive constant. The area of circle B is how many times the area of circle A ?

Answer

- A. 43
- B. 86
- C. 129
- D. 1,849

Correct Answer: D

Rationale

Choice D is correct. The area of a circle can be found by using the formula $A = \pi r^2$, where A is the area and r is the radius of the circle. It's given that the radius of circle A is $3n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(3n)^2$, or $9\pi n^2$. It's also given that the radius of circle B is $129n$. Substituting this value for r into the formula $A = \pi r^2$ gives $A = \pi(129n)^2$, or $16,641\pi n^2$. Dividing the area of circle B by the area of circle A gives $\frac{16,641\pi n^2}{9\pi n^2}$, which simplifies to 1,849. Therefore, the area of circle B is 1,849 times the area of circle A.

Choice A is incorrect. This is how many times greater the radius of circle B is than the radius of circle A.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the coefficient on the term that describes the radius of circle B.

Question ID: 0e489b12

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A rectangle has a length of **64** inches and a width of **32** inches. What is the area, in square inches, of the rectangle?

Correct Answer: 2048

Rationale

The correct answer is **2,048**. The area ***A***, in square inches, of a rectangle is equal to the product of its length ***ℓ***, in inches, and its width ***w***, in inches, or **$A = \ell w$** . It's given that the rectangle has a length of **64** inches and a width of **32** inches. Substituting **64** for ***ℓ*** and **32** for ***w*** in the equation **$A = \ell w$** yields **$A = (64)(32)$** , or **$A = 2,048$** . Therefore, the area, in square inches, of the rectangle is **2,048**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of one side of square M is **6** times the length of one side of square N. The area of square N is **49** square feet. What is the area, in square feet, of square M?

Answer

- A. **1,764**
- B. 504
- C. 294
- D. 252

Correct Answer: A

Rationale

Choice A is correct. The area, **A**, of a square is given by the equation **A = s²**, where **s** is the length of one side of the square. Let **x** represent the length, in feet, of one side of square N. It's given that the area of square N is **49** square feet; therefore, it follows that **49 = x²**. It's also given that the length of one side of square M is **6** times the length of one side of square N. Therefore, the length of one side of square M can be represented by the expression **6x**. Substituting **6x** for **s** in the equation **A = s²** yields **A = (6x)²**, or **A = 36x²**. Since **49 = x²**, substituting **49** for **x²** in the equation **A = 36x²** yields **A = 36(49)**, or **A = 1,764**. Therefore, the area, in square feet, of square M is **1,764**.

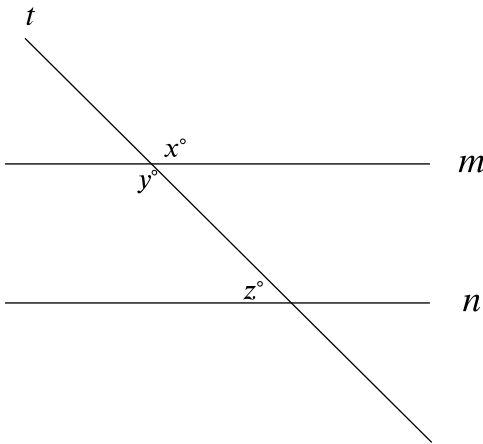
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, lines *m* and *n* are parallel. If $x = 6k + 13$ and $y = 8k - 29$, what is the value of *z*?

Answer

- A. 3
- B. 21
- C. 41
- D. 139

Correct Answer: C

Rationale

Choice C is correct. Vertical angles, which are angles that are opposite each other when two lines intersect, are congruent. The figure shows that lines *t* and *m* intersect. It follows that the angle with measure x° and the angle with measure y° are vertical angles, so $x = y$. It's given that $x = 6k + 13$ and $y = 8k - 29$. Substituting $6k + 13$ for *x* and $8k - 29$ for *y* in the equation $x = y$ yields $6k + 13 = 8k - 29$. Subtracting $6k$ from both sides of this equation yields $13 = 2k - 29$. Adding 29 to both sides of this equation yields $42 = 2k$, or $2k = 42$. Dividing both sides of this equation by 2 yields $k = 21$. It's given that lines *m* and *n* are parallel, and the figure shows that lines *m* and *n* are intersected by a transversal, line *t*. If two parallel lines are intersected by a transversal, then the same-side interior angles are supplementary. It follows that the same-side interior angles with measures y° and z° are supplementary, so $y + z = 180$. Substituting $8k - 29$ for *y* in this equation yields $8k - 29 + z = 180$. Substituting 21 for *k* in this equation yields $8(21) - 29 + z = 180$, or $139 + z = 180$. Subtracting 139 from both sides of this equation yields $z = 41$. Therefore, the value of *z* is 41.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of *k*, not *z*.

Choice D is incorrect. This is the value of *x* or *y*, not *z*.

Question ID: 2f2a9261

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of 45π . If the height of the cylinder is 5, what is the radius of the cylinder?

Answer

- A. 3
- B. 4.5
- C. 9
- D. 40

Correct Answer: A

Rationale

Choice A is correct. The volume of a right circular cylinder with a radius of r is the product of the area of the base, πr^2 , and the height, h . The volume of the right circular cylinder described is 45π and its height is 5. If the radius is r , it follows that $45\pi = \pi(r)^2(5)$. Dividing both sides of this equation by 5π yields $9 = r^2$. Taking the square root of both sides yields $r = 3$ or $r = -3$. Since r represents the radius, the value must be positive. Therefore, the radius is 3.

Choice B is incorrect and may result from finding that the square of the radius is 9, but then from dividing 9 by 2, rather than taking the square root of 9. Choice C is incorrect. This represents the square of the radius. Choice D is incorrect and may result from solving the equation $45\pi = \pi(r)^2(5)$ for r^2 , not r , by dividing by π on both sides and then by subtracting, not dividing, 5 from both sides.

Question ID: edce7254

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

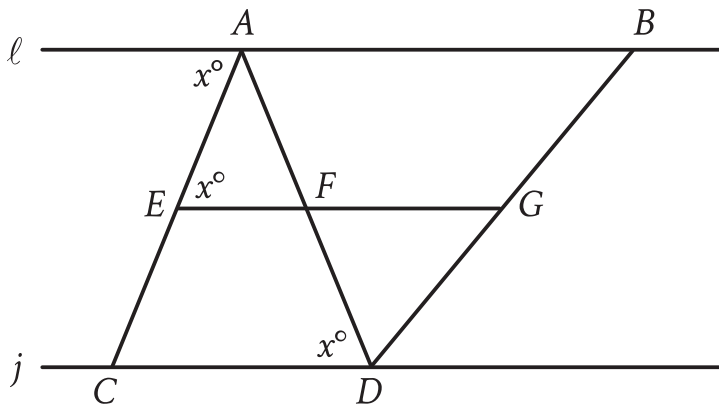
What is the volume, in cubic centimeters, of a right rectangular prism that has a length of 4 centimeters, a width of 9 centimeters, and a height of 10 centimeters?

Rationale

The correct answer is 360. The volume of a right rectangular prism is calculated by multiplying its dimensions: length, width, and height. Multiplying the values given for these dimensions yields a volume of $(4)(9)(10) = 360$ cubic centimeters.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{EG}$ and $\overline{AD}$ intersect at point F , and line **Math output error** is parallel to line j . If $EF = 0.5CD$ and $FG = 2.25CD$, which of the following must be true?

- I. $AB = 2.25EF$
- II. $FG = 0.5AB$
- III. $\frac{EF}{FG} = \frac{AF}{FD}$

Answer

- A. I only
- B. II only
- C. I and II
- D. II and III

Correct Answer: B

Rationale

Choice B is correct. In the figure, $\overline{AE}$ intersects line **Math output error** and $\overline{EG}$, and alternate interior angles are both marked x° . It follows that $\overline{EG}$ is parallel to line **Math output error**. It's given that line **Math output error** is parallel to line j , so $\overline{EG}$ is also parallel to line j . Thus, the following angles also have measure x° : $\angle EFA$, $\angle ACD$, $\angle BAD$, and $\angle GFD$. Triangles AEF and ACD each have two angles with measure x° , so triangles AEF and ACD are similar. It's also given that $EF = 0.5CD$, so $AF = 0.5AD$. It follows that $AF = FD$. Triangles DFG and DAB each have one angle with measure x° and share $\angle FDG$, so triangles DFG and DAB are similar. Since $AF = FD$, it follows that $AD = 2FD$, so $AB = 2FG$. Substituting $2FG$ for AB in statement I yields $2FG = 2.25EF$. It's given that $FG = 2.25CD$. Substituting $2.25CD$ for FG and $0.5CD$ for EF in the equation $2FG = 2.25EF$ yields $2(2.25CD) = 2.25(0.5CD)$, or $4.5CD = 1.125CD$. Thus, statement I is not true. Since $AB = 2FG$, it follows that $0.5AB = FG$, or $FG = 0.5AB$. Therefore, statement II must be true. Substituting $0.5CD$ for EF , $2.25CD$ for FG , and FD for AF in statement III yields $\frac{0.5CD}{2.25CD} = \frac{FD}{FD}$, or $\frac{2}{9} = 1$. Thus, statement III is not true. Therefore, statement II only must be true.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Each base of a right rectangular prism has a length of **19** inches and a width of **8** inches. The prism has a volume of **2,736** cubic inches. What is the height, in inches, of the prism?

Answer

- A. **18**
- B. **27**
- C. **144**
- D. **152**

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length of the base, w is the width of the base, and h is the height of the prism. It's given that each base of a right rectangular prism has a length of **19** inches and a width of **8** inches, and that the prism has a volume of **2,736** cubic inches. Substituting **19** for ℓ , **8** for w , and **2,736** for V in the formula $V = \ell wh$ gives **2,736** $= (19)(8)(h)$, or **2,736** $= 152h$. Dividing each side of this equation by **152** yields **18** $= h$. Therefore, the height, in inches, of the prism is **18**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the area, in square inches, of the base of the prism, not the height, in inches, of the prism.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle XYZ , angle Z is a right angle and the length of $\overline{YZ}$ is **24** units. If $\tan X = \frac{12}{35}$, what is the perimeter, in units, of triangle XYZ ?

Answer

- A. 188
- B. 168
- C. 84
- D. 71

Correct Answer: B

Rationale

Choice B is correct. It's given that angle Z in triangle XYZ is a right angle. Thus, side YZ is the leg opposite angle X and side XZ is the leg adjacent to angle X . The tangent of an acute angle in a right triangle is the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. It follows that $\tan X = \frac{YZ}{XZ}$. It's given that $\tan X = \frac{12}{35}$ and the length of side YZ is **24** units. Substituting $\frac{12}{35}$ for $\tan X$ and **24** for YZ in the equation $\tan X = \frac{YZ}{XZ}$ yields $\frac{12}{35} = \frac{24}{XZ}$. Multiplying both sides of this equation by $35(XZ)$ yields $12(XZ) = 24(35)$, or $12(XZ) = 840$. Dividing both sides of this equation by **12** yields $XZ = 70$. The length XY can be calculated using the Pythagorean theorem, which states that if a right triangle has legs with lengths of a and b and a hypotenuse with length c , then $a^2 + b^2 = c^2$. Substituting **70** for a and **24** for b in this equation yields $70^2 + 24^2 = c^2$, or $5,476 = c^2$. Taking the square root of both sides of this equation yields $\pm 74 = c$. Since the length of the hypotenuse must be positive, $74 = c$. Therefore, the length of XY is **74** units. The perimeter of a triangle is the sum of the lengths of all sides. Thus, $(74 + 70 + 24)$ units, or **168** units, is the perimeter of triangle XYZ .

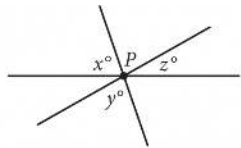
Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This would be the perimeter, in units, for a right triangle where the length of side YZ is **12** units, not **24** units.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, three lines intersect at point P. If $x = 65$ and $y = 75$, what is the value of z ?

Answer

- A. 140
- B. 80
- C. 40
- D. 20

Correct Answer: C

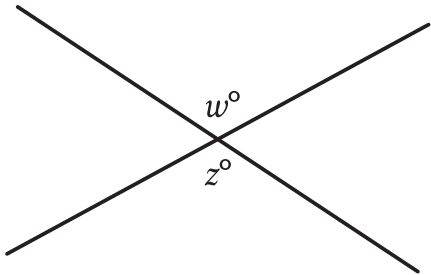
Rationale

Choice C is correct. The angle that is shown as lying between the y° angle and the z° angle is a vertical angle with the x° angle. Since vertical angles are congruent and $x = 65$, the angle between the y° angle and the z° angle measures 65° . Since the 65° angle, the y° angle, and the z° angle are adjacent and form a straight angle, it follows that the sum of the measures of these three angles is 180° , which is represented by the equation $65^\circ + y^\circ + z^\circ = 180^\circ$. It's given that $y = 75$. Substituting 75 for y yields $65^\circ + 75^\circ + z^\circ = 180^\circ$, which can be rewritten as $140^\circ + z^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $z^\circ = 40^\circ$. Therefore, $z = 40$.

Choice A is incorrect and may result from finding the value of $x + y$ rather than z . Choices B and D are incorrect and may result from conceptual or computational errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $w = 136$, what is the value of z ?

Answer

- A. 36
- B. 44
- C. 68
- D. 136

Correct Answer: D

Rationale

Choice D is correct. In the figure shown, the angles with measures w° and z° are vertical angles. Since vertical angles are congruent, $w = z$. Therefore, if $w = 136$, the value of z is 136.

Choice A is incorrect and may result from conceptual or calculation errors.

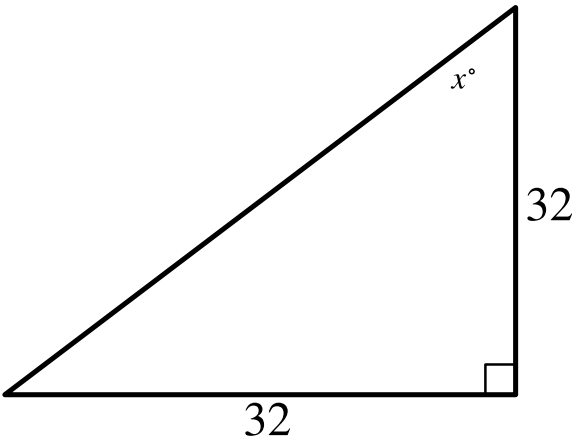
Choice B is incorrect. This is the measure, in degrees, of an angle that's supplementary, not congruent, to the angle with measure w° .

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: b0d50e7b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

Correct Answer: 45

Rationale

The correct answer is **45**. An isosceles right triangle has a right angle and two legs of equal length. In the triangle shown, one angle is a right angle and the two legs each have a length of **32**. Thus, the given triangle is an isosceles right triangle. In an isosceles right triangle, the measures of the two non-right angles are **45°**. It follows that the value of x is **45**.

Question ID: 1b8d255c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Triangles ABC and DEF are similar. Each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . The area of triangle ABC is 270 square inches. What is the area, in square inches, of triangle DEF ?

Correct Answer: 135/8, 16.87, 16.88

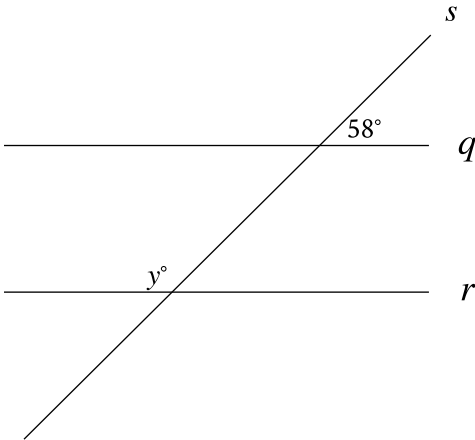
Rationale

The correct answer is $\frac{135}{8}$. It's given that triangles ABC and DEF are similar and each side length of triangle ABC is 4 times the corresponding side length of triangle DEF . For two similar triangles, if each side length of the first triangle is k times the corresponding side length of the second triangle, then the area of the first triangle is k^2 times the area of the second triangle. Therefore, the area of triangle ABC is 4^2 , or 16 , times the area of triangle DEF . It's given that the area of triangle ABC is 270 square inches. Let a represent the area, in square inches, of triangle DEF . It follows that 270 is 16 times a , or $270 = 16a$. Dividing both sides of this equation by 16 yields $\frac{270}{16} = a$, which is equivalent to $\frac{135}{8} = a$. Thus, the area, in square inches, of triangle DEF is $\frac{135}{8}$. Note that 135/8, 16.87, and 16.88 are examples of ways to enter a correct answer.

Question ID: f54b8034

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, line q is parallel to line r , and both lines are intersected by line s . If $y = 2x + 8$, what is the value of x ?

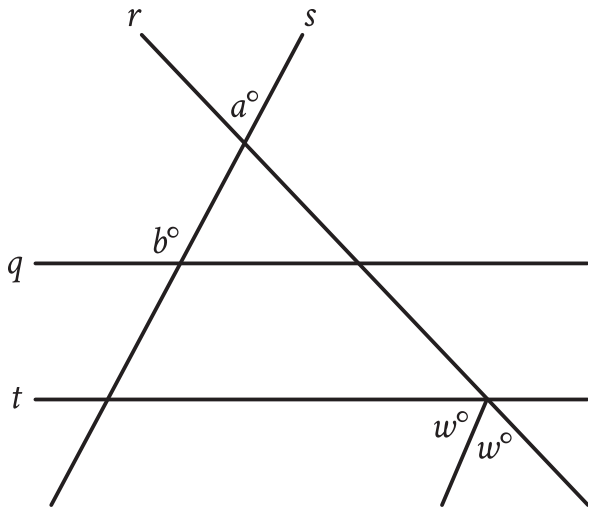
Correct Answer: 57

Rationale

The correct answer is **57**. Based on the figure, the angle with measure y° and the angle vertical to the angle with measure 58° are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure 58° also has measure 58° . It's given that lines q and r are parallel. Therefore, same side interior angles between lines q and r are supplementary. It follows that $y + 58 = 180$. If $y = 2x + 8$, then the value of x can be found by substituting $2x + 8$ for y in the equation $y + 58 = 180$, which yields $(2x + 8) + 58 = 180$, or $2x + 66 = 180$. Subtracting **66** from both sides of this equation yields $2x = 114$. Dividing both sides of this equation by **2** yields $x = 57$. Thus, if $y = 2x + 8$, the value of x is **57**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, parallel lines q and t are intersected by lines r and s . If $a = 43$ and $b = 122$, what is the value of w ?

Correct Answer: 101/2, 50.5

Rationale

The correct answer is $\frac{101}{2}$. In the figure, lines q , r , and s form a triangle. One interior angle of this triangle is vertical to the angle marked a° ; therefore, the interior angle also has measure a° . It's given that $a = 43$. Therefore, the interior angle of the triangle has measure 43° . A second interior angle of the triangle forms a straight line, q , with the angle marked b° . Therefore, the sum of the measures of these two angles is 180° . It's given that $b = 122$. Therefore, the angle marked b° has measure 122° and the second interior angle of the triangle has measure $(180 - 122)^\circ$, or 58° . The sum of the interior angles of a triangle is 180° . Therefore, the measure of the third interior angle of the triangle is $(180 - 43 - 58)^\circ$, or 79° . It's given that parallel lines q and t are intersected by line r . It follows that the triangle's interior angle with measure 79° is congruent to the same side interior angle between lines q and t formed by lines t and r . Since this angle is supplementary to the two angles marked w° , the sum of 79° , w° , and w° is 180° . It follows that $79 + w + w = 180$, or $79 + 2w = 180$. Subtracting 79 from both sides of this equation yields $2w = 101$. Dividing both sides of this equation by 2 yields $w = \frac{101}{2}$. Note that 101/2 and 50.5 are examples of ways to enter a correct answer.

Question ID: afb7b8a8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Trapezoid A and trapezoid B are similar. The length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B. The area of trapezoid A is how many times as large as the area of trapezoid B?

Correct Answer: 1681

Rationale

The correct answer is **1,681**. It's given that trapezoid A and trapezoid B are similar. When two trapezoids are similar, the scale factor between their corresponding side lengths is ***k***, and the scale factor between their areas is ***k*²**. It's given that the length of each side of trapezoid A is **41** times the length of the corresponding side of trapezoid B, so the scale factor between their corresponding side lengths is **41**. It follows that the scale factor between the areas is **41²**, or **1,681**. Therefore, the area of trapezoid A is **1,681** times as large as the area of trapezoid B.

Question ID: bcee5a7b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A rectangle has a length of **56** inches and a width of **28** inches. What is the area, in square inches, of the rectangle?

Answer

- A. **28**
- B. **84**
- C. **168**
- D. **1,568**

Correct Answer: D

Rationale

Choice D is correct. The area, A , of a rectangle can be found using the formula $A = \ell w$, where ℓ represents the length and w represents the width of the rectangle. It's given that a rectangle has a length of **56** inches and a width of **28** inches. Substituting **56** for ℓ and **28** for w in the formula $A = \ell w$ yields $A = (56)(28)$, or $A = 1,568$. Therefore, the area, in square inches, of the rectangle is **1,568**.

Choice A is incorrect. This is the width, in inches, not the area, in square inches, of the rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle RST , angle T is a right angle, point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. If the length of $\overline{RT}$ is 72 units, the length of $\overline{LK}$ is 24 units, and the area of triangle RST is 792 square units, what is the length of $\overline{KT}$, in units?

Correct Answer: 14.66, 14.67, 44/3

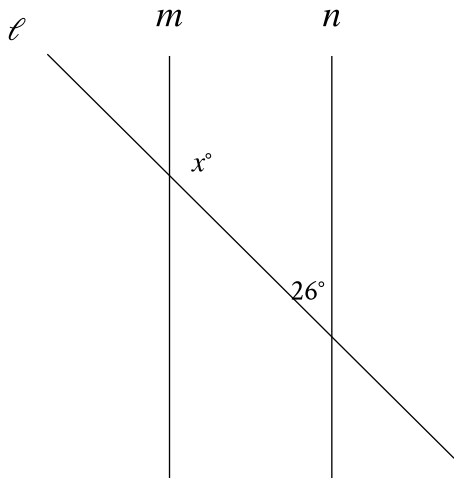
Rationale

The correct answer is $\frac{44}{3}$. It's given that in triangle RST , angle T is a right angle. The area of a right triangle can be found using the formula $A = \frac{1}{2}\ell_1\ell_2$, where A represents the area of the right triangle, ℓ_1 represents the length of one leg of the triangle, and ℓ_2 represents the length of the other leg of the triangle. In triangle RST , the two legs are $\overline{RT}$ and $\overline{ST}$. Therefore, if the length of $\overline{RT}$ is 72 and the area of triangle RST is 792, then $792 = \frac{1}{2}(72)(ST)$, or $792 = (36)(ST)$. Dividing both sides of this equation by 36 yields $22 = ST$. Therefore, the length of $\overline{ST}$ is 22. It's also given that point L lies on $\overline{RS}$, point K lies on $\overline{ST}$, and $\overline{LK}$ is parallel to $\overline{RT}$. It follows that angle LKS is a right angle. Since triangles RST and LSK share angle S and have right angles T and K , respectively, triangles RST and LSK are similar triangles. Therefore, the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is equal to the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$. If the length of $\overline{RT}$ is 72 and the length of $\overline{LK}$ is 24, it follows that the ratio of the length of $\overline{RT}$ to the length of $\overline{LK}$ is $\frac{72}{24}$, or 3, so the ratio of the length of $\overline{ST}$ to the length of $\overline{SK}$ is 3. Therefore, $\frac{22}{SK} = 3$. Multiplying both sides of this equation by SK yields $22 = (3)(SK)$. Dividing both sides of this equation by 3 yields $\frac{22}{3} = SK$. Since the length of $\overline{ST}$, 22, is the sum of the length of $\overline{SK}$, $\frac{22}{3}$, and the length of $\overline{KT}$, it follows that the length of $\overline{KT}$ is $22 - \frac{22}{3}$, or $\frac{44}{3}$. Note that 44/3, 14.66, and 14.67 are examples of ways to enter a correct answer.

Question ID: 99ac3efb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line *m* is parallel to line *n*. What is the value of *x*?

Answer

- A. 13
- B. 26
- C. 52
- D. 154

Correct Answer: D

Rationale

Choice D is correct. The sum of consecutive interior angles between two parallel lines and on the same side of the transversal is 180 degrees. Since it's given that line *m* is parallel to line *n*, it follows that $x + 26 = 180$. Subtracting 26 from both sides of this equation yields 154. Therefore, the value of *x* is 154.

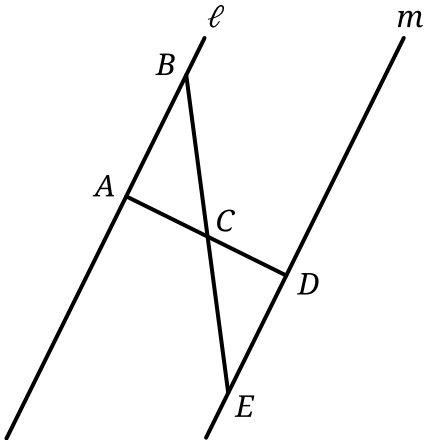
Choice A is incorrect. This is half of the given angle measure.

Choice B is incorrect. This is the value of the given angle measure.

Choice C is incorrect. This is twice the value of the given angle measure.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, segment AD and segment BE intersect at point C . Segment AD is perpendicular to line **Math output error** at point A . Which of the following additional pieces of information is(are) sufficient to determine that triangle ABC is congruent to triangle DEC ?

- I. Segment AD is perpendicular to line m at point D .
- II. $AC = 13$ and $DC = 13$

Answer

- A. I is sufficient by itself, but II is not.
- B. II is sufficient by itself, but I is not.
- C. I is sufficient by itself, and II is sufficient by itself.
- D. Neither I nor II is sufficient by itself.

Correct Answer: D

Rationale

Choice D is correct. In the figure, it's given that segment AD is perpendicular to line **Math output error** at point A . Statement I is that segment AD is perpendicular to line m at point D . It follows that angles BAC and EDC are right angles. Therefore, triangles ABC and DEC are right triangles. At least one pair of congruent corresponding sides is needed to determine triangle congruence by the side-angle-side (SAS) condition or the angle-side-angle (ASA) condition, so statement I is not sufficient to determine that triangle ABC is congruent to triangle DEC . Statement II is that $AC = 13$ and $DC = 13$. It follows that triangles ABC and DEC share one pair of congruent corresponding sides. Two additional pairs of congruent corresponding parts are needed to determine triangle congruence by SAS condition, ASA condition, or side-side-side (SSS) condition, so statement II is not sufficient to determine that triangle ABC is congruent to triangle DEC . Therefore, neither I nor II is sufficient by itself to determine that triangle ABC is congruent to triangle DEC .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

What is the perimeter of an equilateral triangle with a height of $39\sqrt{3}$?

Answer

- A. 117
- B. $117\sqrt{3}$
- C. 234
- D. $1,521\sqrt{3}$

Correct Answer: C

Rationale

Choice C is correct. It’s given that the equilateral triangle has a height of $39\sqrt{3}$. In equilateral triangles, all three angles have a measure of 60° and the height is equal to the length of an altitude from one vertex to the midpoint of the side across from the vertex. Thus, an altitude in this equilateral triangle has a length of $39\sqrt{3}$. The altitude divides the equilateral triangle into two congruent 30-60-90 right triangles, where the altitude is the side across from the 60° angle in each 30-60-90 right triangle. Since the altitude has a length of $39\sqrt{3}$, it follows from the properties of 30-60-90 right triangles that the side across from each 30° angle has a length of 39 and each hypotenuse has a length of $2(39)$, or 78. Therefore, each side of the equilateral triangle has a length of 78. In equilateral triangles, all three sides have equal length. Therefore, the perimeter of this equilateral triangle is $78 + 78 + 78$, or 234.

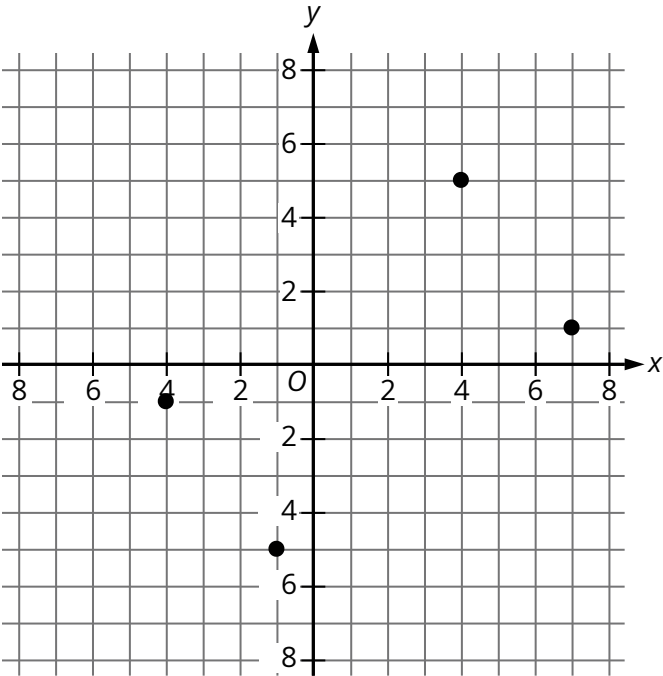
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question



A rectangle is formed by the four points shown. What is the area, in square units, of the rectangle?

Correct Answer: 50

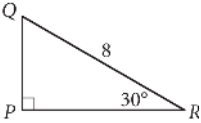
Rationale

The correct answer is 50. The area of a rectangle is **Math output error**, where **Math output error** and w are the lengths of two adjacent sides of the rectangle. It's given that the points $(-4, -1)$, $(4, 5)$, $(7, 1)$, and $(-1, -5)$ form a rectangle. The points $(-4, -1)$ and $(4, 5)$ form a side of the rectangle that is adjacent to the side of the rectangle formed by the points $(4, 5)$ and $(7, 1)$. In the xy -plane, the distance between two points, (x_1, y_1) and (x_2, y_2) , can be found as $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Thus, the length, in units, of the side formed by the points $(-4, -1)$ and $(4, 5)$ can be found as $\sqrt{(4 - (-4))^2 + (5 - (-1))^2}$, which can be simplified to $\sqrt{100}$, or 10. The length, in units, of the side formed by the points $(4, 5)$ and $(7, 1)$ can be found as $\sqrt{(7 - 4)^2 + (1 - 5)^2}$, which can be simplified to $\sqrt{25}$, or 5. Therefore, the area, in square units, of the rectangle formed by the four points shown is $(10)(5)$, or 50.

Question ID: f367a7bb

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



In the right triangle shown above, what is the length of $\overline{PQ}$?

Rationale

The correct answer is 4. Triangle PQR has given angle measures of 30° and 90°, so the third angle must be 60° because the measures of the angles of a triangle sum to 180°. For any special right triangle with angles measuring 30°, 60°, and 90°, the length of the hypotenuse (the side opposite the right angle) is 2x, where x is the length of the side opposite the 30° angle. Segment PQ is opposite the 30° angle. Therefore, 2(PQ) = 8 and PQ = 4.

Question ID: 6f58b48a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Square X has a side length of **12** centimeters. The perimeter of square Y is **2** times the perimeter of square X. What is the length, in centimeters, of one side of square Y?

Answer

- A. **6**
- B. **10**
- C. **14**
- D. **24**

Correct Answer: D

Rationale

Choice D is correct. The perimeter, P , of a square can be found using the formula $P = 4s$, where s is the length of each side of the square. It's given that square X has a side length of **12** centimeters. Substituting **12** for s in the formula for the perimeter of a square yields $P = 4(12)$, or $P = 48$. Therefore, the perimeter of square X is **48** centimeters. It's also given that the perimeter of square Y is **2** times the perimeter of square X. Therefore, the perimeter of square Y is **2(48)**, or **96**, centimeters. Substituting **96** for P in the formula $P = 4s$ gives $96 = 4s$. Dividing both sides of this equation by **4** gives $24 = s$. Therefore, the length of one side of square Y is **24** centimeters.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 7aba3674

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

A rectangle has a length of **13** and a width of **6**. What is the perimeter of the rectangle?

Answer

- A. **12**
- B. **26**
- C. **38**
- D. **52**

Correct Answer: C

Rationale

Choice C is correct. The perimeter of a quadrilateral is the sum of the lengths of its four sides. It's given that the rectangle has a length of **13** and a width of **6**. It follows that the rectangle has two sides with length **13** and two sides with length **6**. Therefore, the perimeter of the rectangle is **13 + 13 + 6 + 6**, or **38**.

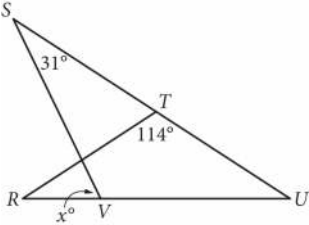
Choice A is incorrect. This is the sum of the lengths of the two sides with length **6**, not the sum of the lengths of all four sides of the rectangle.

Choice B is incorrect. This is the sum of the lengths of the two sides with length **13**, not the sum of the lengths of all four sides of the rectangle.

Choice D is incorrect. This is the perimeter of a rectangle that has four sides with length **13**, not two sides with length **13** and two sides with length **6**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, $RT = TU$. What is the value of x ?

Answer

- A. 72
- B. 66
- C. 64
- D. 58

Correct Answer: C

Rationale

Choice C is correct. Since $RT = TU$, it follows that $\triangle RTU$ is an isosceles triangle with base RU. Therefore, $\angle TRU$ and $\angle TUR$ are the base angles of an isosceles triangle and are congruent. Let the measures of both $\angle TRU$ and $\angle TUR$ be t° . According to the triangle sum theorem, the sum of the measures of the three angles of a triangle is 180° . Therefore, $114^\circ + 2t^\circ = 180^\circ$, so $t = 33$.

Note that $\angle TUR$ is the same angle as $\angle SUV$. Thus, the measure of $\angle SUV$ is 33° . According to the triangle exterior angle theorem, an external angle of a triangle is equal to the sum of the opposite interior angles. Therefore, x° is equal to the sum of the measures of $\angle VSU$ and $\angle SUV$; that is, $31^\circ + 33^\circ = 64^\circ$. Thus, the value of x is 64.

Choice B is incorrect. This is the measure of $\angle STR$, but $\angle STR$ is not congruent to $\angle SVR$. Choices A and D are incorrect and may result from a calculation error.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of **432** cubic centimeters. The area of the base of the cylinder is **24** square centimeters. What is the height, in centimeters, of the cylinder?

Answer

- A. 18
- B. 24
- C. 216
- D. 10,368

Correct Answer: A

Rationale

Choice A is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where πr^2 is the area of the base of the cylinder and h is the height. It's given that a right circular cylinder has a volume of **432** cubic centimeters and the area of the base is **24** square centimeters. Substituting **432** for V and **24** for πr^2 in the formula $V = \pi r^2 h$ yields $432 = 24h$. Dividing both sides of this equation by **24** yields $18 = h$. Therefore, the height of the cylinder, in centimeters, is **18**.

Choice B is incorrect. This is the area of the base, in square centimeters, not the height, in centimeters, of the cylinder.

Choice C is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is **2**, not **24**, cubic centimeters.

Choice D is incorrect. This is the height, in centimeters, of a cylinder if its volume is **432** cubic centimeters and the area of its base is $\frac{1}{24}$, not **24**, cubic centimeters.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The perimeter of triangle ABC is 17 inches, the length of side AB is 4 inches, and the length of side AC is 7 inches. What is the length, in inches, of side BC ?

Answer

- A. 4
- B. 6
- C. 7
- D. 11

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three sides of the triangle. It's given that the lengths of side AB and side AC are 4 inches and 7 inches, respectively. Let x represent the length, in inches, of side BC . The sum of the lengths, in inches, of all three sides of triangle ABC can be represented by the expression $4 + 7 + x$. Since it's given that the perimeter of triangle ABC is 17 inches, it follows that $17 = 4 + 7 + x$, or $17 = 11 + x$. Subtracting 11 from both sides of this equation yields $6 = x$. Therefore, the length, in inches, of side BC is 6 .

Choice A is incorrect. This is the length, in inches, of side AB .

Choice C is incorrect. This is the length, in inches, of side AC .

Choice D is incorrect. This is the sum of the lengths, in inches, of sides AB and AC .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In triangle ABC , angle B is a right angle. The length of side AB is $10\sqrt{37}$ and the length of side BC is $24\sqrt{37}$. What is the length of side AC ?

Answer

- A. $14\sqrt{37}$
- B. $26\sqrt{37}$
- C. $34\sqrt{37}$
- D. $\sqrt{34 \cdot 37}$

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that for a right triangle, $c^2 = a^2 + b^2$, where c represents the length of the hypotenuse and a and b represent the lengths of the legs. It's given that in triangle ABC , angle B is a right angle. Therefore, triangle ABC is a right triangle, where the hypotenuse is side AC and the legs are sides AB and BC . It's given that the lengths of sides AB and BC are $10\sqrt{37}$ and $24\sqrt{37}$, respectively. Substituting these values for a and b in the formula $c^2 = a^2 + b^2$ yields $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$, which is equivalent to $c^2 = 100(37) + 576(37)$, or $c^2 = 676(37)$. Taking the square root of both sides of this equation yields $c = \pm 26\sqrt{37}$. Since c represents the length of the hypotenuse, side AC , c must be positive. Therefore, the length of side AC is $26\sqrt{37}$.

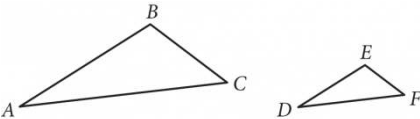
Choice A is incorrect. This is the result of solving the equation $c = 24\sqrt{37} - 10\sqrt{37}$, not $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$.

Choice C is incorrect. This is the result of solving the equation $c = 10\sqrt{37} + 24\sqrt{37}$, not $c^2 = (10\sqrt{37})^2 + (24\sqrt{37})^2$.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figures not drawn to scale.

Triangle ABC and triangle DEF are shown. The relationship between the side lengths of the two triangles is such that

$\frac{AB}{DE} = \frac{BC}{EF} = \frac{AC}{DF} = 3$. If the measure of angle BAC is 20°, what is the measure, in degrees, of angle EDF ? (Disregard the degree symbol when gridding your answer.)

Rationale

The correct answer is 20. By the equality given, the three pairs of corresponding sides of the two triangles are in the same proportion. By the side-side-side (SSS) similarity theorem, triangle ABC is similar to triangle DEF. In similar triangles, the measures of corresponding angles are congruent. Since angle BAC corresponds to angle EDF, these two angles are congruent and their measures are equal. It’s given that the measure of angle BAC is 20°, so the measure of angle EDF is also 20°.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Rectangle $ABCD$ has an area of **280** square centimeters. The length of the rectangle is **10** centimeters. What is the width, in centimeters, of the rectangle?

Answer

- A. 10
- B. 28
- C. 76
- D. 140

Correct Answer: B

Rationale

Choice B is correct. The area A of a rectangle is the product of its length **Math output error** and its width w . It follows that **Math output error**. It's given that rectangle $ABCD$ has an area of **280** square centimeters and the length of the rectangle is **10** centimeters. Substituting **280** for A and **10** for **Math output error** in the equation **Math output error** yields $280 = 10w$. Dividing both sides of this equation by **10** yields $28 = w$. Therefore, the width, in centimeters, of the rectangle is **28**.

Choice A is incorrect. This is the length, in centimeters, not the width, in centimeters, of the rectangle.

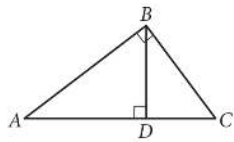
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is half the area, in square centimeters, not the width, in centimeters, of the rectangle.

Question ID: 80219788

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure above, $BD = 6$ and $AD = 8$. What is the length of $\overline{DC}$?

Rationale

The correct answer is 4.5. According to the properties of right triangles, BD divides triangle ABC into two similar triangles, ABD and BCD. The corresponding sides of ABD and BCD are proportional, so the ratio of BD to AD is the same as the ratio of DC to BD. Expressing this information as a proportion gives $\frac{6}{8} = \frac{DC}{6}$. Solving the proportion for DC results in $DC = 4.5$. Note that 4.5 and 9/2 are examples of ways to enter a correct answer.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangles LMN and RST , angles L and R each have measure 60° , $LN = 10$, and $RT = 30$. Which additional piece of information is sufficient to prove that triangle LMN is similar to triangle RST ?

Answer

- A. $MN = 7$ and $ST = 7$
- B. $MN = 7$ and $ST = 21$
- C. The measures of angles M and S are 70° and 60° , respectively.
- D. The measures of angles M and T are 70° and 50° , respectively.

Correct Answer: D

Rationale

Choice D is correct. Two triangles are similar if they have three pairs of congruent corresponding angles. It's given that angles L and R each measure 60° , and so these corresponding angles are congruent. If angle M is 70° , then angle N must be 50° so that the sum of the angles in triangle LMN is 180° . If angle T is 50° , then angle S must be 70° so that the sum of the angles in triangle RST is 180° . Therefore, if the measures of angles M and T are 70° and 50° , respectively, then corresponding angles M and S are both 70° , and corresponding angles N and T are both 50° . It follows that triangles LMN and RST have three pairs of congruent corresponding angles, and so the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle LMN is similar to triangle RST is that the measures of angles M and T are 70° and 50° , respectively.

Choice A is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the two sides given are not proportional and the angle given is not included by the given sides.

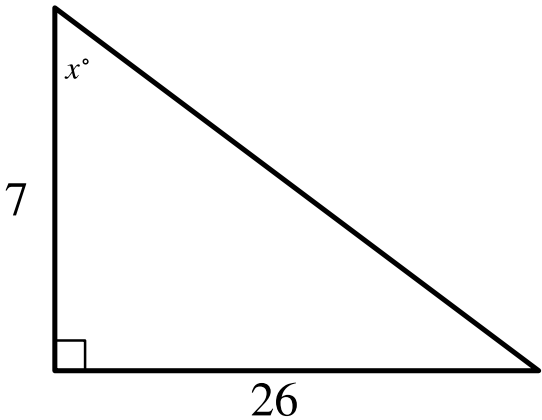
Choice B is incorrect. If the measures of two sides in one triangle are proportional to the corresponding sides in another triangle and the included angles are congruent, then the triangles are similar. However, the angle given is not included between the proportional sides.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 0d85e8d2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\tan x^\circ$?

Answer

- A. $\frac{1}{26}$
- B. $\frac{19}{26}$
- C. $\frac{26}{7}$
- D. $\frac{33}{7}$

Correct Answer: C

Rationale

Choice C is correct. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the side opposite the angle to the length of the shorter side adjacent to the angle. In the triangle shown, the length of the side opposite the angle with measure x° is **26** units and the length of the side adjacent to the angle with measure x° is **7** units. Therefore, the value of $\tan x^\circ$ is $\frac{26}{7}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

The sum of the measures of two angles in a triangle is **64°**. What is the measure of the third angle?

Answer

- A. **26°**
- B. **52°**
- C. **116°**
- D. **128°**

Correct Answer: C

Rationale

Choice C is correct. It’s given that the sum of the measures of two angles in a triangle is **64°**. The sum of the measures of all three angles in a triangle is **180°**. It follows that the measure of the third angle is the difference between **180°** and **64°**, or **180 – 64**, which is **116°**. Therefore, the measure of the third angle is **116°**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 086ccbf9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Each side of a square has a length of **45**. What is the perimeter of this square?

Correct Answer: 180

Rationale

The correct answer is **180**. The perimeter of a polygon is equal to the sum of the lengths of the sides of the polygon. It's given that each side of the square has a length of **45**. Since a square is a polygon with **4** sides, the perimeter of this square is **45 + 45 + 45 + 45**, or **180**.

Question ID: ae2b1ed0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

In a right triangle, the measure of one of the acute angles is **51°**. What is the measure, in degrees, of the other acute angle?

Answer

- A. **6**
- B. **39**
- C. **49**
- D. **51**

Correct Answer: B

Rationale

Choice B is correct. The sum of the measures of the interior angles of a triangle is **180** degrees. Since the triangle is a right triangle, it has one angle that measures **90** degrees. Therefore, the sum of the measures, in degrees, of the remaining two angles is **180 — 90**, or **90**. It’s given that the measure of one of the acute angles in the triangle is **51** degrees. Therefore, the measure, in degrees, of the other acute angle is **90 — 51**, or **39**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure, in degrees, of the acute angle whose measure is given.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. What is the height, in inches, of this pyramid?

Answer

- A. 56
- B. 90
- C. 106
- D. 180

Correct Answer: A

Rationale

Choice A is correct. The total surface area of a right square pyramid is the sum of the combined surface area of its four lateral faces and the area of its square base. It's given that the right square pyramid has a total surface area of **70,560** square inches, and the combined surface area of the four lateral faces of this pyramid is **38,160** square inches. It follows that the area of the square base is the difference between the total surface area and the combined surface area of the four lateral faces, or **(70,560 – 38,160)** square inches, which is equal to **32,400** square inches. The area, ***B***, of the base of a square pyramid is given by the equation **$B = s^2$** , where ***s*** is the side length of the base. Substituting **32,400** for ***B*** in this equation yields **$32,400 = s^2$** . Taking the positive square root of both sides of this equation yields **$180 = s$** . The four lateral faces of a square pyramid are congruent triangles. It follows that the area of each lateral face is **$\frac{38,160}{4}$** , or **9,540**, square inches. The area, ***A***, of each lateral face of a square pyramid is given by the equation **Math output error**, where ***s*** is the side length of the pyramid's base and **Math output error** is the pyramid's slant height. Substituting **9,540** for ***A*** and **180** for ***s*** in this equation yields **Math output error**, or **Math output error**. Dividing both sides of this equation by **90** yields **Math output error**. In the right square pyramid, a right triangle can be drawn, where the lengths of the legs correspond to the pyramid's height, ***h***, and half the side length of its base, **$\frac{1}{2}s$** , and the length of the hypotenuse corresponds to the pyramid's slant height, **Math output error**. It follows that the relationship between ***h***, ***s***, and **Math output error** is given by the equation **Math output error**. Substituting **180** for ***s*** and **106** for **Math output error** in this equation yields **$h^2 + ((\frac{1}{2})(180))^2 = (106)^2$** , or **$h^2 + 8,100 = 11,236$** . Subtracting **8,100** from both sides of this equation yields **$h^2 = 3,136$** . Taking the positive square root of both sides of the equation yields ***h* = 56**. Therefore, the height, in inches, of this pyramid is **56**.

Choice B is incorrect and may result from conceptual or calculation errors.

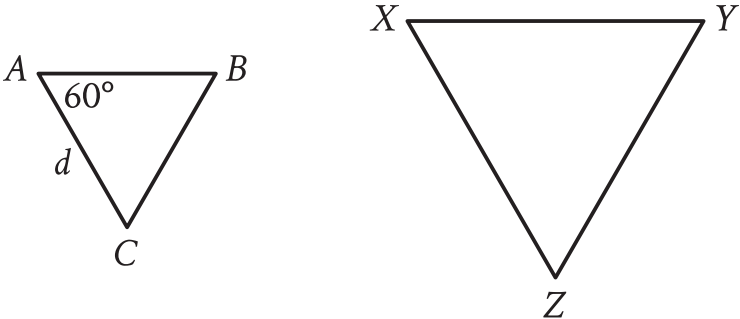
Choice C is incorrect. This is the pyramid's slant height, not its height, in inches.

Choice D is incorrect. This is the side length of the pyramid's base, not its height, in inches.

Question ID: 7cfef47f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figures not drawn to scale.

For the triangles shown, triangle ABC is dilated by a scale factor of 3 to obtain triangle XYZ , where $d = 16$. What is the measure, in degrees, of angle X ?

Answer

- A. 20
- B. 57
- C. 60
- D. 63

Correct Answer: C

Rationale

Choice C is correct. It's given that triangle XYZ is obtained by a dilation of triangle ABC . It follows that triangle ABC is similar to triangle XYZ , where A corresponds to X . Since corresponding angles in similar triangles have the same measure and the measure of angle A is 60 degrees, it follows that the measure of angle X is also 60 degrees.

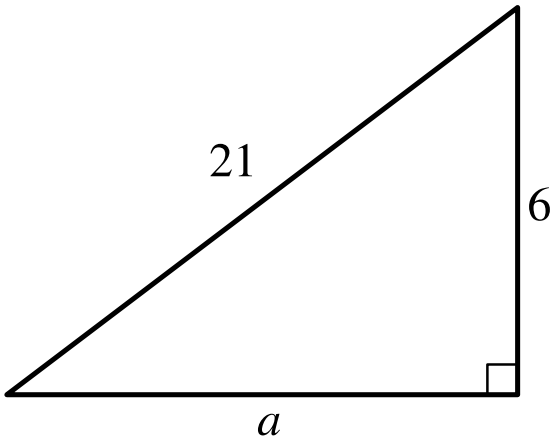
Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

For the triangle shown, which expression represents the value of a ?

- Answer
- A. $\sqrt{21^2 - 6^2}$
 - B. $21^2 - 6^2$
 - C. $\sqrt{21 - 6}$
 - D. $21 - 6$

Correct Answer: A

Rationale

Choice A is correct. For the right triangle shown, the lengths of the legs are a units and 6 units, and the length of the hypotenuse is 21 units. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Therefore, $a^2 + 6^2 = 21^2$. Subtracting 6^2 from both sides of this equation yields $a^2 = 21^2 - 6^2$. Taking the square root of both sides of this equation yields $a = \pm\sqrt{21^2 - 6^2}$. Since a is a length, a must be positive. Therefore, $a = \sqrt{21^2 - 6^2}$. Thus, for the triangle shown, $\sqrt{21^2 - 6^2}$ represents the value of a .

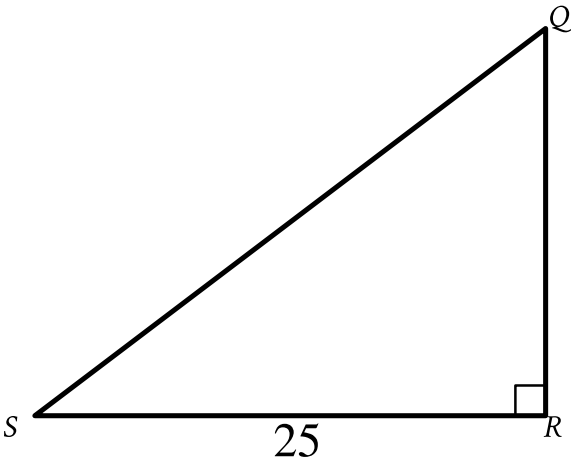
Choice B is incorrect. For the triangle shown, this expression represents the value of a^2 , not a .

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $25 \cos Q$
 - B. $25 \sin Q$
 - C. $\frac{25}{\cos Q}$
 - D. $\frac{25}{\sin Q}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 25. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 25 for RS in this equation, which yields $\sin Q = \frac{25}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 25$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{25}{\sin Q}$. Therefore, the expression $\frac{25}{\sin Q}$ represents the length of $\overline{QS}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has an edge length of **68** inches. A solid sphere with a radius of **34** inches is inside the cube, such that the sphere touches the center of each face of the cube. To the nearest cubic inch, what is the volume of the space in the cube not taken up by the sphere?

Answer

- A. **149,796**
- B. **164,500**
- C. **190,955**
- D. **310,800**

Correct Answer: A

Rationale

Choice A is correct. The volume of a cube can be found by using the formula $V = s^3$, where V is the volume and s is the edge length of the cube. Therefore, the volume of the given cube is $V = 68^3$, or **314,432** cubic inches. The volume of a sphere can be found by using the formula $V = \frac{4}{3}\pi r^3$, where V is the volume and r is the radius of the sphere. Therefore, the volume of the given sphere is $V = \frac{4}{3}\pi(34)^3$, or approximately **164,636** cubic inches. The volume of the space in the cube not taken up by the sphere is the difference between the volume of the cube and volume of the sphere. Subtracting the approximate volume of the sphere from the volume of the cube gives **314,432 – 164,636 = 149,796** cubic inches.

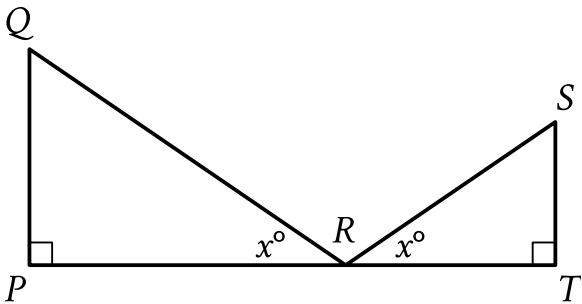
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

$\triangle QPR$ is similar to $\triangle STR$. The lengths represented by $\overline{ST}$, $\overline{QP}$, $\overline{PR}$, and $\overline{QR}$ in the figure are 14, 15, 20, and 25, respectively. What is the length of $\overline{SR}$?

Answer

- A. $\frac{350}{15}$
- B. $\frac{350}{20}$
- C. $\frac{210}{20}$
- D. $\frac{210}{25}$

Correct Answer: A

Rationale

Choice A is correct. The figure shows that angle P in $\triangle QPR$ and angle T in $\triangle STR$ are right angles. It follows that angle P is congruent to angle T . The figure also shows that the measures of angle QRP and angle SRT are both x° . Therefore, angle QRP is congruent to angle SRT . It's given that $\triangle QPR$ is similar to $\triangle STR$. Since angle P is congruent to angle T , and angle QRP is congruent to angle SRT , it follows that $\overline{QR}$ corresponds to $\overline{SR}$, and $\overline{QP}$ corresponds to $\overline{ST}$. Since corresponding sides of similar triangles are proportional, it follows that $\frac{SR}{QR} = \frac{ST}{QP}$. It's also given that the lengths of $\overline{ST}$, $\overline{QP}$, and $\overline{QR}$ are 14, 15, and 25, respectively. Substituting 14 for ST , 15 for QP , and 25 for QR in the equation $\frac{SR}{QR} = \frac{ST}{QP}$ yields $\frac{SR}{25} = \frac{14}{15}$. Multiplying each side of this equation by 25 yields $SR = (\frac{14}{15})(25)$, or $SR = \frac{350}{15}$. Thus, the length of $\overline{SR}$ is $\frac{350}{15}$.

Choice B is incorrect. This is the result of solving the equation $\frac{SR}{25} = \frac{14}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice C is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{20}$, not $\frac{SR}{25} = \frac{14}{15}$.

Choice D is incorrect. This is the result of solving the equation $\frac{SR}{14} = \frac{15}{25}$, not $\frac{SR}{25} = \frac{14}{15}$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Two identical rectangular prisms each have a height of **90 centimeters (cm)**. The base of each prism is a square, and the surface area of each prism is $K \text{ cm}^2$. If the prisms are glued together along a square base, the resulting prism has a surface area of $\frac{92}{47} K \text{ cm}^2$. What is the side length, in **cm**, of each square base?

Answer

- A. 4
- B. 8
- C. 9
- D. 16

Correct Answer: B

Rationale

Choice B is correct. Let x represent the side length, in **cm**, of each square base. If the two prisms are glued together along a square base, the resulting prism has a surface area equal to twice the surface area of one of the prisms, minus the area of the two square bases that are being glued together, which yields $2K - 2x^2 \text{ cm}^2$. It's given that this resulting surface area is equal to $\frac{92}{47} K \text{ cm}^2$, so $2K - 2x^2 = \frac{92}{47} K$. Subtracting $\frac{92}{47} K$ from both sides of this equation yields $2K - \frac{92}{47} K - 2x^2 = 0$. This equation can be rewritten by multiplying $2K$ on the left-hand side by $\frac{47}{47}$, which yields $\frac{94}{47} K - \frac{92}{47} K - 2x^2 = 0$, or $\frac{2}{47} K - 2x^2 = 0$. Adding $2x^2$ to both sides of this equation yields $\frac{2}{47} K = 2x^2$. Multiplying both sides of this equation by $\frac{47}{2}$ yields $K = 47x^2$. The surface area K , in cm^2 , of each rectangular prism is equivalent to the sum of the areas of the two square bases and the areas of the four lateral faces. Since the height of each rectangular prism is **90 cm** and the side length of each square base is $x \text{ cm}$, it follows that the area of each square base is $x^2 \text{ cm}^2$ and the area of each lateral face is $90x \text{ cm}^2$. Therefore, the surface area of each rectangular prism can be represented by the expression $2x^2 + 4(90x)$, or $2x^2 + 360x$. Substituting this expression for K in the equation $K = 47x^2$ yields $2x^2 + 360x = 47x^2$. Subtracting $2x^2$ and $360x$ from both sides of this equation yields $0 = 45x^2 - 360x$. Factoring x from the right-hand side of this equation yields $0 = x(45x - 360)$. Applying the zero product property, it follows that $x = 0$ and $45x - 360 = 0$. Adding 360 to both sides of the equation $45x - 360 = 0$ yields $45x = 360$. Dividing both sides of this equation by 45 yields $x = 8$. Since a side length of a rectangular prism can't be **0**, the length of each square base is **8 cm**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: fe3a4a45

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Triangles EFG and JKL are congruent, where E , F , and G correspond to J , K , and L , respectively. The measure of angle E is 45° and the measure of angle F is 20° . What is the measure of angle J ?

Answer

- A. 20°
- B. 45°
- C. 135°
- D. 160°

Correct Answer: B

Rationale

Choice B is correct. It's given that triangles EFG and JKL are congruent such that angle E corresponds to angle J . Corresponding angles of congruent triangles are congruent, so angle E and angle J must be congruent. Therefore, if the measure of angle E is 45° , then the measure of angle J is also 45° .

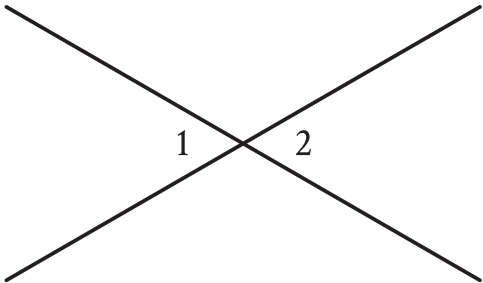
Choice A is incorrect. This is the measure of angle K , not angle J .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. Angle **1** and angle **2** are vertical angles. The measure of angle **1** is **72°**. What is the measure of angle **2**?

Answer

- A. **72°**
- B. 108°
- C. 144°
- D. 288°

Correct Answer: A

Rationale

Choice A is correct. It's given that angle **1** and angle **2** are vertical angles, and the measure of angle **1** is **72°**. Vertical angles have equal measures. Therefore, the measure of angle **2** is **72°**.

Choice B is incorrect. This is the measure of an angle that is supplementary, not congruent, to angle **1**.

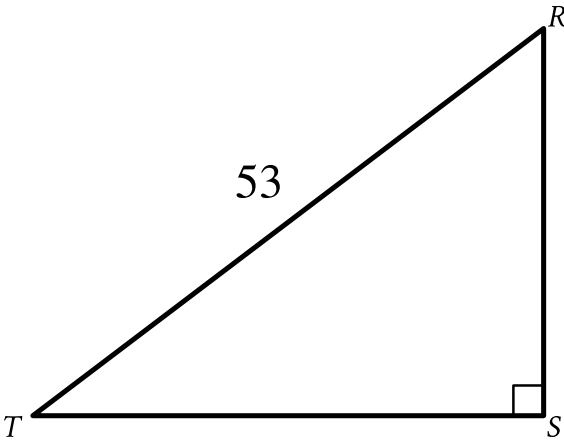
Choice C is incorrect. This is the sum of the measures of angle **1** and angle **2**.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 7a379ded

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, $RS = \sqrt{105}$. What is the value of $\sin R$?

Correct Answer: .9811, 52/53

Rationale

The correct answer is $\frac{52}{53}$. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. The length of the hypotenuse of the right triangle shown is 53. It's given that $RS = \sqrt{105}$. Therefore, the length of one of the legs of the triangle shown is $\sqrt{105}$. Let x represent TS , the length of the other leg of the triangle shown. Therefore, $53^2 = (\sqrt{105})^2 + x^2$, or $2,809 = 105 + x^2$. Subtracting 105 from both sides of this equation yields $2,704 = x^2$. Taking the positive square root of both sides of this equation yields $52 = x$. Therefore, TS , the length of the other leg of the triangle shown, is 52. The sine of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the hypotenuse. The length of the leg opposite angle R is 52, and the length of the hypotenuse is 53. Therefore, the value of $\sin R$ is $\frac{52}{53}$. Note that 52/53 or .9811 are examples of ways to enter a correct answer.

Question ID: 73d06b08

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of a rectangle is **57** square inches. The length of the longest side of the rectangle is **19** inches. What is the length, in inches, of the shortest side of this rectangle?

Correct Answer: 3

Rationale

The correct answer is **3**. The area of a rectangle can be calculated by multiplying the length of its longest side by the length of its shortest side. It's given that the area of the rectangle is **57** square inches and the length of the longest side of the rectangle is **19** inches. Let *x* represent the length, in inches, of the shortest side of this rectangle. It follows that **57 = 19*x***. Dividing both sides of this equation by **19** yields **3 = *x***. Therefore, the length, in inches, of the shortest side of the rectangle is **3**.

Question ID: aa4596b3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The dimensions of a right rectangular prism are 4 inches by 5 inches by 6 inches. What is the surface area, in square inches, of the prism?

Answer

- A. 30
- B. 74
- C. 120
- D. 148

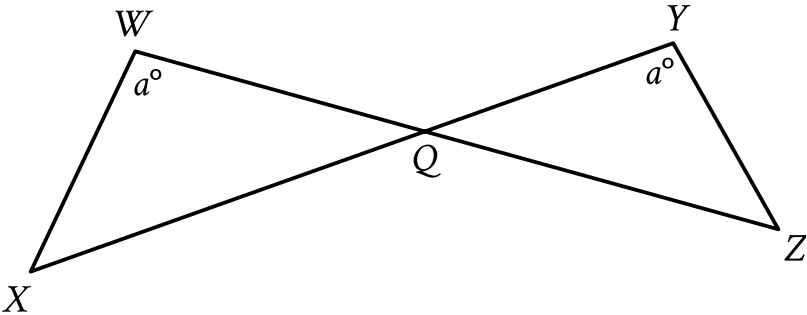
Rationale

Choice D is correct. The surface area is found by summing the area of each face. A right rectangular prism consists of three pairs of congruent rectangles, so the surface area is found by multiplying the areas of three adjacent rectangles by 2 and adding these products. For this prism, the surface area is equal to $2(4 \cdot 5) + 2(5 \cdot 6) + 2(4 \cdot 6)$, or $2(20) + 2(30) + 2(24)$, which is equal to 148.

Choice A is incorrect. This is the area of one of the faces of the prism. Choice B is incorrect and may result from adding the areas of three adjacent rectangles without multiplying by 2. Choice C is incorrect. This is the volume, in cubic inches, of the prism.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q , $YQ = 21$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

Answer

- A. 18
- B. 36
- C. 120
- D. 200

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . It follows that $\angle WQX$ and $\angle YQZ$ are vertical angles, which means they are congruent to each other. The figure shows that $\angle W$ and $\angle Y$ both have a measure of a° , so they are congruent to each other. Therefore, $\triangle QWX$ and $\triangle QYZ$ are similar triangles, where $\overline{WQ}$ corresponds to $\overline{YQ}$ and $\overline{WX}$ corresponds to $\overline{YZ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{YQ} = \frac{WX}{WQ}$. It's given that $YQ = 21$, $WQ = 70$, and $WX = 60$. Substituting 21 for YQ , 70 for WQ , and 60 for WX in the equation $\frac{YZ}{YQ} = \frac{WX}{WQ}$ yields $\frac{YZ}{21} = \frac{60}{70}$, or $\frac{YZ}{21} = \frac{6}{7}$. Multiplying each side of this equation by 21 yields $YZ = 18$. Therefore, the length of $\overline{YZ}$ is 18.

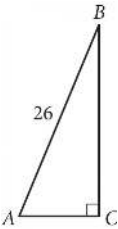
Choice B is incorrect. This is the length of $\overline{QZ}$, not $\overline{YZ}$.

Choice C is incorrect. This is the length of $\overline{XQ}$, not $\overline{YZ}$.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Triangle ABC above is a right triangle, and $\sin(B) = \frac{5}{13}$. What is the length of side $\overline{BC}$?

Rationale

The correct answer is 24. The sine of an acute angle in a right triangle is equal to the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the sine of angle B, or $\sin(B)$, is equal to the ratio of the length of side $\overline{AC}$ to the length of side $\overline{AB}$. It's given that the length of side $\overline{AB}$ is 26 and that $\sin(B) = \frac{5}{13}$. Therefore, $\frac{5}{13} = \frac{AB}{26}$. Multiplying both sides of this equation by 26 yields $AC = 10$.

By the Pythagorean Theorem, the relationship between the lengths of the sides of triangle ABC is as follows: $26^2 = 10^2 + BC^2$, or $676 = 100 + BC^2$. Subtracting 100 from both sides of $676 = 100 + BC^2$ yields $576 = BC^2$. Taking the square root of both sides of $576 = BC^2$ yields $24 = BC$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

What is the area of a rectangle with a length of **4 centimeters (cm)** and a width of **2 cm**?

Answer

- A. **6 cm²**
- B. **8 cm²**
- C. **12 cm²**
- D. **36 cm²**

Correct Answer: B

Rationale

Choice B is correct. The area of a rectangle with length *ℓ* and width *w* can be found using the formula $A = \ell w$. It’s given that the rectangle has a length of **4 cm** and a width of **2 cm**. Therefore, the area of this rectangle is **(4 cm)(2 cm)**, or **8 cm²**.

Choice A is incorrect. This is the sum, **in cm**, of the length and width of the rectangle, not the area, **in cm²**.

Choice C is incorrect. This is the perimeter, **in cm**, of the rectangle, not the area, **in cm²**.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question ID: 3b4dfb61

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Square A has side lengths that are **166** times the side lengths of square B. The area of square A is **k** times the area of square B. What is the value of **k** ?

Correct Answer: 27556

Rationale

The correct answer is **27,556**. The area of a square is **s^2** , where **s** is the side length of the square. Let **x** represent the length of each side of square B. Substituting **x** for **s** in **s^2** yields **x^2** . It follows that the area of square B is **x^2** . It's given that square A has side lengths that are **166** times the side lengths of square B. Since **x** represents the length of each side of square B, the length of each side of square A can be represented by the expression **$166x$** . It follows that the area of square A is **$(166x)^2$** , or **$27,556x^2$** . It's given that the area of square A is **k** times the area of square B. Since the area of square A is equal to **$27,556x^2$** , and the area of square B is equal to **x^2** , an equation representing the given statement is **$27,556x^2 = kx^2$** . Since **x** represents the length of each side of square B, the value of **x** must be positive. Therefore, the value of **x^2** is also positive, so it does not equal **0**. Dividing by **x^2** on both sides of the equation **$27,556x^2 = kx^2$** yields **$27,556 = k$** . Therefore, the value of **k** is **27,556**.

Question ID: dae2f1d9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

For two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. The measures, in degrees, of $\angle Q$ and $\angle R$ are $x + 61$ and $4x + 4$, respectively. What is the value of x ?

Answer

- A. 5
- B. 19
- C. 23
- D. 29

Correct Answer: A

Rationale

Choice A is correct. It's given that for two acute angles, $\angle Q$ and $\angle R$, $\cos(Q) = \sin(R)$. For two acute angles, if the sine of one angle is equal to the cosine of the other angle, the angles are complementary. It follows that $\angle Q$ and $\angle R$ are complementary. That is, the sum of the measures of the angles is 90 degrees. It's given that the measure of $\angle Q$ is $x + 61$ degrees and the measure of $\angle R$ is $4x + 4$ degrees. It follows that $(x + 61) + (4x + 4) = 90$. By combining like terms, this equation can be rewritten as $5x + 65 = 90$. Subtracting 65 from each side of this equation yields $5x = 25$. Dividing each side of this equation by 5 yields $x = 5$.

Choice B is incorrect. This would be the value of x if $\cos(Q) = \cos(R)$ rather than $\cos(Q) = \sin(R)$.

Choice C is incorrect. This would be the value of x if $\cos(Q) = -\cos(R)$ rather than $\cos(Q) = \sin(R)$ and if $\angle R$ were obtuse rather than acute.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Two lines intersect at exactly one point, forming two acute angles and two obtuse angles. The measure of one of these angles is $(9x - 140)^\circ$. Which of the following could **NOT** be the sum of the measures of any two of these angles?

Answer

- A. $(-18x + 280)^\circ$
- B. $(-18x + 640)^\circ$
- C. $(18x - 280)^\circ$
- D. 180°

Correct Answer: A

Rationale

Choice A is correct. It’s given that two lines intersect at exactly one point, forming two acute angles and two obtuse angles. When two lines intersect, opposite angles are congruent and adjacent angles are supplementary. It’s given that the measure of one of the angles is $(9x - 140)^\circ$, so the measure of the opposite angle is $(9x - 140)^\circ$, and the measure of each adjacent angle is $(180 - (9x - 140))^\circ$, or $(-9x + 320)^\circ$. The possible sums of the measures of any two of the angles are $(9x - 140)^\circ + (9x - 140)^\circ$, or $(18x - 280)^\circ$, $(-9x + 320)^\circ + (-9x + 320)^\circ$, or $(-18x + 640)^\circ$, and $(9x - 140)^\circ + (-9x + 320)^\circ$, or 180° . Of the given choices, only $(-18x + 280)^\circ$ is not equivalent to any of these expressions and could not be the sum of the measures of any two of the angles.

Choice B is incorrect. This is the sum of the measures of two angles that are adjacent to the given angle.

Choice C is incorrect. This is the sum of the measure of the given angle and the measure of its opposite angle.

Choice D is incorrect. This is the sum of the measure of the given angle and the measure of an angle adjacent to the given angle.

Question ID: 559405f5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

In triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . What is the measure of $\angle J$, in degrees? (Disregard the degree symbol when entering your answer.)

Correct Answer: 84

Rationale

The correct answer is **84**. The sum of the measures of the interior angles of a triangle is 180° . It's given that in triangle JKL , the measures of $\angle K$ and $\angle L$ are each 48° . Adding the measures, in degrees, of $\angle K$ and $\angle L$ gives $48 + 48$, or **96**. Therefore, the measure of $\angle J$, in degrees, is $180 - 96$, or **84**.

Question ID: ddb05c9c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. What is the radius, in inches, of this cylinder?

Correct Answer: 14

Rationale

The correct answer is **14**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that the right circular cylinder has a height of **16** inches and a volume of **$3,136\pi$** cubic inches. Substituting **16** for h and **$3,136\pi$** for V in the formula $V = \pi r^2 h$ yields **$3,136\pi = \pi r^2 (16)$** , or **$3,136\pi = 16\pi r^2$** . Dividing both sides of this equation by **16π** yields **$196 = r^2$** . Since r represents the radius of the cylinder, the value of r is positive. Taking the positive square root of both sides of this equation yields **$r = 14$** . Therefore, the radius, in inches, of the cylinder is **14**.

Question ID: d1b668a8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A cube has an edge length of **41** inches. What is the volume, in cubic inches, of the cube?

Answer

- A. **164**
- B. **1,681**
- C. **10,086**
- D. **68,921**

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a cube can be found using the formula $V = s^3$, where s is the edge length of the cube. It's given that a cube has an edge length of **41** inches. Substituting **41** inches for s in this equation yields $V = 41^3$ cubic inches, or $V = \mathbf{68,921}$ cubic inches. Therefore, the volume of the cube is **68,921** cubic inches.

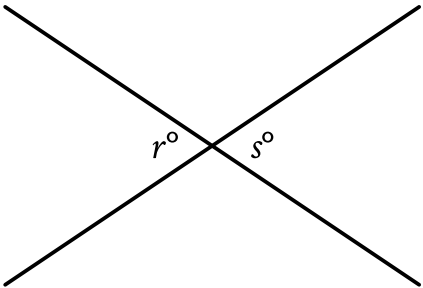
Choice A is incorrect. This is the perimeter, in inches, of the cube.

Choice B is incorrect. This is the area, in square inches, of a face of the cube.

Choice C is incorrect. This is the surface area, in square inches, of the cube.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, two lines intersect at a point. If $r = 53$, what is the value of s ?

Answer

- A. 37
- B. 53
- C. 143
- D. 233

Correct Answer: B

Rationale

Choice B is correct. It's given that two lines intersect at a point. The angles with measures of r° and s° are vertical angles, so it follows that they are equal in measure. Therefore, $s = r$. If $r = 53$, it follows that $s = 53$.

Choice A is incorrect. This is the measure, in degrees, of the complement of an angle with a measure of 53° , not the value of s .

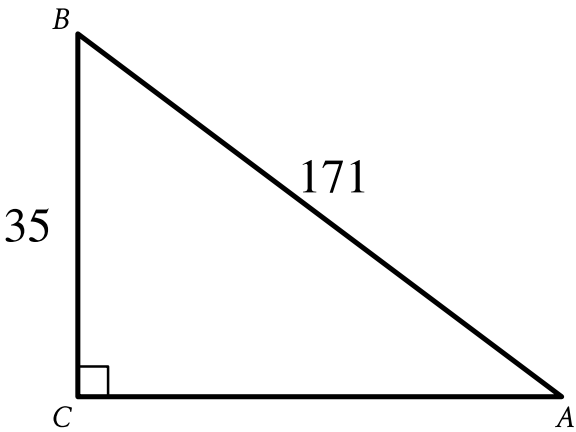
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 1c28e76c

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin A$?

Answer

- A. $\frac{1}{171}$
- B. $\frac{35}{171}$
- C. $\frac{171}{35}$
- D. 171

Correct Answer: B

Rationale

Choice B is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. The hypotenuse of a right triangle is the side opposite the right angle. In right triangle ABC , side BC is the side opposite angle A and side AB is the hypotenuse. It's given that the length of side BC is 35 units and the length of side AB is 171 units. Therefore, the value of $\sin A$ is $\frac{35}{171}$.

Choice A is incorrect and may result from conceptual or calculation errors.

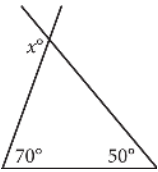
Choice C is incorrect. This is the ratio of the length of the hypotenuse to the length of the side opposite angle A rather than the ratio of the length of the side opposite angle A to the length of the hypotenuse.

Choice D is incorrect. This is the length of the hypotenuse rather than $\sin A$.

Question ID: 7e9a72e3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



In the figure above, two sides of a triangle are extended. What is the value of x ?

Answer

- A. 110
- B. 120
- C. 130
- D. 140

Correct Answer: B

Rationale

Choice B is correct. The sum of the interior angles of a triangle is 180°. The measures of the two interior angles of the given triangle are shown. Therefore, the measure of the third interior angle is 180° – 70° – 50° = 60°. The angles of measures x° and 60° are supplementary, so their sum is 180°. Therefore, x = 180 – 60 = 120.

Choice A is incorrect and may be the result of misinterpreting x° as supplementary to 70°. Choice C is incorrect and may be the result of misinterpreting x° as supplementary to 50°. Choice D is incorrect and may be the result of a calculation error.

Question ID: 8bfaf601

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A square is inscribed in a circle. The radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. What is the side length, in inches, of the square?

Answer

- A. 20
- B. $\frac{20\sqrt{2}}{2}$
- C. $20\sqrt{2}$
- D. 40

Correct Answer: A

Rationale

Choice A is correct. When a square is inscribed in a circle, a diagonal of the square is a diameter of the circle. It's given that a square is inscribed in a circle and the length of a radius of the circle is $\frac{20\sqrt{2}}{2}$ inches. Therefore, the length of a diameter of the circle is $2\left(\frac{20\sqrt{2}}{2}\right)$ inches, or $20\sqrt{2}$ inches. It follows that the length of a diagonal of the square is $20\sqrt{2}$ inches. A diagonal of a square separates the square into two right triangles in which the legs are the sides of the square and the hypotenuse is a diagonal. Since a square has 4 congruent sides, each of these two right triangles has congruent legs and a hypotenuse of length $20\sqrt{2}$ inches. Since each of these two right triangles has congruent legs, they are both 45-45-90 triangles. In a 45-45-90 triangle, the length of the hypotenuse is $\sqrt{2}$ times the length of a leg. Let s represent the length of a leg of one of these 45-45-90 triangles. It follows that $20\sqrt{2} = \sqrt{2}(s)$. Dividing both sides of this equation by $\sqrt{2}$ yields $20 = s$. Therefore, the length of a leg of one of these 45-45-90 triangles is 20 inches. Since the legs of these two 45-45-90 triangles are the sides of the square, it follows that the side length of the square is 20 inches.

Choice B is incorrect. This is the length of a radius, in inches, of the circle.

Choice C is incorrect. This is the length of a diameter, in inches, of the circle.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

A line intersects two parallel lines, forming four acute angles and four obtuse angles. The measure of one of these eight angles is $(7x - 250)^\circ$. The sum of the measures of four of the eight angles is k° . Which of the following could NOT be equivalent to k , for all values of x ?

Answer

- A. $-14x + 1,540$
- B. $14x - 320$
- C. $-28x + 1,720$
- D. 360

Correct Answer: A

Rationale

Choice A is correct. It's given that a line intersects two parallel lines, forming four acute angles and four obtuse angles. Since there are two parallel lines intersected by a transversal, all four acute angles have the same measure and all four obtuse angles have the same measure. Additionally, each acute angle is supplementary to each obtuse angle. It's given that the measure of one of these eight angles is $(7x - 250)^\circ$. It follows that a supplementary angle has measure $(180 - (7x - 250))^\circ$, or $(-7x + 430)^\circ$. It's also given that the sum of the measures of four of the eight angles is k° . It follows that the possible values of k are $4(7x - 250)$; $(7x - 250) + 3(-7x + 430)$; $2(7x - 250) + 2(-7x + 430)$; $3(7x - 250) + (-7x + 430)$; and $4(-7x + 430)$. These values are equivalent to $28x - 1,000$; $-14x + 1,040$; 360 ; $14x - 320$; and $-28x + 1,720$, respectively. It follows that of the given choices, only $-14x + 1,540$ could NOT be equivalent to k , for all values of x .

Choice B is incorrect. This is the sum of three angles with measure $(7x - 250)^\circ$ and one angle with measure $(-7x + 430)^\circ$.

Choice C is incorrect. This is the sum of four angles with measure $(-7x + 430)^\circ$.

Choice D is incorrect. This is the sum of two angles with measure $(7x - 250)^\circ$ and two angles with measure $(-7x + 430)^\circ$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A sphere has a radius of $\frac{17}{5}$ feet. What is the volume, in cubic feet, of the sphere?

Answer

- A. $\frac{5\pi}{17}$
- B. $\frac{68\pi}{15}$
- C. $\frac{32\pi}{5}$
- D. $\frac{19,652\pi}{375}$

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a sphere can be found using the formula $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. It's given that the sphere has a radius of $\frac{17}{5}$ feet. Substituting $\frac{17}{5}$ for r in the formula $V = \frac{4}{3}\pi r^3$ yields $V = \frac{4}{3}\pi\left(\frac{17}{5}\right)^3$, which is equivalent to $V = \frac{4}{3}\pi\left(\frac{4,913}{125}\right)$, or $V = \frac{19,652\pi}{375}$. Therefore, the volume, in cubic feet, of the sphere is $\frac{19,652\pi}{375}$.

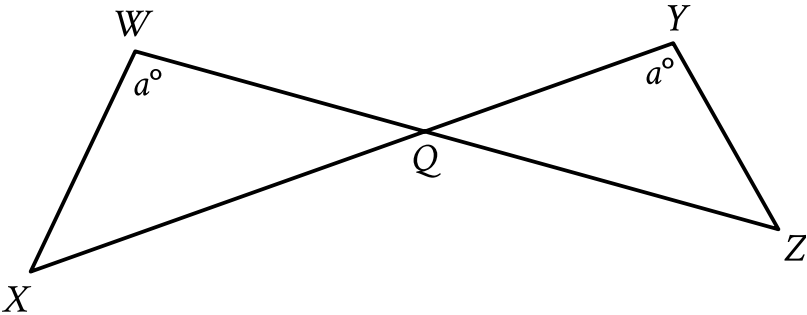
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the volume, in cubic feet, of a sphere with a radius of $\sqrt[3]{\frac{17}{5}}$ feet.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{WZ}$ and $\overline{XY}$ intersect at point Q . $YQ = 63$, $WQ = 70$, $WX = 60$, and $XQ = 120$. What is the length of $\overline{YZ}$?

Correct Answer: 54

Rationale

The correct answer is **54**. The figure shown includes two triangles, triangle WQX and triangle YQZ , such that angle WQX and angle YQZ are vertical angles. It follows that angle WQX is congruent to angle YQZ . It's also given in the figure that the measures of angle W and angle Y are a° . Therefore angle W is congruent to angle Y . Since triangle WQX and triangle YQZ have two pairs of congruent angles, triangle WQX is similar to triangle YQZ by the angle-angle similarity postulate, where $\overline{YZ}$ corresponds to $\overline{WX}$, and $\overline{YQ}$ corresponds to $\overline{WQ}$. Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{YZ}{WX} = \frac{YQ}{WQ}$. It's given that $YQ = 63$, $WQ = 70$, and $WX = 60$. Substituting **63** for YQ , **70** for WQ , and **60** for WX in the equation $\frac{YZ}{WX} = \frac{YQ}{WQ}$ yields $\frac{YZ}{60} = \frac{63}{70}$. Multiplying each side of this equation by **60** yields $YZ = (\frac{63}{70})(60)$, or $YZ = 54$. Therefore, the length of $\overline{YZ}$ is **54**.

Question ID: 711269b9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

What is the area of a rectangle with a length of **17 centimeters (cm)** and a width of **7 cm**?

Answer

- A. **24 cm²**
- B. **48 cm²**
- C. **119 cm²**
- D. **576 cm²**

Correct Answer: C

Rationale

Choice C is correct. The area of a rectangle with length *l* and width *w* can be found using the formula ***A = lw***. It’s given that the rectangle has a length of **17 cm** and a width of **7 cm**. Therefore, the area of this rectangle is ***A = 17(7)***, or **119 cm²**.

Choice A is incorrect. This is the sum of the length and width of the rectangle, not the area.

Choice B is incorrect. This is the perimeter of the rectangle, not the area.

Choice D is incorrect. This is the sum of the length and width of the rectangle squared, not the area.

Question ID: b68f4360

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A right triangle has legs with lengths of **24** centimeters and **21** centimeters. If the length of this triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$, where d is an integer, what is the value of d ?

Correct Answer: 113

Rationale

The correct answer is **113**. It's given that the legs of a right triangle have lengths **24** centimeters and **21** centimeters. In a right triangle, the square of the length of the hypotenuse is equal to the sum of the squares of the lengths of the two legs. It follows that if h represents the length, in centimeters, of the hypotenuse of the right triangle, $h^2 = 24^2 + 21^2$. This equation is equivalent to $h^2 = 1,017$. Taking the square root of each side of this equation yields $h = \sqrt{1,017}$. This equation can be rewritten as $h = \sqrt{9 \cdot 113}$, or $h = \sqrt{9} \cdot \sqrt{113}$. This equation is equivalent to $h = 3\sqrt{113}$. It's given that the length of the triangle's hypotenuse, in centimeters, can be written in the form $3\sqrt{d}$. It follows that the value of d is **113**.

Question ID: 3817a518

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

In right triangle QRS , the length of hypotenuse SQ is $4\sqrt{82}$ and the length of side QR is 4. What is the length of side RS ?

Correct Answer: 36

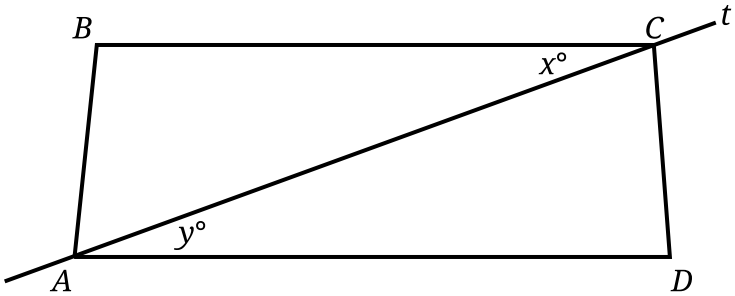
Rationale

The correct answer is **36**. It's given that side SQ is the hypotenuse of right triangle QRS . It's also given that the length of side SQ is $4\sqrt{82}$ and the length of side QR is 4. It follows that sides QR and RS are the legs of triangle QRS . The Pythagorean theorem states that for a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. Let b represent the length of side RS . By the Pythagorean theorem, $4^2 + b^2 = (4\sqrt{82})^2$, or $16 + b^2 = 1,312$. Subtracting 16 from both sides of this equation yields $b^2 = 1,296$. Taking the positive square root of both sides of this equation yields $b = 36$. Therefore, the length of side RS is 36.

Question ID: 70b2c8a2

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, $\overline{BC}$ is parallel to $\overline{AD}$ and $x = 22$. Line t passes through points A and C . What is the value of y ?

Answer

- A. 22
- B. 44
- C. 68
- D. 158

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{BC}$ is parallel to $\overline{AD}$. Since line t intersects $\overline{BC}$ and $\overline{AD}$, it's a transversal. The angles in the figure marked as x° and y° are on opposite sides of the transversal and lie between the parallel segments, so they are alternate interior angles. When two parallel segments are intersected by a transversal, alternate interior angles have equal measure. It's also given that $x = 22$. Therefore, the value of y is 22.

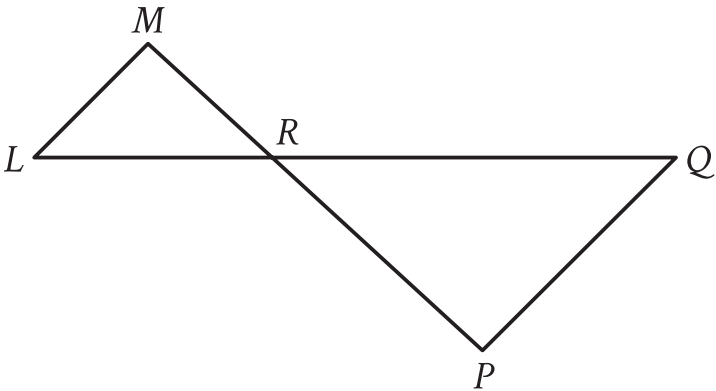
Choice B is incorrect. This is the value of $2x$, not the value of y .

Choice C is incorrect. This is the complement of 22, not the value of y .

Choice D is incorrect. This is the supplement of 22, not the value of y .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. What is the length of $\overline{LQ}$?

Answer

- A. $\frac{119}{11}$
- B. $\frac{77}{6}$
- C. $\frac{113}{6}$
- D. $\frac{119}{6}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that angle MRL and angle PRQ are vertical angles. Since vertical angles are congruent, angle MRL and angle PRQ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, alternate interior angles MLR and PQR are congruent. Since triangles LMR and PQR have two pairs of congruent angles, the triangles are similar. Sides LR and MR in triangle LMR correspond to sides RQ and RP , respectively, in triangle PQR . Since the lengths of corresponding sides in similar triangles are proportional, it follows that $\frac{RQ}{LR} = \frac{RP}{MR}$. It's given that the lengths of $\overline{MR}$, $\overline{LR}$, and $\overline{RP}$ are 6, 7, and 11, respectively. Substituting 6 for MR , 7 for LR and 11 for RP in the equation $\frac{RQ}{LR} = \frac{RP}{MR}$ yields $\frac{RQ}{7} = \frac{11}{6}$. Multiplying each side of this equation by 7 yields $RQ = (\frac{11}{6})(7)$, or $RQ = \frac{77}{6}$. It's given that $\overline{LQ}$ intersects $\overline{MP}$ at point R , so $LQ = LR + RQ$. Substituting 7 for LR and $\frac{77}{6}$ for RQ in this equation yields $LQ = 7 + \frac{77}{6}$, or $LQ = \frac{119}{6}$. Therefore, the length of $\overline{LQ}$ is $\frac{119}{6}$.

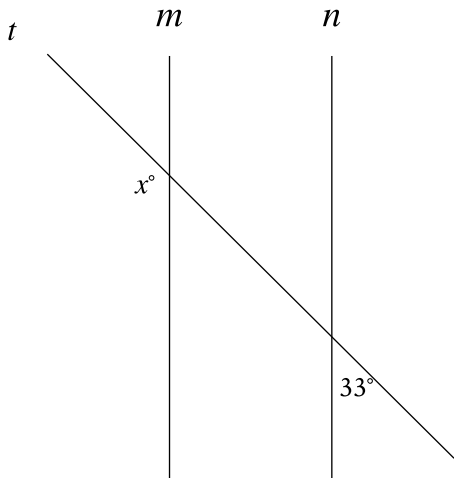
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the length of $\overline{RQ}$, not $\overline{LQ}$.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line *m* is parallel to line *n*, and line *t* intersects both lines. What is the value of *x*?

Answer

- A. 33
- B. 57
- C. 123
- D. 147

Correct Answer: D

Rationale

Choice D is correct. It’s given that line *m* is parallel to line *n*, and line *t* intersects both lines. It follows that line *t* is a transversal. When two lines are parallel and intersected by a transversal, exterior angles on the same side of the transversal are supplementary. Thus, $x + 33 = 180$. Subtracting 33 from both sides of this equation yields $x = 147$. Therefore, the value of *x* is 147.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 931fecbf

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The area of a square is **64** square inches. What is the side length, in inches, of this square?

Answer

- A. **8**
- B. **16**
- C. **64**
- D. **128**

Correct Answer: A

Rationale

Choice A is correct. It's given that the area of a square is **64** square inches. The area **A**, in square inches, of a square is given by the formula **A** = **s**², where **s** is the side length, in inches, of the square. Substituting **64** for **A** in this formula yields **64** = **s**². Taking the positive square root of both sides of this equation yields **8** = **s**. Thus, the side length, in inches, of this square is **8**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the area, in square inches, of the square, not the side length, in inches, of the square.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Circle *K* has a radius of 4 millimeters (mm). Circle *L* has an area of $100\pi \text{ mm}^2$. What is the total area, in mm^2 , of circles *K* and *L*?

Answer

- A. 14π
- B. 28π
- C. 56π
- D. 116π

Correct Answer: D

Rationale

Choice D is correct. The area, *A*, of a circle is given by the formula $A = \pi r^2$, where *r* represents the radius of the circle. It's given that circle *K* has a radius of 4 millimeters (mm). Substituting 4 for *r* in the formula $A = \pi r^2$ yields $A = \pi(4)^2$, or $A = 16\pi$. Therefore, the area of circle *K* is $16\pi \text{ mm}^2$. It's given that circle *L* has an area of $100\pi \text{ mm}^2$. Therefore, the total area, in mm^2 , of circles *K* and *L* is $16\pi + 100\pi$, or 116π .

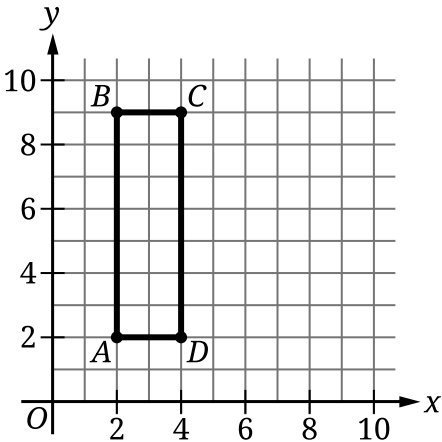
Choice A is incorrect. This is the sum of the radii, in mm, of circles *K* and *L* multiplied by π , not the total area, in mm^2 , of the circles.

Choice B is incorrect. This is the sum of the diameters, in mm, of circles *K* and *L* multiplied by π , not the total area, in mm^2 , of the circles.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question



Rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. What is the area, in square units, of rectangle $ABCD$?

Answer

- A. 5
- B. 9
- C. 14
- D. 28

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a rectangle is given by the formula $A = \ell w$, where ℓ is the length and w is the width of the rectangle. It's given that rectangle $ABCD$ shown has a length of **7** units and a width of **2** units. Substituting **7** for ℓ and **2** for w in the formula $A = \ell w$ yields $A = (7)(2)$, or $A = 14$. Therefore, the area, in square units, of rectangle $ABCD$ is **14**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9edff5b0

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A circle has a radius of **2.1** inches. The area of the circle is **$b\pi$** square inches, where **b** is a constant. What is the value of **b** ?

Correct Answer: 4.41, 441/100

Rationale

The correct answer is **4.41**. The area, **A** , of a circle is given by the formula **$A = \pi r^2$** , where **r** is the radius of the circle. It's given that the area of the circle is **$b\pi$** square inches, where **b** is a constant, and the radius of the circle is **2.1** inches. Substituting **$b\pi$** for **A** and **2.1** for **r** in the formula **$A = \pi r^2$** yields **$b\pi = \pi(2.1^2)$** . Dividing both sides of this equation by **π** yields **$b = 4.41$** . Therefore, the value of **b** is **4.41**.

Question ID: 1d2456f6

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A triangle has a base length of **40** centimeters and a height of **90** centimeters. What is the area, in square centimeters, of the triangle?

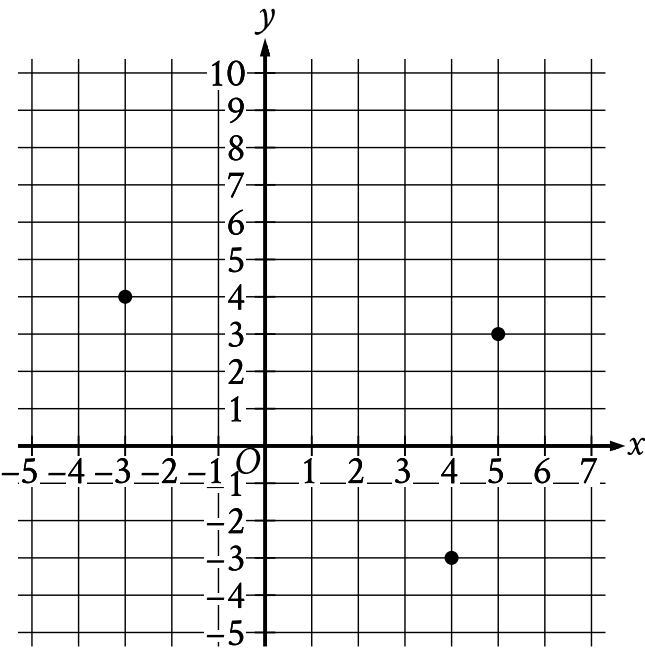
Correct Answer: 1800

Rationale

The correct answer is **1,800**. The area, ***A***, of a triangle can be found using the formula **$A = \frac{1}{2}bh$** , where ***b*** is the base length of the triangle and ***h*** is the height of the triangle. It's given that the triangle has a base length of **40** centimeters and a height of **90** centimeters. Substituting **40** for ***b*** and **90** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$A = \frac{1}{2}(40)(90)$** , or **$A = 1,800$** . Therefore, the area, in square centimeters, of the triangle is **1,800**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question



What is the area, in square units, of the triangle formed by connecting the three points shown?

Correct Answer: 24.5, 49/2

Rationale

The correct answer is **24.5**. It's given that a triangle is formed by connecting the three points shown, which are $(-3, 4)$, $(5, 3)$, and $(4, -3)$. Let this triangle be triangle A. The area of triangle A can be found by calculating the area of the rectangle that circumscribes it and subtracting the areas of the three triangles that are inside the rectangle but outside triangle A. The rectangle formed by the points $(-3, 4)$, $(5, 4)$, $(5, -3)$, and $(-3, -3)$ circumscribes triangle A. The width, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(5, -3)$. This distance is $4 - (-3)$, or 7. The length, in units, of this rectangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of the rectangle is $(7)(8)$, or 56. One of the triangles that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(5, 4)$, and $(5, 3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(5, 3)$. This distance is $4 - 3$, or 1. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, 4)$ and $(-3, 4)$. This distance is $5 - (-3)$, or 8. It follows that the area, in square units, of this triangle is $\frac{1}{2}(8)(1)$, or 4. A second triangle that lies inside the rectangle but outside triangle A is formed by the points $(4, -3)$, $(5, 3)$, and $(5, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(5, 3)$ and $(5, -3)$. This distance is $3 - (-3)$, or 6. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(5, -3)$ and $(4, -3)$. This distance is $5 - 4$, or 1. It follows that the area, in square units, of this triangle is $\frac{1}{2}(1)(6)$, or 3. The third triangle that lies inside the rectangle but outside triangle A is formed by the points $(-3, 4)$, $(-3, -3)$, and $(4, -3)$. The length, in units, of a base of this triangle can be found by calculating the distance between the points $(4, -3)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. The corresponding height, in units, of this triangle can be found by calculating the distance between the points $(-3, 4)$ and $(-3, -3)$. This distance is $4 - (-3)$, or 7. It follows that the area, in square units, of this triangle is $\frac{1}{2}(7)(7)$, or 24.5. Thus, the area, in square

units, of the triangle formed by connecting the three points shown is **56** — **4** — **3** — **24.5**, or **24.5**. Note that 24.5 and 49/2 are examples of ways to enter a correct answer.

Question ID: 8b3c6ded

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Triangle R has an area of **80 square centimeters (cm^2)**. Square S has side lengths of **4 cm**. What is the total area of triangle R and square S, **in cm^2** ?

Answer

- A. **42**
- B. **44**
- C. **84**
- D. **96**

Correct Answer: D

Rationale

Choice D is correct. It’s given that triangle R has an area of **80 cm^2** . The area of a square is ℓ^2 , where ℓ is the side length of the square. It’s given that square S has side lengths of **4 cm**. It follows that the area, in **cm^2** , of square S is **4²**, or **16**. Therefore, the total area, in **cm^2** , of triangle R and square S is **80 + 16**, or **96**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 6a1afc91

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right rectangular prism has a length of **28 centimeters (cm)**, a width of **15 cm**, and a height of **16 cm**. What is the surface area, **in cm²**, of the right rectangular prism?

Correct Answer: 2216

Rationale

The correct answer is **2,216**. The surface area of a prism is the sum of the areas of all its faces. A right rectangular prism consists of six rectangular faces, where opposite faces are congruent. It's given that this prism has a length of **28 cm**, a width of **15 cm**, and a height of **16 cm**. Thus, for this prism, there are two faces with area **(28) (15) cm²**, two faces with area **(28) (16) cm²**, and two faces with area **(15) (16) cm²**. Therefore, the surface area, **in cm²**, of the right rectangular prism is **2 (28) (15) + 2 (28) (16) + 2 (15) (16)**, or **2,216**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Parallelogram $ABCD$ is similar to parallelogram $PQRS$. The length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$. The area of parallelogram $ABCD$ is **5** square centimeters. What is the area, in square centimeters, of parallelogram $PQRS$?

Answer

- A. **7**
- B. **10**
- C. **20**
- D. **25**

Correct Answer: C

Rationale

Choice C is correct. It's given that parallelogram $ABCD$ is similar to parallelogram $PQRS$. When two parallelograms are similar, if the scale factor between their corresponding side lengths is k , the scale factor between their areas is k^2 . It's given that the length of each side of parallelogram $PQRS$ is **2** times the length of its corresponding side of parallelogram $ABCD$, so the scale factor between their corresponding side lengths is **2**. It follows that the scale factor between their areas is **2²**, or **4**. It's given that the area, in square centimeters, of parallelogram $ABCD$ is **5**. It follows that the area, in square centimeters, of parallelogram $PQRS$ is **5(4)**, or **20**.

Choice A is incorrect and may result from conceptual or calculation errors.

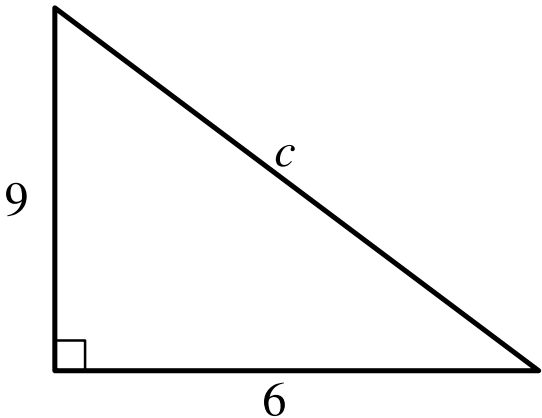
Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 0d8f5be1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the right triangle shown, which of the following is closest to the value of c ?

- Answer**
- A. 7.5
 - B. 10.8
 - C. 15
 - D. 58.5

Correct Answer: B

Rationale

Choice B is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length t and legs with lengths r and s , then $r^2 + s^2 = t^2$. It's given in the right triangle shown that the legs have lengths of 9 and 6 and the hypotenuse has a length of c . Substituting 9 for r , 6 for s , and c for t in $r^2 + s^2 = t^2$ yields $9^2 + 6^2 = c^2$, or $117 = c^2$. Taking the square root of both sides of this equation yields $\pm\sqrt{117} = c$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{117}$, which is approximately equal to 10.8167. Of the choices, 10.8 is the closest to the value of c .

Choice A is incorrect and may result from conceptual or calculation errors.

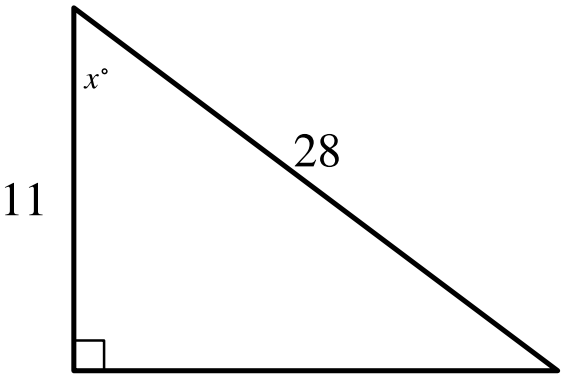
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 76956d0b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\cos x^\circ$?

Correct Answer: .3928, .3929, 11/28

Rationale

The correct answer is $\frac{11}{28}$. The cosine of an acute angle in a right triangle is defined as the ratio of the length of the leg adjacent to the angle to the length of the hypotenuse. In the triangle shown, the length of the leg adjacent to the angle with measure x° is 11 units and the length of the hypotenuse is 28 units. Therefore, the value of $\cos x^\circ$ is $\frac{11}{28}$. Note that 11/28, .3928, .3929, 0.392, and 0.393 are examples of ways to enter a correct answer.

Question ID: bde22993

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A triangle has a base length of **56** centimeters and a height of **112** centimeters. What is the area, in square centimeters, of the triangle?

Answer

- A. **168**
- B. **1,568**
- C. **3,136**
- D. **6,272**

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the base length and h is the height of the triangle. It's given that a triangle has a base length of **56** centimeters and a height of **112** centimeters. Substituting **56** for b and **112** for h in the formula $A = \frac{1}{2}bh$ yields $A = (\frac{1}{2})(56)(112)$, or $A = 3,136$. Therefore, the area, in square centimeters, of the triangle is **3,136**.

Choice A is incorrect. This is the value of $56 + 112$, not $(\frac{1}{2})(56)(112)$.

Choice B is incorrect. This is the value of $(\frac{1}{4})(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Choice D is incorrect. This is the value of $(56)(112)$, not $(\frac{1}{2})(56)(112)$.

Question ID: 05a132a1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question

A right triangle has legs with lengths of **28** centimeters and **20** centimeters. What is the length of this triangle's hypotenuse, in centimeters?

Answer

- A. $8\sqrt{6}$
- B. $4\sqrt{74}$
- C. 48
- D. 1,184

Correct Answer: B

Rationale

Choice B is correct. The Pythagorean theorem states that in a right triangle, the sum of the squares of the lengths of the two legs is equal to the square of the length of the hypotenuse. It's given that the right triangle has legs with lengths of **28** centimeters and **20** centimeters. Let c represent the length of this triangle's hypotenuse, in centimeters. Therefore, by the Pythagorean theorem, $28^2 + 20^2 = c^2$, or $1,184 = c^2$. Taking the positive square root of both sides of this equation yields $\sqrt{1,184} = c$, or $4\sqrt{74} = c$. Therefore, the length of this triangle's hypotenuse, in centimeters, is $4\sqrt{74}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the square of the length of the triangle's hypotenuse.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Rectangles $ABCD$ and $EFGH$ are similar. The length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. The area of $ABCD$ is **54** square units. What is the area, in square units, of $EFGH$?

Answer

- A. **9**
- B. **36**
- C. **324**
- D. **1,944**

Correct Answer: D

Rationale

Choice D is correct. The area of a rectangle is given by bh , where b is the length of the base of the rectangle and h is its height. Let x represent the length, in units, of the base of rectangle $ABCD$, and let y represent its height, in units. Substituting x for b and y for h in the formula bh yields xy . Therefore, the area, in square units, of $ABCD$ can be represented by the expression xy . It's given that the length of each side of $EFGH$ is **6** times the length of the corresponding side of $ABCD$. Therefore, the length, in units, of the base of $EFGH$ can be represented by the expression $6x$, and its height, in units, can be represented by the expression $6y$. Substituting $6x$ for b and $6y$ for h in the formula bh yields $(6x)(6y)$, which is equivalent to $36xy$. Therefore, the area, in square units, of $EFGH$ can be represented by the expression $36xy$. It's given that the area of $ABCD$ is **54** square units. Since xy represents the area, in square units, of $ABCD$, substituting **54** for xy in the expression $36xy$ yields $36(54)$, or **1,944**. Therefore, the area, in square units, of $EFGH$ is **1,944**.

Choice A is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{1}{6}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice B is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{\frac{2}{3}}$, not **6**, times the length of the corresponding side of $ABCD$.

Choice C is incorrect. This is the area of a rectangle where the length of each side of the rectangle is $\sqrt{6}$, not **6**, times the length of the corresponding side of $ABCD$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of the base of a triangle is **8** inches, and the height of the triangle is **12** inches. What is the area, in square inches, of the triangle?

Answer

- A. **20**
- B. **48**
- C. **60**
- D. **96**

Correct Answer: B

Rationale

Choice B is correct. The formula for the area, A , of a triangle is $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the length of the base of a triangle is **8** inches and the height of the triangle is **12** inches. Substituting **8** for b and **12** for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(8)(12)$, or $A = 48$. Therefore, the area, in square inches, of the triangle is **48**.

Choice A is incorrect. This is the sum of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the product of the length of the base of the triangle and the height of the triangle, not the area of the triangle.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Triangles PQR and LMN are graphed in the xy -plane. Triangle PQR has vertices P , Q , and R at $(4, 5)$, $(4, 7)$, and $(6, 5)$, respectively. Triangle LMN has vertices L , M , and N at $(4, 5)$, $(4, 7 + k)$, and $(6 + k, 5)$, respectively, where k is a positive constant. If the measure of $\angle Q$ is t° , what is the measure of $\angle N$?

Answer

- A. $(90 - (t - k))^\circ$
- B. $(90 - (t + k))^\circ$
- C. $(90 - t)^\circ$
- D. $(90 + k)^\circ$

Correct Answer: C

Rationale

Choice C is correct. Since $P = (4, 5)$ and $Q = (4, 7)$, side PQ is parallel to the y -axis and has a length of 2 . Since $P = (4, 5)$ and $R = (6, 5)$, side PR is parallel to the x -axis and has a length of 2 . Therefore, triangle PQR is a right isosceles triangle, where $\angle P$ has measure 90° and $\angle Q$ and $\angle R$ each have measure 45° . It follows that if the measure of $\angle Q$ is t° , then $t = 45$. Since $L = (4, 5)$ and $M = (4, 7 + k)$, side LM is parallel to the y -axis and has a length of $k + 2$. Since $L = (4, 5)$ and $N = (6 + k, 5)$, side LN is parallel to the x -axis and has a length of $k + 2$. Therefore, triangle LMN is a right isosceles triangle, where $\angle L$ has measure 90° and $\angle M$ and $\angle N$ each have measure 45° . Of the given choices, only $(90 - t)^\circ$ is equal to 45° , so the measure of $\angle N$ is $(90 - t)^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9a6b3320

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC , the measure of angle A is 52° and $AC = 30$. In triangle PQR , the measure of angle P is 52° and $PR = 120$. Which additional piece of information is sufficient to prove that triangle ABC is similar to triangle PQR ?

Answer

- A. $AB = 50$ and $PQ = 50$.
- B. $AB = 50$ and $QR = 200$.
- C. The measures of angle B and angle R are 32° and 96° , respectively.
- D. The measures of angle B and angle Q are 52° and 32° , respectively.

Correct Answer: C

Rationale

Choice C is correct. It's given that the measure of angle A is 52° and the measure of angle P is 52° . Therefore, angle A is congruent to angle P . Choice C states that the measures of angle B and angle R are 32° and 96° , respectively. The triangle sum theorem states that the measures of the angles in a triangle sum to 180° . In triangle ABC , if the measure of angle B is 32° , it follows that the measure of angle C is $(180 - (52 + 32))^\circ$, or 96° . In triangle PQR , if the measure of angle R is 96° , it follows that the measure of angle Q is $(180 - (52 + 96))^\circ$, or 32° . Therefore, in triangles ABC and PQR , angle B is congruent to angle Q , and angle C is congruent to angle R . If two triangles have three pairs of congruent angles, then the triangles are similar. Therefore, the additional piece of information that is sufficient to prove that triangle ABC is similar to triangle PQR is that the measures of angle B and angle R are 32° and 96° , respectively.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. If the measure of angle B is 52° , it follows that the measure of angle C is $(180 - (52 + 52))^\circ$, or 76° . If the measure of angle Q is 32° , which isn't equal to any angle measure in triangle ABC , the triangles can't be similar.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

Which of the following expressions is equivalent to $(\sin 24^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\sin 66^\circ)$?

Answer

- A. $2(\cos 66^\circ)(\sin 24^\circ)$
- B. $2(\cos 66^\circ) + 2(\cos 24^\circ)$
- C. $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$
- D. $(\cos 66^\circ)^2 + (\sin 24^\circ)^2$

Correct Answer: C

Rationale

Choice C is correct. The sine of an angle is equal to the cosine of its complementary angle. Since angles with measures 24° and 66° are complementary to each other, $\sin 24^\circ$ is equal to $\cos 66^\circ$ and $\sin 66^\circ$ is equal to $\cos 24^\circ$. Substituting $\cos 66^\circ$ for $\sin 24^\circ$ and $\cos 24^\circ$ for $\sin 66^\circ$ in the given expression yields $(\cos 66^\circ)(\cos 66^\circ) + (\cos 24^\circ)(\cos 24^\circ)$, or $(\cos 66^\circ)^2 + (\cos 24^\circ)^2$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A manufacturer determined that right cylindrical containers with a height that is 4 inches longer than the radius offer the optimal number of containers to be displayed on a shelf. Which of the following expresses the volume, V , in cubic inches, of such containers, where r is the radius, in inches?

Answer

- A. $V = 4\pi^3$
- B. $V = \pi(2r)^3$
- C. $V = \pi^2 + 4\pi$
- D. $V = \pi^3 + 4\pi^2$

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right cylinder is given by the formula $V = \pi r^2 h$, where r represents the radius of the base of the cylinder and h represents the height. Since the height is 4 inches longer than the radius, the expression $r + 4$ represents the height of each cylindrical container. It follows that the volume of each container is represented by the equation $V = \pi r^2 (r + 4)$. Distributing the expression πr^2 into each term in the parentheses yields $V = \pi r^3 + 4\pi r^2$.

Choice A is incorrect and may result from representing the height as $4r$ instead of $r + 4$. Choice B is incorrect and may result from representing the height as $2r$ instead of $r + 4$. Choice C is incorrect and may result from representing the volume of a right cylinder as $V = \pi r h$ instead of $V = \pi r^2 h$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . In triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . Which of the following additional pieces of information is needed to determine whether triangle DEF is similar to triangle RST ?

Answer

- A. The measure of angle F
- B. The measure of angle T
- C. The measure of angle F and the measure of angle T
- D. No additional information is needed.

Correct Answer: D

Rationale

Choice D is correct. When two angles of one triangle are congruent to two angles of another triangle, the triangles are similar. It's given that in triangle DEF , the measure of angle D is 47° and the measure of angle E is 97° . It's also given that in triangle RST , the measure of angle R is 47° and the measure of angle S is 97° . It follows that angle D is congruent to angle R and that angle E is congruent to angle S . Therefore, triangle DEF is similar to triangle RST and no additional information is needed.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. What is the area of the triangle, in square units?

Answer

- A. $8\sqrt{2} + \sqrt{80}$
- B. 12
- C. $24\sqrt{80}$
- D. 24

Correct Answer: B

Rationale

Choice B is correct. The area, A , of a triangle can be found using the formula $A = \frac{1}{2}bh$, where b is the length of the base of the triangle and h is the height of the triangle. It's given that the triangle is a right triangle. Therefore, its base and height can be represented by the two legs. It's also given that the triangle has sides of length $2\sqrt{2}$, $6\sqrt{2}$, and $\sqrt{80}$ units. Since $\sqrt{80}$ units is the greatest of these lengths, it's the length of the hypotenuse. Therefore, the two legs have lengths $2\sqrt{2}$ and $6\sqrt{2}$ units. Substituting these values for b and h in the formula $A = \frac{1}{2}bh$ gives $A = \frac{1}{2}(2\sqrt{2})(6\sqrt{2})$, which is equivalent to $A = 6\sqrt{4}$ square units, or $A = 12$ square units.

Choice A is incorrect. This expression represents the perimeter, rather than the area, of the triangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 429a418a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

Rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, what is the area, in square inches, of the resulting figure?

Answer

- A. **92**
- B. **84**
- C. **80**
- D. **52**

Correct Answer: D

Rationale

Choice D is correct. It's given that rectangle P has an area of **72** square inches. If a rectangle with an area of **20** square inches is removed from rectangle P, the area, in square inches, of the resulting figure is **72 — 20**, or **52**.

Choice A is incorrect and may result from conceptual or calculation errors.

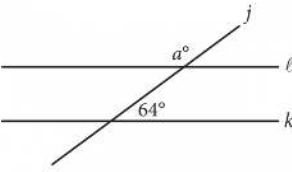
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: a61f9290

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure above, lines ℓ and k are parallel. What is the value of a ?

Answer

- A. 26
- B. 64
- C. 116
- D. 154

Correct Answer: C

Rationale

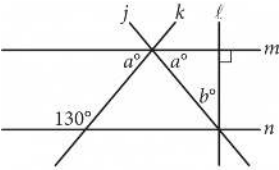
Choice C is correct. Since lines ℓ and k are parallel, corresponding angles formed by the intersection of line j with lines ℓ and k are congruent. Therefore, the angle with measure a° must be the supplement of the angle with measure 64° . The sum of two supplementary angles is 180° , so $a = 180 - 64 = 116$.

Choice A is incorrect and likely results from thinking the angle with measure a° is the complement of the angle with measure 64° . Choice B is incorrect and likely results from thinking the angle with measure a° is congruent to the angle with measure 64° . Choice D is incorrect and likely results from a conceptual or computational error.

Question ID: f95cb45a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.
In the figure above, lines m and n are parallel. What is the value of b ?

Answer

- A. 40
- B. 50
- C. 65
- D. 80

Correct Answer: A

Rationale

Choice A is correct. Given that lines m and n are parallel, the angle marked 130° must be supplementary to the leftmost angle marked a° because they are same-side interior angles. Therefore, $130^\circ + a^\circ = 180^\circ$, which yields $a = 50^\circ$. Lines  and m intersect at a right angle, so lines j, , and m form a right triangle where the two acute angles are a° and b°. The acute angles of a right triangle are complementary, so $a^\circ + b^\circ = 90^\circ$, which yields $50^\circ + b^\circ = 90^\circ$, and $b = 40$.

Choice B is incorrect. This is the value of a, not b. Choice C is incorrect and may be the result of dividing 130° by 2. Choice D is incorrect and may be the result of multiplying b by 2.

Question ID: 0bd5d440

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The volume of right circular cylinder A is 22 cubic centimeters. What is the volume, in cubic centimeters, of a right circular cylinder with twice the radius and half the height of cylinder A?

Answer

- A. 11
- B. 22
- C. 44
- D. 66

Correct Answer: C

Rationale

Choice C is correct. The volume of right circular cylinder A is given by the expression $\pi r^2 h$, where r is the radius of its circular base and h is its height. The volume of a cylinder with twice the radius and half the height of cylinder A is given by $\pi (2r)^2 \left(\frac{1}{2}h\right)$, which is equivalent to $4\pi r^2 \left(\frac{1}{2}h\right) = 2\pi r^2 h$. Therefore, the volume is twice the volume of cylinder A, or $2 \times 22 = 44$.

Choice A is incorrect and likely results from not multiplying the radius of cylinder A by 2. Choice B is incorrect and likely results from not squaring the 2 in 2r when applying the volume formula. Choice D is incorrect and likely results from a conceptual error.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The lengths of two sides of a triangle are **4** centimeters and **6** centimeters. If the perimeter of the triangle is **18** centimeters, what is the length, in centimeters, of the third side of this triangle?

Answer

- A. **2**
- B. **8**
- C. **10**
- D. **24**

Correct Answer: B

Rationale

Choice B is correct. The perimeter of a triangle is the sum of the lengths of all three of its sides. It's given that the lengths of two sides of a triangle are **4** centimeters and **6** centimeters. Let *x* represent the length, in centimeters, of the third side of this triangle. The sum of the lengths, in centimeters, of all three sides of the triangle can be represented by the expression **4 + 6 + *x***. Since it's given that the perimeter of the triangle is **18** centimeters, it follows that **4 + 6 + *x* = 18**, or **10 + *x* = 18**. Subtracting **10** from both sides of this equation yields ***x* = 8**. Therefore, the length, in centimeters, of the third side of this triangle is **8**.

Choice A is incorrect. If the length of the third side of this triangle were **2** centimeters, the perimeter, in centimeters, of the triangle would be **4 + 6 + 2**, or **12**, not **18**.

Choice C is incorrect. If the length of the third side of this triangle were **10** centimeters, the perimeter, in centimeters, of the triangle would be **4 + 6 + 10**, or **20**, not **18**.

Choice D is incorrect. If the length of the third side of this triangle were **24** centimeters, the perimeter, in centimeters, of the triangle would be **4 + 6 + 24**, or **34**, not **18**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In convex pentagon $ABCDE$, segment AB is parallel to segment DE . The measure of angle B is 139 degrees, and the measure of angle D is 174 degrees. What is the measure, in degrees, of angle C ?

Correct Answer: 47

Rationale

The correct answer is **47**. It's given that the measure of angle B is 139 degrees. Therefore, the exterior angle formed by extending segment AB at point B has measure $180 - 139$, or **41**, degrees. It's given that segment AB is parallel to segment DE . Extending segment BC at point C and extending segment DE at point D until the two segments intersect results in a transversal that intersects two parallel line segments. One of these intersection points is point B , and let the other intersection point be point X . Since segment AB is parallel to segment DE , alternate interior angles are congruent. Angle CXD and the exterior angle formed by extending segment AB at point B are alternate interior angles. Therefore, the measure of angle CXD is **41** degrees. It's given that the measure of angle D in pentagon $ABCDE$ is 174 degrees. Therefore, angle CDX has measure $180 - 174$, or **6**, degrees. Since angle C in pentagon $ABCDE$ is an exterior angle of triangle CDX , it follows that the measure of angle C is the sum of the measures of angles CDX and CXD . Therefore, the measure, in degrees, of angle C is $6 + 41$, or **47**.

Alternate approach: A line can be created that's perpendicular to segments AB and DE and passes through point C . Extending segments AB and DE at points B and D , respectively, until they intersect this line yields two right triangles. Let these intersection points be point X and point Y , and the two right triangles be triangle BXC and triangle DYC . It's given that the measure of angle B is 139 degrees. Therefore, angle CBX has measure $180 - 139$, or **41**, degrees. Since the measure of angle CBX is **41** degrees and the measure of angle BXC is **90** degrees, it follows that the measure of angle XCB is $180 - 90 - 41$, or **49**, degrees. It's given that the measure of angle D is 174 degrees. Therefore, angle YDC has measure $180 - 174$, or **6**, degrees. Since the measure of angle YDC is **6** degrees and the measure of angle CYD is **90** degrees, it follows that the measure of angle DCY is $180 - 90 - 6$, or **84**, degrees. Since angles XCB , DCY , and angle C in pentagon $ABCDE$ form segment XY , it follows that the sum of the measures of those angles is **180** degrees. Therefore, the measure, in degrees, of angle C is $180 - 49 - 84$, or **47**.

Question ID: b32fec92

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A hemisphere is half of a sphere. If a hemisphere has a radius of **27** inches, which of the following is closest to the volume, in cubic inches, of this hemisphere?

Answer

- A. 1,500
- B. 6,100
- C. 30,900
- D. 41,200

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a sphere is given by $V = \frac{4}{3}\pi r^3$, where r is the radius of the sphere. Since a hemisphere is half of a sphere, it follows that the volume, V , of a hemisphere is given by $V = \left(\frac{1}{2}\right)\left(\frac{4}{3}\right)\pi r^3$, or $V = \frac{2}{3}\pi r^3$. Substituting **27** for r in this formula yields $V = \frac{2}{3}\pi(27)^3$, which gives $V = 13,122\pi$, or V is approximately equal to **41,223.98**. Therefore, the choice that is closest to the volume, in cubic inches, of this hemisphere is **41,200**.

- Choice A is incorrect and may result from conceptual or calculation errors.
- Choice B is incorrect and may result from conceptual or calculation errors.
- Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 10faca1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. The length of each side of triangle T is greater than the length of each side of triangle S. Which of the following could be the value of k ?

Answer

- A. $\frac{29}{28}$
- B. 1
- C. $\frac{28}{29}$
- D. 0

Correct Answer: A

Rationale

Choice A is correct. It's given that each side of equilateral triangle S is multiplied by a scale factor of k to create equilateral triangle T. Since the length of each side of triangle T is greater than the length of each side of triangle S, the scale factor of k must be greater than 1. Of the given choices, only $\frac{29}{28}$ is greater than 1.

Choice B is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of 1, the length of each side of triangle T would be equal to the length of each side of triangle S.

Choice C is incorrect. If each side of equilateral triangle S is multiplied by a scale factor of $\frac{28}{29}$, the length of each side of triangle T would be less than the length of each side of triangle S.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 807408a5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right square prism has a height of **14** units. The volume of the prism is **2,016** cubic units. What is the length, in units, of an edge of the base?

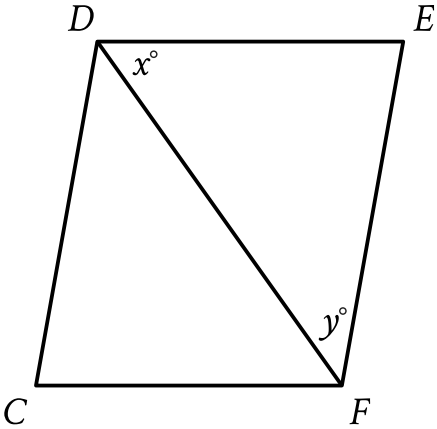
Correct Answer: 12

Rationale

The correct answer is **12**. The volume, V , of a right square prism can be calculated using the formula $V = s^2h$, where s represents the length of an edge of the base and h represents the height of the prism. It's given that the volume of the prism is **2,016** cubic units and the height is **14** units. Substituting **2,016** for V and **14** for h in the formula $V = s^2h$ yields $2,016 = (s^2)(14)$. Dividing both sides of this equation by **14** yields $144 = s^2$. Taking the square root of both sides of this equation yields two solutions: $-12 = s$ and $12 = s$. The length can't be negative, so $12 = s$. Therefore, the length, in units, of an edge of the base is **12**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, $CD = EF$ and $FC = DE$. If $x = 65$ and $y = 30$, what is the measure of $\angle CDE$?

Answer

- A. 30°
- B. 65°
- C. 95°
- D. 130°

Correct Answer: C

Rationale

Choice C is correct. It's given that $CD = EF$ and $FC = DE$, and the figure shows that $\overline{DF}$ is a side of both $\triangle CDF$ and $\triangle EFD$. Since $\triangle CDF$ and $\triangle EFD$ have three pairs of congruent sides, the triangles are congruent, where $\angle CFD$ corresponds to $\angle EDF$ and $\angle CDF$ corresponds to $\angle EFD$. It's given that the measure of $\angle EDF$ is 65° and the measure of $\angle EFD$ is 30° . Since corresponding angles of congruent triangles are congruent, $\angle CFD$ is congruent to $\angle EDF$ and $\angle CDF$ is congruent to $\angle EFD$. Therefore, the measure of $\angle CFD$ is 65° and the measure of $\angle CDF$ is 30° . Since $\angle CDE$ is made up of $\angle CDF$ and $\angle EDF$, it follows that the measure of $\angle CDE$ is equal to the sum of the measures of $\angle CDF$ and $\angle EDF$. Therefore, the measure of $\angle CDE$ is equal to $(30 + 65)^\circ$, or 95° .

Choice A is incorrect. This is the measure of $\angle CDF$, not the measure of $\angle CDE$.

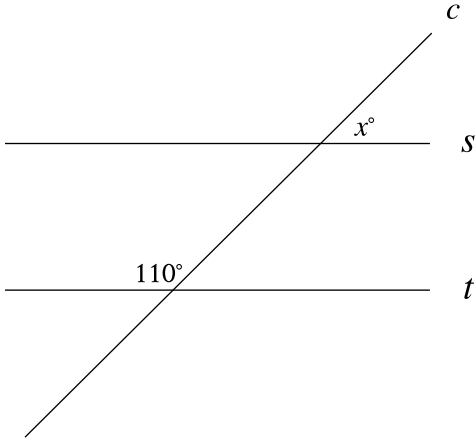
Choice B is incorrect. This is the measure of $\angle CFD$, not the measure of $\angle CDE$.

Choice D is incorrect. This is the sum of the measures of $\angle CFD$ and $\angle EDF$, not $\angle CDF$ and $\angle EDF$.

Question ID: 266cd290

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line *c* intersects parallel lines *s* and *t*. What is the value of *x*?

Correct Answer: 70

Rationale

The correct answer is **70**. Based on the figure, the angle with measure **110°** and the angle vertical to the angle with measure ***x*°** are same side interior angles. Since vertical angles are congruent, the angle vertical to the angle with measure ***x*°** also has measure ***x*°**. It's given that lines *s* and *t* are parallel. Therefore, same side interior angles between lines *s* and *t* are supplementary. It follows that ***x* + 110 = 180**. Subtracting **110** from both sides of this equation yields ***x* = 70**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

In triangle ABC , $AB = 4,680$ millimeters (mm) and $BC = 4,680$ mm. Which statement is sufficient to prove that triangle ABC is equilateral?

Answer

- A. $AC = 4,680$ mm
- B. $AC = 468$ mm
- C. $AC = 46.8$ mm
- D. $AC = 4.68$ mm

Correct Answer: A

Rationale

Choice A is correct. In an equilateral triangle, all three sides have the same length. It's given that in triangle ABC , $AB = 4,680$ mm and $BC = 4,680$ mm. Therefore, if $AC = 4,680$ mm, then all three sides of triangle ABC have the same length, so triangle ABC is equilateral. Therefore, $AC = 4,680$ mm is sufficient to prove that triangle ABC is equilateral.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The length of each edge of a box is **29** inches. Each side of the box is in the shape of a square. The box does not have a lid. What is the exterior surface area, in square inches, of this box without a lid?

Correct Answer: 4205

Rationale

The correct answer is **4,205**. The exterior surface area of a figure is the sum of the areas of all its faces. It's given that the box does not have a lid and that each side of the box is in the shape of a square. Therefore, the box consists of **5** congruent square faces. It's also given that the length of each edge is **29** inches. Let *s* represent the length of an edge of a square. It follows that the area of a square is equal to *s*². Therefore, the area of each of the **5** square faces is equal to **29²**, or **841**, square inches. Since the box consists of **5** congruent square faces, it follows that the sum of the areas of all its faces, or the exterior surface area of this box without a lid, is **5(841)**, or **4,205**, square inches.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

The width of a rectangle is **7** centimeters. The length of the rectangle is **40** centimeters longer than the width. What is the area, in square centimeters, of this rectangle?

Answer

- A. **7**
- B. **14**
- C. **54**
- D. **329**

Correct Answer: D

Rationale

Choice D is correct. It’s given that the width of this rectangle is **7** centimeters and that the length of this rectangle is **40** centimeters longer than the width. Therefore, the length of this rectangle is **7 + 40**, or **47**, centimeters. The area of a rectangle can be found by multiplying its length and its width. Therefore the area, in square centimeters, of this rectangle is **(7) (47)**, or **329**.

Choice A is incorrect. This is the width, in centimeters, not the area, in square centimeters, of this rectangle.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right rectangular prism has a base area of $24t$ square centimeters (cm^2). The length of the base of the rectangular prism is $\frac{8}{3}$ cm, and the height of the rectangular prism is 15 cm. Which expression represents the surface area, in cm^2 , of the right rectangular prism?

Answer

- A. $48t + 160$
- B. $318t + 80$
- C. $1,968t + 80$
- D. $360t$

Correct Answer: B

Rationale

Choice B is correct. It's given that the base area of the right rectangular prism is $24t \text{ cm}^2$ and the length of the base is $\frac{8}{3}$ cm. Dividing the base area by the length yields the width: $\frac{24t}{\frac{8}{3}}$, or $(24t)(\frac{3}{8})$, or $9t$ cm. It's also given that the height of the prism is 15 cm. A right rectangular prism has two rectangular bases and four rectangular lateral faces. The total area of the two bases is $2(24t)$, or $48t \text{ cm}^2$. The four lateral faces include two with dimensions $\frac{8}{3}$ cm by 15 cm and two with dimensions $9t$ cm by 15 cm. The total area of these four lateral faces is $2(\frac{8}{3})(15) + 2(9t)(15) \text{ cm}^2$, which is equivalent to $80 + 270t \text{ cm}^2$. Adding this to the total area of the two bases yields $48t + 270t + 80$, or $318t + 80 \text{ cm}^2$. Thus, the expression, which represents the surface area, in cm^2 , of the right rectangular prism is $318t + 80$.

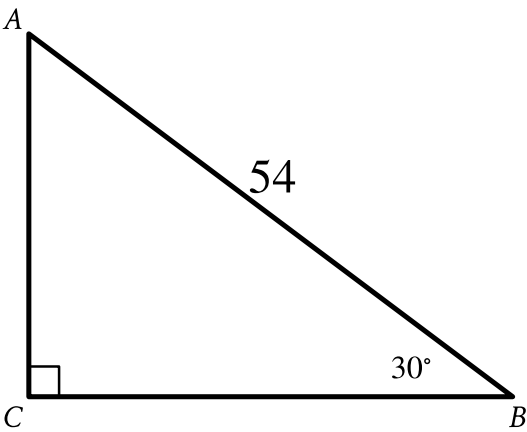
Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the volume, in cm^3 , not the surface area, in cm^2 .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

Right triangle ABC is shown. What is the value of $\tan A$?

Answer

- A. $\frac{\sqrt{3}}{54}$
- B. $\frac{1}{\sqrt{3}}$
- C. $\sqrt{3}$
- D. $27\sqrt{3}$

Correct Answer: C

Rationale

Choice C is correct. In the triangle shown, the measure of angle B is 30° and angle C is a right angle, which means that it has a measure of 90° . Since the sum of the angles in a triangle is equal to 180° , the measure of angle A is equal to $180^\circ - (30 + 90)^\circ$, or 60° . In a right triangle whose acute angles have measures 30° and 60° , the lengths of the legs can be represented by the expressions x , $x\sqrt{3}$, and $2x$, where x is the length of the leg opposite the angle with measure 30° , $x\sqrt{3}$ is the length of the leg opposite the angle with measure 60° , and $2x$ is the length of the hypotenuse. In the triangle shown, the hypotenuse has a length of 54 . It follows that $2x = 54$, or $x = 27$. Therefore, the length of the leg opposite angle B is 27 and the length of the leg opposite angle A is $27\sqrt{3}$. The tangent of an acute angle in a right triangle is defined as the ratio of the length of the leg opposite the angle to the length of the leg adjacent to the angle. The length of the leg opposite angle A is $27\sqrt{3}$ and the length of the leg adjacent to angle A is 27 . Therefore, the value of $\tan A$ is $\frac{27\sqrt{3}}{27}$, or $\sqrt{3}$.

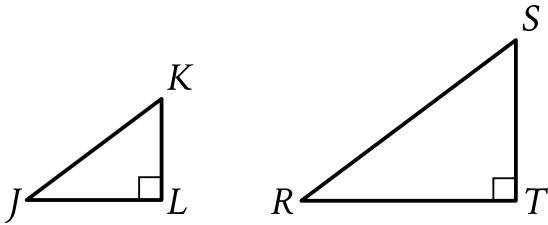
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the value of $\frac{1}{\tan A}$, not the value of $\tan A$.

Choice D is incorrect. This is the length of the leg opposite angle A , not the value of $\tan A$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . The length of $\overline{JK}$ is 15, and the perimeter of triangle JKL is 36. The length of $\overline{RS}$ is 135. What is the perimeter of triangle RST ?

Answer

- A. 324
- B. 540
- C. 2,916
- D. 8,100

Correct Answer: A

Rationale

Choice A is correct. It's given that triangle JKL is similar to triangle RST , where J corresponds to R and K corresponds to S . It follows that $\overline{JK}$ corresponds to $\overline{RS}$. If two triangles are similar, then the scale factor between their perimeters is equal to the scale factor between the lengths of their corresponding sides. It's given that the length of $\overline{JK}$ is 15 and the length of $\overline{RS}$ is 135. Therefore, the scale factor from the length of $\overline{JK}$ to the length of $\overline{RS}$ is $\frac{135}{15}$, or 9. It's given that the perimeter of triangle JKL is 36. Let p represent the perimeter of triangle RST . It follows that $\frac{p}{36} = 9$. Multiplying each side of this equation by 36 yields $p = 324$. Therefore, the perimeter of triangle RST is 324.

Choice B is incorrect and may result from conceptual or calculation errors.

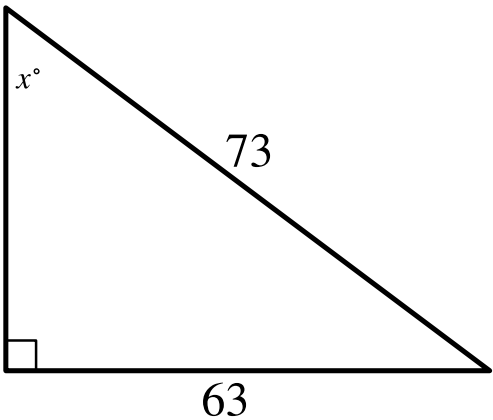
Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: f6da58ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the right triangle shown, what is the value of $\sin x^\circ$?

Answer

- A. $\frac{1}{73}$
- B. $\frac{10}{73}$
- C. $\frac{63}{73}$
- D. $\frac{136}{73}$

Correct Answer: C

Rationale

Choice C is correct. The sine of an acute angle in a right triangle is the ratio of the length of the side opposite that angle to the length of the hypotenuse. In the right triangle shown, it's given that the length of the side opposite the angle with measure x° is 63 units and the length of the hypotenuse is 73 units. Therefore, the value of $\sin x^\circ$ is $\frac{63}{73}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 9ab65399

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. What is the volume, in cubic meters, of the prism?

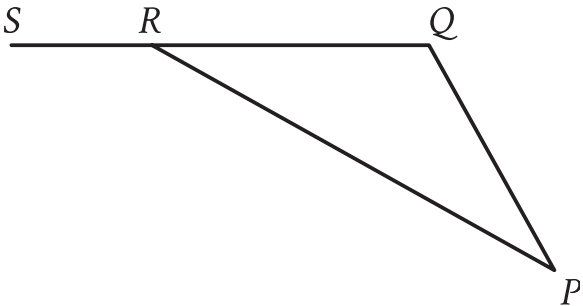
Correct Answer: 880

Rationale

The correct answer is **880**. The volume, V , of a right rectangular prism is given by the formula $V = \ell wh$, where ℓ is the length, w is the width, and h is the height of the prism. It's given that a right rectangular prism has a length of **11** meters, a width of **8** meters, and a height of **10** meters. Substituting **11** for ℓ , **8** for w , and **10** for h in the formula $V = \ell wh$ yields $V = (11)(8)(10)$, or $V = 880$. Therefore, the volume, in cubic meters, of the prism is **880**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In triangle PQR , $\overline{QR}$ is extended to point S . The measure of $\angle PQR$ is 132° , and the measure of $\angle PRS$ is 163° . What is the measure of $\angle QPR$?

Answer

- A. 48°
- B. 31°
- C. 24°
- D. 17°

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, since $\overline{QS}$ is a line segment, the sum of the measures of $\angle PRS$ and $\angle PRQ$ is 180° . It's given that the measure of $\angle PRS$ is 163° . Thus, the measure of $\angle PRQ$ is $(180 - 163)^\circ$, or 17° . The sum of the measures of the interior angles of a triangle is 180° . It's given that the measure of $\angle PQR$ is 132° . Therefore, the measure of $\angle QPR$ is $(180 - 17 - 132)^\circ$, or 31° .

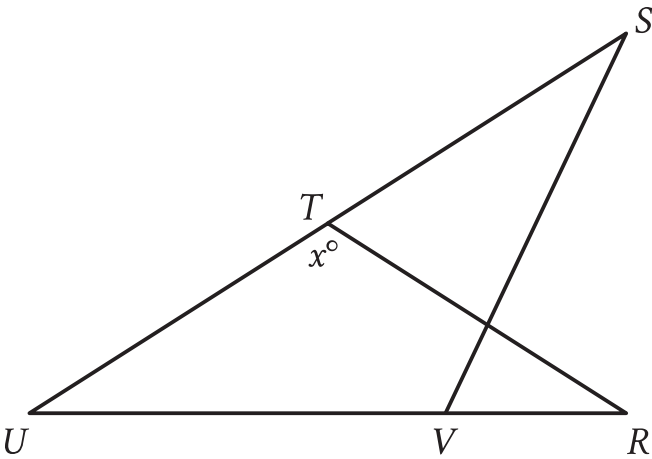
Choice A is incorrect. This is the measure of the supplement of $\angle PQR$, not the measure of $\angle QPR$.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the measure of $\angle PRQ$, not the measure of $\angle QPR$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $RT = TU$, the measure of angle VST is 29° , and the measure of angle RVS is 41° . What is the value of x ?

Correct Answer: 156

Rationale

The correct answer is **156**. In the figure shown, the sum of the measures of angle UVS and angle RVS is 180° . It's given that the measure of angle RVS is 41° . Therefore, the measure of angle UVS is $(180 - 41)^\circ$, or 139° . The sum of the measures of the interior angles of a triangle is 180° . In triangle UVS , the measure of angle UVS is 139° and it's given that the measure of angle VST is 29° . Thus, the measure of angle VUS is $(180 - 139 - 29)^\circ$, or 12° . It's given that $RT = TU$. Therefore, triangle TUR is an isosceles triangle and the measure of VUS is equal to the measure of angle TRU . In triangle TUR , the measure of angle VUS is 12° and the measure of angle TRU is 12° . Thus, the measure of angle UTR is $(180 - 12 - 12)^\circ$, or 156° . The figure shows that the measure of angle UTR is x° , so the value of x is **156**.

Question ID: 36cce7f3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. The surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. What is the value of n ?

Answer

- A. $\frac{9}{7,744}$
- B. $\frac{27}{3,872}$
- C. $\frac{3}{88}$
- D. $\frac{9}{44}$

Correct Answer: A

Rationale

Choice A is correct. Since all cubes are similar to each other, cube X and cube Y are similar figures. In similar figures, if lengths differ by a scale factor of k , then areas differ by a scale factor of k^2 . It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. It follows that the surface area, in square inches, of cube Y is $\left(\frac{3}{88}\right)^2$, or $\frac{9}{7,744}$, times the surface area, in square inches, of cube X. Therefore, the value of n is $\frac{9}{7,744}$.

Alternate approach: The surface area of a cube is $6s^2$, where s is the edge length. Let x represent the edge length, in inches, of cube X. Substituting x for s in $6s^2$ yields $6x^2$. Therefore, the surface area, in square inches, of cube X can be expressed as $6x^2$. It's given that the edge length, in inches, of cube Y is $\frac{3}{88}$ the edge length, in inches, of cube X. Therefore, the edge length, in inches, of cube Y can be expressed as $\left(\frac{3}{88}\right)x$, or $\frac{3x}{88}$. Substituting $\frac{3x}{88}$ for s in $6s^2$ yields $6\left(\frac{3x}{88}\right)^2$, which is equivalent to $6\left(\frac{9x^2}{7,744}\right)$, or $\frac{27x^2}{3,872}$. Therefore, the surface area, in square inches, of cube Y can be expressed as $\frac{27x^2}{3,872}$. It's given that the surface area, in square inches, of cube Y is n times the surface area, in square inches, of cube X. It follows that $\frac{27x^2}{3,872} = n(6x^2)$. Dividing both sides of this equation by $6x^2$ yields $\frac{27x^2}{3,872(6x^2)} = n$, which is equivalent to $\frac{27x^2}{23,232x^2} = n$, or $\frac{9}{7,744} = n$. Therefore, the value of n is $\frac{9}{7,744}$.

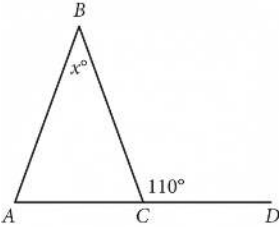
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



In the given figure, $\overline{AC}$ extends to point D. If the measure of $\angle BAC$ is equal to the measure of $\angle BCA$, what is the value of x ?

Answer

- A. 110
- B. 70
- C. 55
- D. 40

Correct Answer: D

Rationale

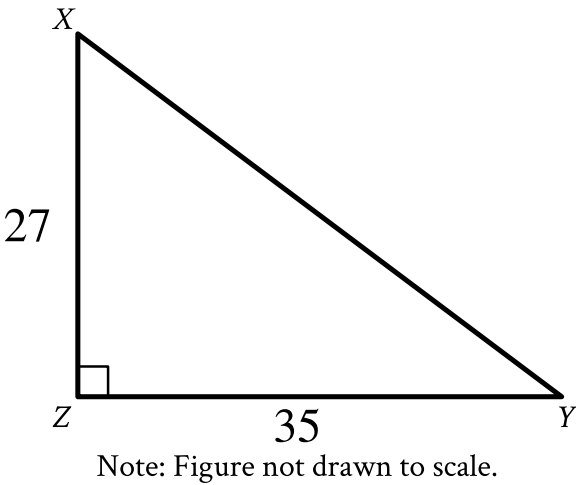
Choice D is correct. Since $\angle BCD$ and $\angle BCA$ form a linear pair of angles, their measures sum to 180° . It's given that the measure of $\angle BCD$ is 110° . Therefore, $110^\circ + \angle BCA = 180^\circ$. Subtracting 110° from both sides of this equation gives the measure of $\angle BCA$ as 70° . It's also given that the measure of $\angle BAC$ is equal to the measure of $\angle BCA$. Thus, the measure of $\angle BAC$ is also 70° . The measures of the interior angles of a triangle sum to 180° . Thus, $70^\circ + 70^\circ + x^\circ = 180^\circ$. Combining like terms on the left-hand side of this equation yields $140^\circ + x^\circ = 180^\circ$. Subtracting 140° from both sides of this equation yields $x^\circ = 40^\circ$, or $x = 40$.

Choice A is incorrect. This is the value of the measure of $\angle BCD$. Choice B is incorrect. This is the value of the measure of each of the other two interior angles, $\angle BCA$ and $\angle BAC$. Choice C is incorrect and may result from an error made when identifying the relationship between the exterior angle of a triangle and the interior angles of the triangle.

Question ID: 7dd7bbe3

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Triangle XYZ shown is a right triangle. Which of the following has the same value as $\sin X$?

Answer

- A. $\tan X$
- B. $\tan Y$
- C. $\cos X$
- D. $\cos Y$

Correct Answer: D

Rationale

Choice D is correct. The sine of an angle is equal to the cosine of its complementary angle. In the triangle shown, angle Z is a right angle; thus, angles X and Y are complementary angles. Therefore, $\cos Y$ has the same value as $\sin X$.

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Question ID: 60a7638d

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z respectively. The length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC . The measure of side AB is 16. What is the measure of side XY ?

Answer

- A. 14
- B. 16
- C. 18
- D. 32

Correct Answer: D

Rationale

Choice D is correct. It's given that triangle ABC is similar to triangle XYZ , such that A , B , and C correspond to X , Y , and Z , respectively. Therefore, side AB corresponds to side XY . Since the length of each side of triangle XYZ is 2 times the length of its corresponding side in triangle ABC , it follows that the measure of side XY is 2 times the measure of side AB . Thus, since the measure of side AB is 16, then the measure of side XY is $2(16)$, or 32.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the measure of side AB , not side XY .

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The circumference of the base of a right circular cylinder is 20π meters, and the height of the cylinder is 6 meters. What is the volume, in cubic meters, of the cylinder?

Answer

- A. 60π
- B. 120π
- C. 600π
- D. $2,400\pi$

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of 6 meters. Therefore, $h = 6$. It's also given that the right circular cylinder has a base with a circumference of 20π meters. The circumference, C , of a circle is given by $C = 2\pi r$, where r is the radius of the circle. Substituting 20π for C in the formula $C = 2\pi r$ yields $20\pi = 2\pi r$. Dividing each side of this equation by 2π yields $10 = r$. Substituting 10 for r and 6 for h in the formula $V = \pi r^2 h$ yields $V = \pi(10)^2(6)$, or $V = 600\pi$. Therefore, the volume, in cubic meters, of the cylinder is 600π .

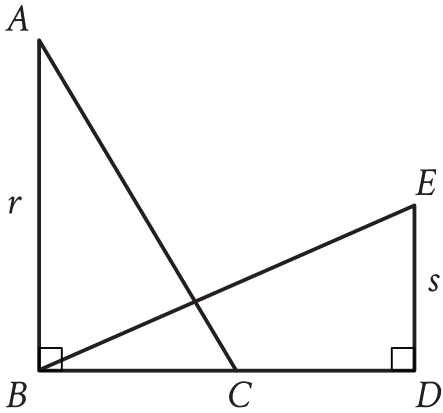
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the lateral surface area, not the volume, of the cylinder.

Choice D is incorrect. This is the result of using the diameter, not the radius, for the value of r in the formula $V = \pi r^2 h$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, $\triangle ABC$ is congruent to $\triangle BDE$. If $r = 54$ and $s = 26$, what is the length of $\overline{CD}$?

Answer

- A. 26
- B. 27
- C. 28
- D. 54

Correct Answer: C

Rationale

Choice C is correct. It's given that $\triangle ABC$ is congruent to $\triangle BDE$. Corresponding sides of congruent triangles are equal in length. It's given that r , which represents the length of $\overline{AB}$, is equal to 54 and s , which represents the length of $\overline{DE}$, is equal to 26. Since $\overline{AB}$ and $\overline{DE}$ aren't equal in length, it follows that $\overline{AB}$ corresponds to $\overline{BD}$ and $\overline{BC}$ corresponds to $\overline{DE}$. Therefore, the length of $\overline{BD}$ is 54 and the length of $\overline{BC}$ is 26. The points B , C , and D lie on a straight line, so the length of $\overline{BD}$ can be expressed as the sum of the lengths of $\overline{BC}$ and $\overline{CD}$. This relationship can be written as $BD = BC + CD$. Substituting 54 for BD and 26 for BC in this equation yields $54 = 26 + CD$. Subtracting 26 from both sides of this equation yields $CD = 28$. Therefore, the length of $\overline{CD}$ is 28.

Choice A is incorrect. This is the length of $\overline{BC}$, not $\overline{CD}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the length of $\overline{BD}$, not $\overline{CD}$.

Question ID: 6fffff4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a surface area of 54 square meters. What is the volume, in cubic meters, of the cube?

Answer

- A. 18
- B. 27
- C. 36
- D. 81

Correct Answer: B

Rationale

Choice B is correct. The surface area of a cube with side length s is equal to $6s^2$. Since the surface area is given as 54 square meters, the equation $54 = 6s^2$ can be used to solve for s . Dividing both sides of the equation by 6 yields $9 = s^2$. Taking the square root of both sides of this equation yields $3 = s$ and $-3 = s$. Since the side length of a cube must be a positive value, $s = -3$ can be discarded as a possible solution, leaving $s = 3$. The volume of a cube with side length s is equal to s^3 . Therefore, the volume of this cube, in cubic meters, is 3^3 , or 27.

Choices A, C, and D are incorrect and may result from calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Right triangles LMN and PQR are similar, where L and M correspond to P and Q , respectively. Angle M has a measure of 53° . What is the measure of angle Q ?

Answer

- A. 37°
- B. 53°
- C. 127°
- D. 143°

Correct Answer: B

Rationale

Choice B is correct. It's given that triangle LMN is similar to triangle PQR . Corresponding angles of similar triangles are congruent. Since angle M and angle Q correspond to each other, they must be congruent. Therefore, if the measure of angle M is 53° , then the measure of angle Q is also 53° .

Choice A is incorrect and may result from concluding that angle M and angle Q are complementary rather than congruent.

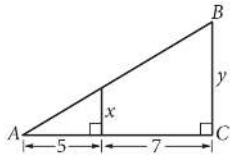
Choice C is incorrect and may result from concluding that angle M and angle Q are supplementary rather than congruent.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 7c3910ec

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

The area of triangle ABC above is at least 48 but no more than 60. If y is an integer, what is one possible value of x ?

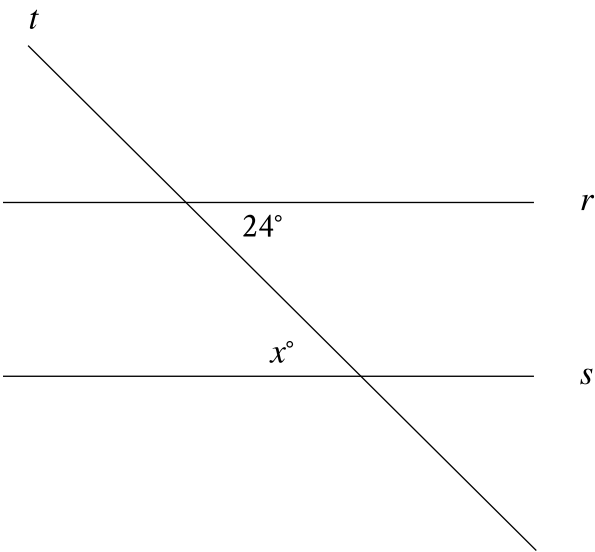
Rationale

The correct answer is either $\frac{10}{3}$, $\frac{15}{4}$, or $\frac{25}{6}$. The area of triangle ABC can be expressed as $\frac{1}{2}(5+7)y$ or $6y$. It's given that the area of triangle ABC is at least 48 but no more than 60. It follows that $48 \leq 6y \leq 60$. Dividing by 6 to isolate y in this compound inequality yields $8 \leq y \leq 10$. Since y is an integer, $y = 8, 9, \text{ or } 10$. In the given figure, the two right triangles shown are similar because they have two pairs of congruent angles: their respective right angles and angle A. Therefore, the following proportion is true: $\frac{x}{y} = \frac{5}{12}$. Substituting 8 for y in the proportion results in $\frac{x}{8} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{10}{3}$. Substituting 9 for y in the proportion results in $\frac{x}{9} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{15}{4}$. Substituting 10 for y in the proportion results in $\frac{x}{10} = \frac{5}{12}$. Cross multiplying and solving for x yields $\frac{25}{6}$. Note that 10/3, 15/4, 25/6, 3.333, 3.75, 4.166, and 4.167 are examples of ways to enter a correct answer.

Question ID: 5b5932a9

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line t intersects lines r and s . Which additional piece of information is sufficient to prove that lines r and s are parallel?

Answer

- A. $x < 24$
- B. $x = 24$
- C. $x > 24$
- D. $x - 24 = 360$

Correct Answer: B

Rationale

Choice B is correct. In the figure shown, transversal t intersects lines r and s . Two lines intersected by a transversal are parallel if and only if a pair of alternate interior angles have equal measure. The angle marked 24° and the angle marked x° form a pair of alternate interior angles. Therefore, $x = 24$ is sufficient to prove that lines r and s are parallel.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Triangle ABC and triangle DEF are similar triangles, where $\overline{AB}$ and $\overline{DE}$ are corresponding sides. If $DE = 2AB$ and the perimeter of triangle ABC is 20, what is the perimeter of triangle DEF ?

Answer

- A. 10
- B. 40
- C. 80
- D. 120

Correct Answer: B

Rationale

Choice B is correct. Since triangles ABC and DEF are similar and $DE = 2AB$, the length of each side of triangle DEF is two times the length of its corresponding side in triangle ABC. Therefore, the perimeter of triangle DEF is two times the perimeter of triangle ABC. Since the perimeter of triangle ABC is 20, the perimeter of triangle DEF is 40.

Choice A is incorrect. This is half, not two times, the perimeter of triangle ABC. Choice C is incorrect. This is two times the perimeter of triangle DEF rather than two times the perimeter of triangle ABC. Choice D is incorrect. This is six times, not two times, the perimeter of triangle ABC.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

In $\triangle RST$, the measure of $\angle R$ is 63° . Which of the following could be the measure, in degrees, of $\angle S$?

Answer

- A. 116
- B. 118
- C. 126
- D. 180

Correct Answer: A

Rationale

Choice A is correct. The sum of the measures of the angles of a triangle is 180° . Therefore, the sum of the measures of $\angle R$, $\angle S$, and $\angle T$ is 180° . It's given that the measure of $\angle R$ is 63° . It follows that the sum of the measures of $\angle S$ and $\angle T$ is $(180 - 63)^\circ$, or 117° . Therefore, the measure of $\angle S$, in degrees, must be less than 117 . Of the given choices, only 116 is less than 117 . Thus, the measure, in degrees, of $\angle S$ could be 116 .

Choice B is incorrect. If the measure of $\angle S$ is 118° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

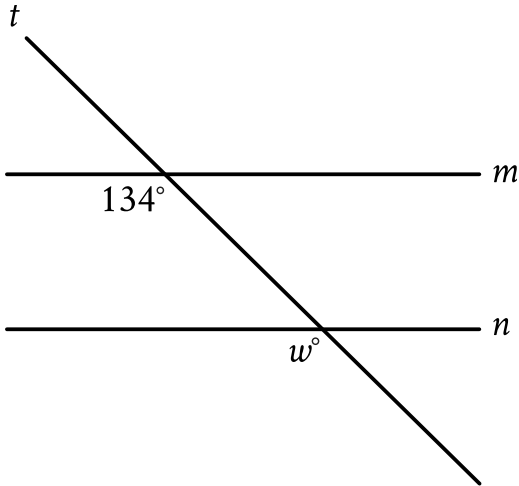
Choice C is incorrect. If the measure of $\angle S$ is 126° , then the sum of the measures of the angles of the triangle is greater than, not equal to, 180° .

Choice D is incorrect. This is the sum of the measures of the angles of a triangle, in degrees.

Question ID: 08de986a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line m is parallel to line n . What is the value of w ?

Answer

- A. 13
- B. 34
- C. 66
- D. 134

Correct Answer: D

Rationale

Choice D is correct. It's given that lines m and n are parallel. Since line t intersects both lines m and n , it's a transversal. The angles in the figure marked as 134° and w° are on the same side of the transversal, where one is an interior angle with line m as a side, and the other is an exterior angle with line n as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that $w^\circ = 134^\circ$. Therefore, the value of w is 134.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC, the measure of angle A is 50° . If triangle ABC is isosceles, which of the following is NOT a possible measure of angle B ?

Answer

- A. 50°
- B. 65°
- C. 80°
- D. 100°

Correct Answer: D

Rationale

Choice D is correct. The sum of the three interior angles in a triangle is 180° . It's given that angle A measures 50° . If angle B measured 100° , the measure of angle C would be $180^\circ - (50^\circ + 100^\circ) = 30^\circ$. Thus, the measures of the angles in the triangle would be 50° , 100° , and 30° . However, an isosceles triangle has two angles of equal measure. Therefore, angle B can't measure 100° .

Choice A is incorrect. If angle B has measure 50° , then angle C would measure $180^\circ - (50^\circ + 50^\circ) = 80^\circ$, and 50° , 50° , and 80° could be the angle measures of an isosceles triangle. Choice B is incorrect. If angle B has measure 65° , then angle C would measure $180^\circ - (65^\circ + 50^\circ) = 65^\circ$, and 50° , 65° , and 65° could be the angle measures of an isosceles triangle. Choice C is incorrect. If angle B has measure 80° , then angle C would measure $180^\circ - (80^\circ + 50^\circ) = 50^\circ$, and 50° , 80° , and 50° could be the angle measures of an isosceles triangle.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A rectangle has a length of **3** units and a width of **39** units. Which expression gives the area, in square units, of this rectangle?

Answer

- A. **$2(3 + 39)$**
- B. **$2(3 \cdot 39)$**
- C. **$3 + 39$**
- D. **$3 \cdot 39$**

Correct Answer: D

Rationale

Choice D is correct. The area of a rectangle is given by ℓw , where ℓ is the length of the rectangle and w is the width of the rectangle. It's given that a rectangle has a length of **3** units and a width of **39** units. It follows that the area of the rectangle is **$3 \cdot 39$** square units. Therefore, the expression that gives the area, in square units, of this rectangle, is **$3 \cdot 39$** .

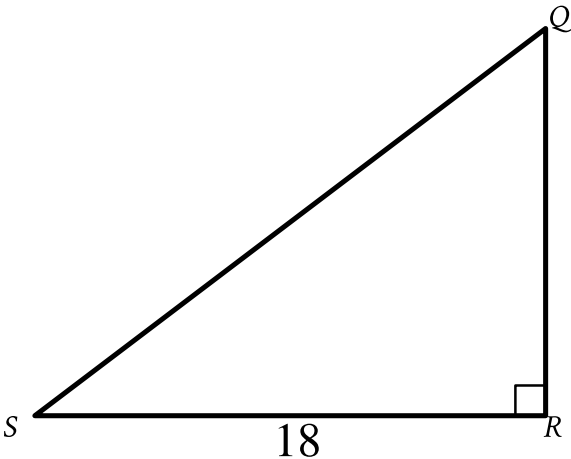
Choice A is incorrect. This expression gives the perimeter, in units, of this rectangle.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In triangle QRS shown, $QR < RS$. Which expression represents the length of $\overline{QS}$?

- Answer
- A. $18 \cos Q$
 - B. $18 \sin Q$
 - C. $\frac{18}{\cos Q}$
 - D. $\frac{18}{\sin Q}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that triangle QRS is a right triangle. Each of the given choices is an expression containing $\sin Q$ or $\cos Q$. For an acute angle in a right triangle, the sine of the angle is the length of the opposite leg divided by the length of the hypotenuse, and the cosine of the angle is the length of the adjacent leg divided by the length of the hypotenuse. It follows that $\sin Q = \frac{RS}{QS}$ and $\cos Q = \frac{QR}{QS}$, where $QR < RS$. It's given in the figure that the length of side RS is 18. Since only the equation $\sin Q = \frac{RS}{QS}$ contains RS , an expression representing QS can be found by substituting 18 for RS in this equation, which yields $\sin Q = \frac{18}{QS}$. Multiplying both sides of this equation by QS yields $QS \cdot \sin Q = 18$. Dividing both sides of this equation by $\sin Q$ yields $QS = \frac{18}{\sin Q}$. Therefore, the expression $\frac{18}{\sin Q}$ represents the length of $\overline{QS}$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

What is the area, in square centimeters, of a rectangle with a length of **36** centimeters and a width of **34** centimeters?

Answer

- A. **70**
- B. **140**
- C. **1,156**
- D. **1,224**

Correct Answer: D

Rationale

Choice D is correct. The area **A** , in square centimeters, of a rectangle can be found using the formula **$A = \ell w$** , where **ℓ** is the length, in centimeters, of the rectangle and **w** is its width, in centimeters. It's given that the rectangle has a length of **36** centimeters and a width of **34** centimeters. Substituting **36** for **ℓ** and **34** for **w** in the formula **$A = \ell w$** yields **$A = 36(34)$** , or **$A = 1,224$** . Therefore, the area, in square centimeters, of this rectangle is **1,224**.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, in centimeters, not the area, in square centimeters, of the rectangle.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 694b2d44

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

At a certain time and day, the Washington Monument in Washington, DC, casts a shadow that is 300 feet long. At the same time, a nearby cherry tree casts a shadow that is 16 feet long. Given that the Washington Monument is approximately 555 feet tall, which of the following is closest to the height, in feet, of the cherry tree?

Answer

- A. 10
- B. 20
- C. 30
- D. 35

Rationale

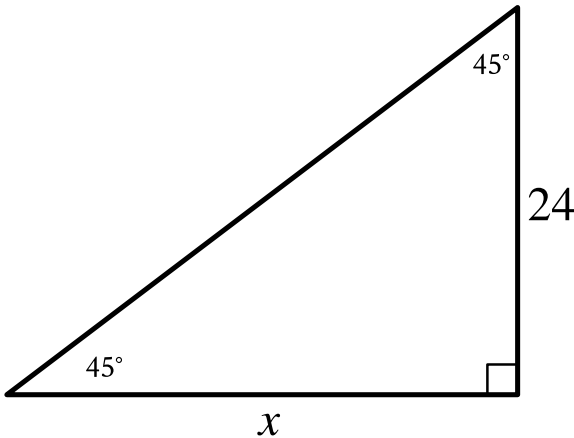
Choice C is correct. There is a proportional relationship between the height of an object and the length of its shadow. Let c represent the height, in feet, of the cherry tree. The given relationship can be expressed by the proportion $\frac{555}{300} = \frac{c}{16}$. Multiplying both sides of this equation by 16 yields $c = 29.6$. This height is closest to the value given in choice C, 30.

Choices A, B, and D are incorrect and may result from calculation errors.

Question ID: 240a55b5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of x ?

Answer

- A. 24
- B. 45
- C. 48
- D. 69

Correct Answer: A

Rationale

Choice A is correct. Since the two acute angles have the same measure and the third angle is a right angle, the triangle shown is an isosceles right triangle. In an isosceles right triangle, the two legs have the same length. The figure shows that the length of one leg of the triangle is 24 and the length of the other leg of the triangle is x . It follows that the value of x is 24.

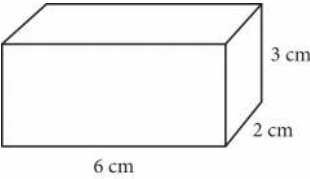
Choice B is incorrect. This is the measure, in degrees, of one of the angles shown.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question



The figure shows the lengths, in centimeters (cm), of the edges of a right rectangular prism. The volume V of a right rectangular prism is ℓwh , where ℓ is the length of the prism, w is the width of the prism, and h is the height of the prism. What is the volume, in cubic centimeters, of the prism?

Answer

- A. 36
- B. 24
- C. 12
- D. 11

Correct Answer: A

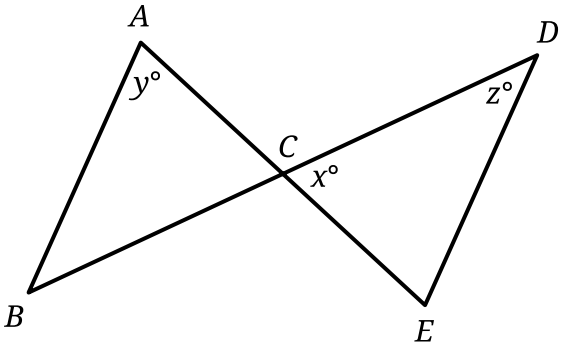
Rationale

Choice A is correct. It’s given that the volume of a right rectangular prism is ℓwh . The prism shown has a length of 6 cm, a width of 2 cm, and a height of 3 cm. Thus, $\ell wh = (6)(2)(3)$, or 36 cubic centimeters.

Choice B is incorrect. This is the volume of a rectangular prism with edge lengths of 6, 2, and 2. Choice C is incorrect and may result from only finding the product of the length and width of the base of the prism. Choice D is incorrect and may result from finding the sum, not the product, of the edge lengths of the prism.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure, $\overline{AE}$ intersects $\overline{DB}$ at point C and $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . If $x = 63$ and $y = 2x - 47$, what is the value of z ?

Answer

- A. 38
- B. 47
- C. 63
- D. 79

Correct Answer: A

Rationale

Choice A is correct. It's given that $\overline{AE}$ intersects $\overline{DB}$ at point C and that $\overline{AB}$ is parallel to $\overline{DE}$, creating triangles ABC and CDE . Since $\overline{AB}$ is parallel to $\overline{DE}$, angles BAC and DEC are opposite interior angles and are congruent. It follows that the measure of angle DEC is y° . This means the three interior angles of triangle CDE have measures of x° , z° , and y° , respectively. It's given that $x = 63$ and $y = 2x - 47$. Substituting 63 for x in the equation $y = 2x - 47$ yields $y = 2(63) - 47$, or $y = 79$. Therefore, the three interior angles of triangle CDE have measures 63° , z° , and 79° , respectively. The sum of the interior angles of a triangle is 180° . Therefore, $63 + z + 79 = 180$, or $z + 142 = 180$. Subtracting 142 from both sides of this equation yields $z = 38$. Therefore, the value of z is 38 .

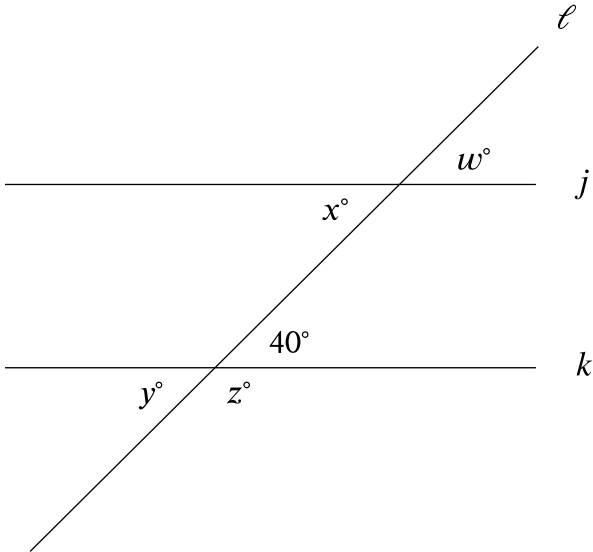
Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the value of x , not z .

Choice D is incorrect. This is the value of y , not z .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.

In the figure shown, line ℓ intersects lines j and k . Which additional piece of information is sufficient to prove that lines j and k are parallel?

- Answer
- A. $w = 40$
 - B. $x = 140$
 - C. $y = 40$
 - D. $z = 140$

Correct Answer: A

Rationale

Choice A is correct. In the figure shown, lines j and k are parallel if and only if a pair of corresponding angles are congruent. It's given that one angle has a measure of 40° and that the corresponding angle has a measure of w° . Therefore, $w = 40$ is sufficient to prove that lines j and k are parallel.

Choice B is incorrect. The angle measuring x° and the angle measuring 40° are alternate interior angles. Thus, if lines j and k are parallel, x is equal to 40 , not 140 .

Choice C is incorrect. The angle measuring y° and the angle measuring 40° are vertical angles. Thus, $y = 40$, whether lines j and k are parallel or not.

Choice D is incorrect. The angle measuring z° is supplementary to the angle measuring 40° . Thus, $z = 180 - 40$, or $z = 140$, whether lines j and k are parallel or not.

Question ID: 8debb46f

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Easy

Question

What is the area, in square centimeters, of a rectangle with a length of **34 centimeters** (**cm**) and a width of **29 cm**?

Correct Answer: 986

Rationale

The correct answer is **986**. The area, **A** , of a rectangle is given by **Math output error**, where **Math output error** is the length of the rectangle and **w** is its width. It's given that the length of the rectangle is **34** centimeters (cm) and the width is **29** cm. Substituting **34** for **Math output error** and **29** for **w** in the equation **Math output error** yields **$A = (34)(29)$** , or **$A = 986$** . Therefore, the area, in square centimeters, of this rectangle is **986**.

Question ID: ff91d3f5

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

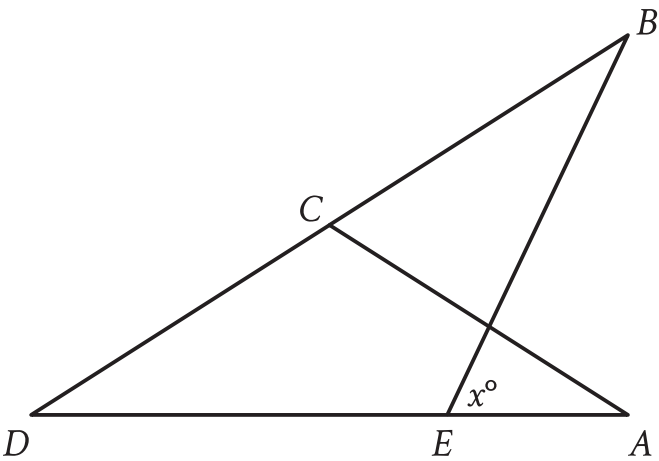
A triangle with angle measures 30° , 60° , and 90° has a perimeter of $18 + 6\sqrt{3}$. What is the length of the longest side of the triangle?

Rationale

The correct answer is 12. It is given that the triangle has angle measures of 30° , 60° , and 90° , and so the triangle is a special right triangle. The side measures of this type of special triangle are in the ratio $2:1:\sqrt{3}$. If x is the measure of the shortest leg, then the measure of the other leg is $\sqrt{3}x$ and the measure of the hypotenuse is $2x$. The perimeter of the triangle is given to be $18 + 6\sqrt{3}$, and so the equation for the perimeter can be written as $2x + x + \sqrt{3}x = 18 + 6\sqrt{3}$. Combining like terms and factoring out a common factor of x on the left-hand side of the equation gives $(3 + \sqrt{3})x = 18 + 6\sqrt{3}$. Rewriting the right-hand side of the equation by factoring out 6 gives $(3 + \sqrt{3})x = 6(3 + \sqrt{3})$. Dividing both sides of the equation by the common factor $(3 + \sqrt{3})$ gives $x = 6$. The longest side of the right triangle, the hypotenuse, has a length of $2x$, or $2(6)$, which is 12.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $AC = CD$. The measure of angle EBC is 45° , and the measure of angle ACD is 104° . What is the value of x ?

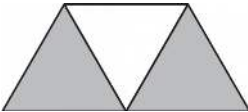
Correct Answer: 83

Rationale

The correct answer is **83**. It's given that in the figure, $AC = CD$. Thus, triangle ACD is an isosceles triangle and the measure of angle CDA is equal to the measure of angle CAD . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle ACD is 180° . It's given that the measure of angle ACD is 104° . It follows that the sum of the measures of angles CDA and CAD is $(180 - 104)^\circ$, or 76° . Since the measure of angle CDA is equal to the measure of angle CAD , the measure of angle CDA is half of 76° , or 38° . The sum of the measures of the interior angles of triangle BDE is 180° . It's given that the measure of angle EBC is 45° . Since the measure of angle BDE , which is the same angle as angle CDA , is 38° , it follows that the measure of angle DEB is $(180 - 45 - 38)^\circ$, or 97° . Since angle DEB and angle AEB form a straight line, the sum of the measures of these angles is 180° . It's given in the figure that the measure of angle AEB is x° . It follows that $97 + x = 180$. Subtracting **97** from both sides of this equation yields $x = \mathbf{83}$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



A graphic designer is creating a logo for a company. The logo is shown in the figure above. The logo is in the shape of a trapezoid and consists of three congruent equilateral triangles. If the perimeter of the logo is 20 centimeters, what is the combined area of the shaded regions, in square centimeters, of the logo?

Answer

- A. $2\sqrt{3}$
- B. $4\sqrt{3}$
- C. $8\sqrt{3}$
- D. 16

Correct Answer: C

Rationale

Choice C is correct. It's given that the logo is in the shape of a trapezoid that consists of three congruent equilateral triangles, and that the perimeter of the trapezoid is 20 centimeters (cm). Since the perimeter of the trapezoid is the sum of the lengths of 5 of the sides of the triangles, the length of each side of an equilateral triangle is $\frac{20}{5} = 4$ cm. Dividing up one equilateral triangle into two right triangles yields a pair of congruent 30°-60°-90° triangles. The shorter leg of each right triangle is half the length of the side of an equilateral triangle, or 2 cm. Using the Pythagorean Theorem, $a^2 + b^2 = c^2$, the height of the equilateral triangle can be found. Substituting $a = 2$ and $c = 4$ and solving for b yields $\sqrt{4^2 - 2^2} = \sqrt{12} = 2\sqrt{3}$ cm. The area of one equilateral triangle is $\frac{1}{2}bh$, where $b = 2$ and $h = 2\sqrt{3}$. Therefore, the area of one equilateral triangle is $\frac{1}{2}(4)(2\sqrt{3}) = 4\sqrt{3}$ cm². The shaded area consists of two such triangles, so its area is $(2)(4)\sqrt{3} = 8\sqrt{3}$ cm².

Alternate approach: The area of a trapezoid can be found by evaluating the expression $\frac{1}{2}(b_1 + b_2)h$, where b_1 is the length of one base, b_2 is the length of the other base, and h is the height of the trapezoid. Substituting $b_1 = 8$, $b_2 = 4$, and $h = 2\sqrt{3}$ yields the expression $\frac{1}{2}(8 + 4)(2\sqrt{3})$, or $\frac{1}{2}(12)(2\sqrt{3})$, which gives an area of $12\sqrt{3}$ cm² for the trapezoid. Since two-thirds of the trapezoid is shaded, the area of the shaded region is $\frac{2}{3} \times 12\sqrt{3} = 8\sqrt{3}$.

Choice A is incorrect. This is the height of the trapezoid. Choice B is incorrect. This is the area of one of the equilateral triangles, not two. Choice D is incorrect and may result from using a height of 4 for each triangle rather than the height of $2\sqrt{3}$.

Question ID: c0b26ea8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

The perimeter of an equilateral triangle is **852** centimeters. The three vertices of the triangle lie on a circle. The radius of the circle is $w\sqrt{3}$ centimeters. What is the value of w ?

Correct Answer: 284/3, 94.66, 94.67

Rationale

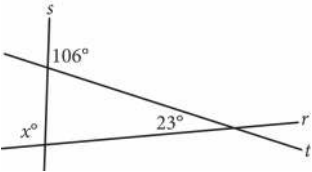
The correct answer is $\frac{284}{3}$. Since the perimeter of a triangle is the sum of the lengths of its sides, and the given triangle is equilateral, the length of each side is $\frac{852}{3}$, or **284** centimeters (cm). Right triangle AMO can be formed, where M is the midpoint of one of the triangle's sides, A is one of this side's endpoints, and O is the center of the circle. It follows that AM is $\frac{284}{2}$, or **142**, cm. Additionally, triangle AMO has angles measuring 30° , 60° , and 90° , where the measure of angle OMA is 90° and the measure of angle OAM is 30° . It follows that the length of side MO is half the length of hypotenuse AO , and the length of side AM is $\sqrt{3}$ times the length of side MO . It's given that $AO = w\sqrt{3}$ cm. Therefore, $MO = \frac{w\sqrt{3}}{2}$ cm and $AM = \frac{w\sqrt{3}\sqrt{3}}{2}$ cm, which is equivalent to $AM = \frac{3w}{2}$ cm. Since $AM = 142$ cm, it follows that $\frac{3w}{2} = 142$. Multiplying both sides of this equation by **2** yields $3w = 284$. Dividing both sides of this equation by **3** yields $w = \frac{284}{3}$. Note that 284/3, 94.66, and 94.67 are examples of ways to enter a correct answer.

Question ID: 5e7aa61a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Intersecting lines r, s, and t are shown below.



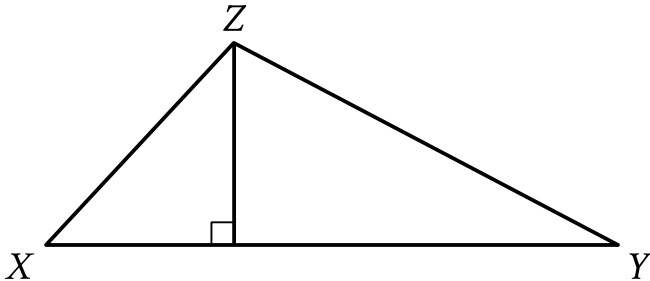
What is the value of x ?

Rationale

The correct answer is 97. The intersecting lines form a triangle, and the angle with measure of x° is an exterior angle of this triangle. The measure of an exterior angle of a triangle is equal to the sum of the measures of the two nonadjacent interior angles of the triangle. One of these angles has measure of 23° and the other, which is supplementary to the angle with measure 106° , has measure of $180^\circ - 106^\circ = 74^\circ$. Therefore, the value of x is $23 + 74 = 97$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the figure shown, the measure of angle X is 54° . The length of $\overline{XY}$ is 26 units and the length of $\overline{XZ}$ is 19 units. What is the area, in square units, of triangle XYZ ?

- Answer
- A. 247
 - B. 494
 - C. $247 \sin 54^\circ$
 - D. $494 \sin 54^\circ$

Correct Answer: C

Rationale

Choice C is correct. The area, A , of a triangle is given by the formula $A = \frac{1}{2}bh$, where b is the length of a base of the triangle and h is its corresponding height. It's given that the length of $\overline{XY}$ is 26 units. This length represents a base of the triangle, so $b = 26$. Let h represent its corresponding height. It's given that the measure of angle X is 54° and the length of $\overline{XZ}$ is 19 units. It follows that $\sin 54^\circ = \frac{h}{19}$. Multiplying both sides of this equation by 19 yields $19 \sin 54^\circ = h$. Substituting 26 for b and $19 \sin 54^\circ$ for h in the formula $A = \frac{1}{2}bh$ yields $A = \frac{1}{2}(26)(19 \sin 54^\circ)$, or $A = 247 \sin 54^\circ$. Therefore, the area, in square units, of triangle XYZ is $247 \sin 54^\circ$.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. What is the slant height, in centimeters, of this cone?

Answer

- A. 12
- B. 36
- C. 77
- D. 85

Correct Answer: D

Rationale

Choice D is correct. The volume, V , of a right circular cone is given by the formula $V = \frac{1}{3}\pi r^2 h$, where πr^2 is the area of the circular base of the cone and h is the height. It's given that this right circular cone has a volume of $71,148\pi$ cubic centimeters and the area of its base is $5,929\pi$ square centimeters. Substituting $71,148\pi$ for V and $5,929\pi$ for πr^2 in the formula $V = \frac{1}{3}\pi r^2 h$ yields $71,148\pi = (\frac{1}{3})(5,929\pi)(h)$. Dividing each side of this equation by $5,929\pi$ yields $12 = \frac{h}{3}$. Multiplying each side of this equation by 3 yields $36 = h$. Let s represent the slant height, in centimeters, of this cone. A right triangle is formed by the radius, r , height, h , and slant height, s , of this cone, where r and h are the legs of the triangle and s is the hypotenuse. Using the Pythagorean theorem, the equation $r^2 + h^2 = s^2$ represents this relationship. Because $5,929\pi$ is the area of the base and the area of the base is πr^2 , it follows that $5,929\pi = \pi r^2$. Dividing both sides of this equation by π yields $5,929 = r^2$. Substituting 5,929 for r^2 and 36 for h in the equation $r^2 + h^2 = s^2$ yields $5,929 + 36^2 = s^2$, which is equivalent to $5,929 + 1,296 = s^2$, or $7,225 = s^2$. Taking the positive square root of both sides of this equation yields $85 = s$. Therefore, the slant height of the cone is 85 centimeters.

Choice A is incorrect. This is one-third of the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice B is incorrect. This is the height, in centimeters, not the slant height, in centimeters, of this cone.

Choice C is incorrect. This is the radius, in centimeters, of the base, not the slant height, in centimeters, of this cone.

Question ID: 2635fa9a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The length of each side of a square is **94** centimeters (cm). Which expression gives the area, in **cm²**, of the square?

Answer

- A. **2 · 94**
- B. **2 · 94 · 94**
- C. **4 · 94**
- D. **94 · 94**

Correct Answer: D

Rationale

Choice D is correct. The area of a square is given by ***s*²**, where ***s*** is the length of each side of the square. It's given that the length of each side of a square is **94 cm**. It follows that the area, in **cm²**, of the square is **(94)²**, or **94 · 94**. Therefore, the expression that gives the area, in **cm²**, of the square is **94 · 94**.

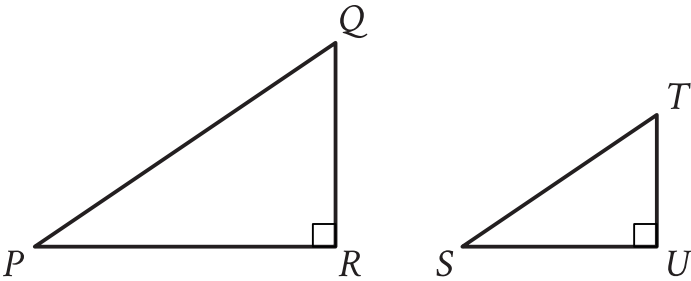
Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression gives the perimeter, in **cm**, of the square.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figures not drawn to scale.

Right triangles PQR and STU are similar, where P corresponds to S . If the measure of angle Q is 18° , what is the measure of angle S ?

Answer

- A. 18°
- B. 72°
- C. 82°
- D. 162°

Correct Answer: B

Rationale

Choice B is correct. In similar triangles, corresponding angles are congruent. It's given that right triangles PQR and STU are similar, where angle P corresponds to angle S . It follows that angle P is congruent to angle S . In the triangles shown, angle R and angle U are both marked as right angles, so angle R and angle U are corresponding angles. It follows that angle Q and angle T are corresponding angles, and thus, angle Q is congruent to angle T . It's given that the measure of angle Q is 18° , so the measure of angle T is also 18° . Angle U is a right angle, so the measure of angle U is 90° . The sum of the measures of the interior angles of a triangle is 180° . Thus, the sum of the measures of the interior angles of triangle STU is 180 degrees. Let s represent the measure, in degrees, of angle S . It follows that $s + 18 + 90 = 180$, or $s + 108 = 180$. Subtracting 108 from both sides of this equation yields $s = 72$. Therefore, if the measure of angle Q is 18 degrees, then the measure of angle S is 72 degrees.

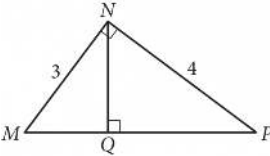
Choice A is incorrect. This is the measure of angle T .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is the sum of the measures of angle S and angle U .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



In the figure above, what is the length of $\overline{NQ}$?

Answer

- A. 2.2
- B. 2.3
- C. 2.4
- D. 2.5

Correct Answer: C

Rationale

Choice C is correct. First, $\overline{MP}$ is the hypotenuse of right $\triangle MNP$, whose legs have lengths 3 and 4. Therefore, $(MP)^2 = 3^2 + 4^2$, so $(MP)^2 = 25$ and $MP = 5$. Second, because $\angle MNP$ corresponds to $\angle NQP$ and because $\angle MPN$ corresponds to $\angle NPQ$, $\triangle MNP$ is similar to $\triangle NQP$. The ratio of corresponding sides of similar triangles is constant, so $\frac{NQ}{MN} = \frac{NP}{MP}$. Since $MP = 5$ and it's given that $MN = 3$ and $NP = 4$, $\frac{NQ}{3} = \frac{4}{5}$. Solving for NQ results in $NQ = \frac{12}{5}$, or 2.4.

Choices A, B, and D are incorrect and may result from setting up incorrect ratios.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

The base of a right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters. The height of this pyramid is **7** centimeters. What is the volume, in cubic centimeters, of this pyramid?

Correct Answer: 126

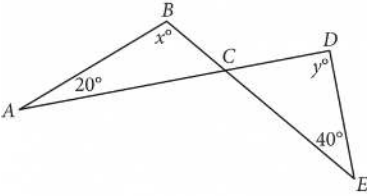
Rationale

The correct answer is **126**. It's given that the base of the right rectangular pyramid has a length of **27** centimeters and a width of **2** centimeters and that the height of the pyramid is **7** centimeters. The volume, V , of a right rectangular pyramid is given by the formula **Math output error**, where **Math output error** and w are the length and width, respectively, of the pyramid's base and h is the height of the pyramid. Substituting **27** for **Math output error**, **2** for w , and **7** for h in this formula yields $V = \frac{1}{3}(27)(2)(7)$, or $V = 126$. Therefore, the volume, in cubic centimeters, of the pyramid is **126**.

Question ID: c3a4b363

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question



Note: Figure not drawn to scale.
In the figure above, $\overline{AD}$ intersects $\overline{BE}$ at C. If $x = 100$, what is the value of y ?

Answer

- A. 100
- B. 90
- C. 80
- D. 60

Correct Answer: C

Rationale

Choice C is correct. It's given that $x = 100$; therefore, substituting 100 for x in triangle ABC gives two known angle measures for this triangle. The sum of the measures of the interior angles of any triangle equals 180° . Subtracting the two known angle measures of triangle ABC from 180° gives the third angle measure: $180^\circ - 100^\circ - 20^\circ = 60^\circ$. This is the measure of angle BCA. Since vertical angles are congruent, the measure of angle DCE is also 60° . Subtracting the two known angle measures of triangle CDE from 180° gives the third angle measure: $180^\circ - 60^\circ - 40^\circ = 80^\circ$. Therefore, the value of y is 80.

Choice A is incorrect and may result from a calculation error. Choice B is incorrect and may result from classifying angle CDE as a right angle. Choice D is incorrect and may result from finding the measure of angle BCA or DCE instead of the measure of angle CDE.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

Quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. The measure of angle P is 30° , the measure of angle Q is 50° , and the measure of angle R is 70° . The length of each side of $P'Q'R'S'$ is **3** times the length of each corresponding side of $PQRS$. What is the measure of angle P' ?

Answer

- A. 10°
- B. 30°
- C. 40°
- D. 90°

Correct Answer: B

Rationale

Choice B is correct. It's given that quadrilateral $P'Q'R'S'$ is similar to quadrilateral $PQRS$, where P, Q, R , and S correspond to P', Q', R' , and S' , respectively. Since corresponding angles of similar quadrilaterals are congruent, it follows that the measure of angle P is equal to the measure of angle P' . It's given that the measure of angle P is 30° . Therefore, the measure of angle P' is 30° .

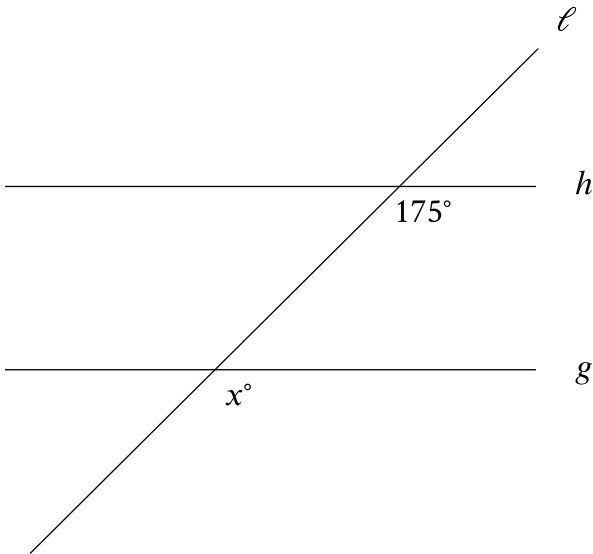
Choice A is incorrect. This is $\frac{1}{3}$ the measure of angle P' .

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect. This is **3** times the measure of angle P' .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure shown, line g is parallel to line h . Line **Math output error** intersects both lines. What is the value of x ?

Correct Answer: 175

Rationale

The correct answer is **175**. It's given that line g is parallel to line h . Since line **Math output error** intersects both lines, it's a transversal. The angles in the figure marked as **175°** and x° are on the same side of the transversal, where one is an interior angle with line h as a side, and the other is an exterior angle with line g as a side, respectively. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles have equal measure. Therefore, the value of x is **175**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Rectangle $ABCD$ is similar to rectangle $EFGH$. The area of rectangle $ABCD$ is **648** square inches, and the area of rectangle $EFGH$ is **72** square inches. The length of the longest side of rectangle $ABCD$ is **36** inches. What is the length, in inches, of the longest side of rectangle $EFGH$?

Answer

- A. **4**
- B. **9**
- C. **12**
- D. **36**

Correct Answer: C

Rationale

Choice C is correct. It's given that rectangle $ABCD$ is similar to rectangle $EFGH$. Therefore, if the length of each side of rectangle $ABCD$ is k times the length of the corresponding side of rectangle $EFGH$, then the area of rectangle $ABCD$ is k^2 times the area of rectangle $EFGH$. It's given that the area of rectangle $ABCD$ is **648** square inches and the area of rectangle $EFGH$ is **72** square inches. It follows that $k^2 = \frac{648}{72}$, or $k^2 = 9$. Taking the square root of each side of this equation yields $k = \sqrt{9}$, or $k = 3$. It follows that the length of each side of rectangle $ABCD$ is **3** times the length of the corresponding side of rectangle $EFGH$. It's given that the length of the longest side of rectangle $ABCD$ is **36** inches. Therefore, **36** inches is **3** times the length of the longest side of rectangle $EFGH$, and the longest side of rectangle $EFGH$ is equal to $\frac{36}{3}$, or **12**, inches.

Choice A is incorrect. This is the length, in inches, of the longest side of a rectangle with side lengths that are $\frac{1}{9}$ the corresponding side lengths of rectangle $ABCD$, rather than a rectangle with an area that is $\frac{1}{9}$ the area of rectangle $ABCD$.

Choice B is incorrect. This is the factor by which the area of rectangle $ABCD$ is larger than the area of rectangle $EFGH$, not the length, in inches, of the longest side of rectangle $EFGH$.

Choice D is incorrect. This is the length, in inches, of the longest side of rectangle $ABCD$, not rectangle $EFGH$.

Question ID: 1878bade

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 27° . What is the measure, in degrees, of angle F ?

Correct Answer: 63

Rationale

The correct answer is **63**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 27° . It follows that the measure of angle D is also 27° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $27 + 90 + x = 180$, or $117 + x = 180$. Subtracting **117** from both sides of this equation yields $x = 63$. Therefore, the measure, in degrees, of angle F is **63**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question

A rectangle is inscribed in a circle, such that each vertex of the rectangle lies on the circumference of the circle. The diagonal of the rectangle is twice the length of the shortest side of the rectangle. The area of the rectangle is $1,089\sqrt{3}$ square units. What is the length, in units, of the diameter of the circle?

Correct Answer: 66

Rationale

The correct answer is **66**. It's given that each vertex of the rectangle lies on the circumference of the circle. Therefore, the length of the diameter of the circle is equal to the length of the diagonal of the rectangle. The diagonal of a rectangle forms a right triangle with the shortest and longest sides of the rectangle, where the shortest side and the longest side of the rectangle are the legs of the triangle and the diagonal of the rectangle is the hypotenuse of the triangle. Let s represent the length, in units, of the shortest side of the rectangle. Since it's given that the diagonal is twice the length of the shortest side, $2s$ represents the length, in units, of the diagonal of the rectangle. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $a^2 + b^2 = c^2$. Substituting s for a and $2s$ for c in this equation yields $s^2 + b^2 = (2s)^2$, or $s^2 + b^2 = 4s^2$. Subtracting s^2 from both sides of this equation yields $b^2 = 3s^2$. Taking the positive square root of both sides of this equation yields $b = s\sqrt{3}$. Therefore, the length, in units, of the rectangle's longest side is $s\sqrt{3}$. The area of a rectangle is the product of the length of the shortest side and the length of the longest side. The lengths, in units, of the shortest and longest sides of the rectangle are represented by s and $s\sqrt{3}$, and it's given that the area of the rectangle is $1,089\sqrt{3}$ square units. It follows that $1,089\sqrt{3} = s(s\sqrt{3})$, or $1,089\sqrt{3} = s^2\sqrt{3}$. Dividing both sides of this equation by $\sqrt{3}$ yields $1,089 = s^2$. Taking the positive square root of both sides of this equation yields $33 = s$. Since the length, in units, of the diagonal is represented by $2s$, it follows that the length, in units, of the diagonal is $2(33)$, or **66**. Since the length of the diameter of the circle is equal to the length of the diagonal of the rectangle, the length, in units, of the diameter of the circle is **66**.

Question ID: 7339a4db

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A right prism has a square base with side lengths of **10** inches. The height of the prism is **58** inches. What is the volume, in cubic inches, of the prism?

Correct Answer: 5800

Rationale

The correct answer is **5,800**. The volume, V , of a right prism with a square base is given by the formula $V = s^2h$, where s is the side length of the square base and h is the height of the prism. It's given that a right prism has a square base with side lengths of **10** inches. It's also given that the height of the prism is **58** inches. Substituting **10** for s and **58** for h in the formula $V = s^2h$ yields $V = (10)^2(58)$, or $V = 5,800$. Therefore, the volume, in cubic inches, of the prism is **5,800**.

Question ID: 5368ebc1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

Square A has side lengths that are **246** times the side lengths of square B. The area of square A is ***k*** times the area of square B. What is the value of ***k***?

Answer

- A. **60,516**
- B. **492**
- C. **246**
- D. **123**

Correct Answer: A

Rationale

Choice A is correct. The area of a square is ***s*²**, where ***s*** is the side length of the square. Therefore, the area of square B is ***b*²**, where ***b*** is the side length of square B. It's given that square A has side lengths that are **246** times the side lengths of square B. Therefore, the side length of square A can be represented by the expression **246*b***. It follows that the area of square A is **(246*b*)²**, or **60,516*b*²**. It's given that the area of square A is ***k*** times the area of square B, so **60,516*b*² = *k**b*²**. Therefore, the value of ***k*** is **60,516**.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

For triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . Which additional piece(s) of information is(are) sufficient to prove that triangle ABC is similar to triangle FGH ?

- I. The measure of angle A is equal to the measure of angle F .
- II. The measure of angle C is equal to the measure of angle H .
- III. The length of $\overline{FH}$ is 3 times the length of $\overline{AC}$.

Answer

- A. I is sufficient, II is sufficient, and III is sufficient.
- B. I is sufficient and II is sufficient, but III is not.
- C. II is sufficient and III is sufficient, but I is not.
- D. III is sufficient, but I is not and II is not.

Correct Answer: B

Rationale

Choice B is correct. It's given that in triangles ABC and FGH , angles B and G each measure 51° , the length of $\overline{AB}$ is 6 , and the length of $\overline{FG}$ is 18 . For statement I, if the measure of angle A is equal to the measure of angle F , then triangles ABC and FGH have two pairs of congruent angles, angles A and F and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement I is sufficient. For statement II, if the measure of angle C is equal to the measure of angle H , then triangles ABC and FGH have 2 pairs of congruent angles, angles C and H and angles B and G . When 2 pairs of angles in 2 triangles are congruent, the triangles are similar, so statement II is sufficient. For statement III, if the length of $\overline{FH}$ is 3 times the length of $\overline{AC}$, then the ratio of the length of $\overline{FH}$ to the length of $\overline{AC}$ is 3 . It's given that the length of $\overline{FG}$ is 18 and the length of $\overline{AB}$ is 6 . Therefore, the ratio of the length of $\overline{FG}$ to the length of $\overline{AB}$ is also 3 , so there are two pairs of sides in proportion. However, angle B isn't between sides AB and AC , and angle G isn't between sides FG and FH . Since the given angle measures aren't between the proportional sides, this information isn't sufficient to prove that the triangles are similar. Therefore, statements I and II are each sufficient, but statement III isn't sufficient.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: 42f21cda

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

In triangle ABC , the measure of angle B is 52° and the measure of angle C is 17° . What is the measure of angle A ?

Answer

- A. 21°
- B. 35°
- C. 69°
- D. 111°

Correct Answer: D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . Adding the measures of angles B and C gives $52 + 17 = 69^\circ$. Therefore, the measure of angle A is $180 - 69 = 111^\circ$.

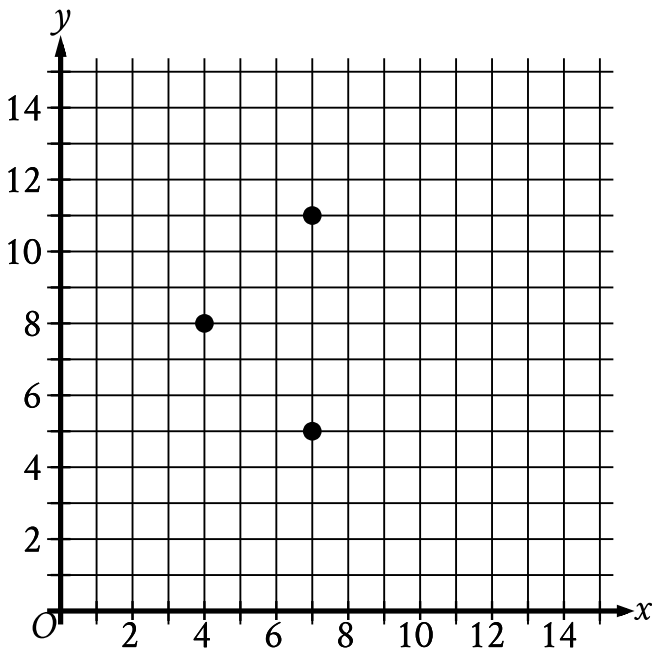
Choice A is incorrect and may result from subtracting the sum of the measures of angles B and C from 90° , instead of from 180° .

Choice B is incorrect and may result from subtracting the measure of angle C from the measure of angle B .

Choice C is incorrect and may result from adding the measures of angles B and C but not subtracting the result from 180° .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The three points shown define a circle. The circumference of this circle is $k\pi$, where k is a constant. What is the value of k ?

Answer

- A. 3
- B. 6
- C. 7
- D. 9

Correct Answer: B

Rationale

Choice B is correct. It's given that the three points shown define a circle, so the center of that circle is an equal distance from each of the three points. The point $(7, 8)$ is halfway between the points $(7, 5)$ and $(7, 11)$ and is a distance of 3 units from each of those two points. The point $(7, 8)$ is also a distance of 3 units from $(4, 8)$. Because the point $(7, 8)$ is the same distance from all three given points, it must be the center of the circle. The radius of a circle is the distance from the center to any point on the circle. Since that distance is 3 , it follows that the radius of the circle is 3 . The circumference of a circle with radius r is equal to $2\pi r$. It follows that the circumference of the circle is $2(\pi)(3)$, or 6π . It's given that the circumference of the circle is $k\pi$. Therefore, the value of k is 6 .

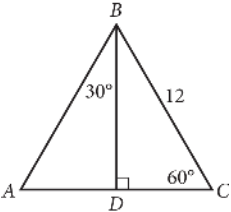
Choice A is incorrect. This is the radius of the circle, not the value of k in the expression $k\pi$.

Choice C is incorrect. This is the x-coordinate of the center of the circle, not the value of k in the expression $k\pi$.

Choice D is incorrect. This is the value of k for which $k\pi$ represents the area of the circle, in square units, not the circumference of the circle, in units.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



In $\triangle ABC$ above, what is the length of $\overline{AD}$?

Answer

- A. 4
- B. 6
- C. $6\sqrt{2}$
- D. $6\sqrt{3}$

Correct Answer: B

Rationale

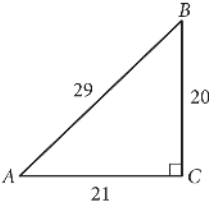
Choice B is correct. Triangles ADB and CDB are both $30^\circ - 60^\circ - 90^\circ$ triangles and share $\overline{BD}$. Therefore, triangles ADB and CDB are congruent by the angle-side-angle postulate. Using the properties of $30^\circ - 60^\circ - 90^\circ$ triangles, the length of $\overline{AD}$ is half the length of hypotenuse $\overline{AB}$. Since the triangles are congruent, $AB = BC = 12$. So the length of $\overline{AD}$ is $\frac{12}{2} = 6$.

Alternate approach: Since angle CBD has a measure of 30° , angle ABC must have a measure of 60° . It follows that triangle ABC is equilateral, so side AC also has length 12. It also follows that the altitude BD is also a median, and therefore the length of AD is half of the length of AC, which is 6.

Choice A is incorrect. If the length of $\overline{AD}$ were 4, then the length of $\overline{AB}$ would be 8. However, this is incorrect because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12. Choices C and D are also incorrect. Following the same procedures as used to test choice A gives $\overline{AB}$ a length of $12\sqrt{2}$ for choice C and $12\sqrt{3}$ for choice D. However, these results cannot be true because $\overline{AB}$ is congruent to $\overline{BC}$, which has a length of 12.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Medium

Question



In the figure above, what is the value of $\tan(A)$?

Answer

- A. $\frac{20}{29}$
- B. $\frac{21}{29}$
- C. $\frac{20}{21}$
- D. $\frac{21}{20}$

Correct Answer: C

Rationale

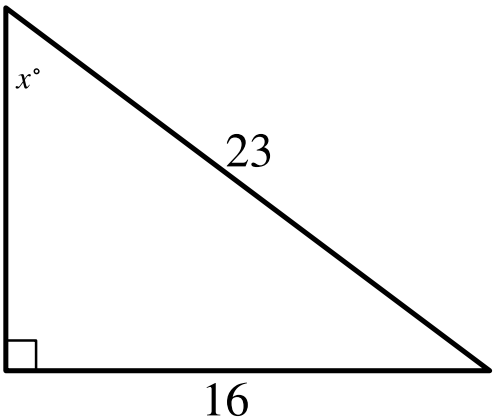
Choice C is correct. Angle A is an acute angle in a right triangle, so the value of $\tan(A)$ is equivalent to the ratio of the length of the side opposite angle A, 20, to the length of the nonhypotenuse side adjacent to angle A, 21. Therefore, $\tan(A) = \frac{20}{21}$.

Choice A is incorrect. This is the value of $\sin(A)$. Choice B is incorrect. This is the value of $\cos(A)$. Choice D is incorrect. This is the value of $\tan(B)$.

Question ID: 7eb63aa4

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Hard

Question



Note: Figure not drawn to scale.

In the triangle shown, what is the value of $\sin x^\circ$?

Correct Answer: .6956, .6957, 16/23

Rationale

The correct answer is $\frac{16}{23}$. In a right triangle, the sine of an acute angle is defined as the ratio of the length of the side opposite the angle to the length of the hypotenuse. In the triangle shown, the length of the side opposite the angle with measure x° is 16 units and the length of the hypotenuse is 23 units. Therefore, the value of $\sin x^\circ$ is $\frac{16}{23}$. Note that 16/23, .6956, .6957, 0.695, and 0.696 are examples of ways to enter a correct answer.

Question ID: 375b20d8

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question

In $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . What is the measure of $\angle Z$?

Answer

- A. 58°
- B. 74°
- C. 122°
- D. 212°

Correct Answer: A

Rationale

Choice A is correct. The triangle angle sum theorem states that the sum of the measures of the interior angles of a triangle is 180° . It's given that in $\triangle XYZ$, the measure of $\angle X$ is 24° and the measure of $\angle Y$ is 98° . It follows that the measure of $\angle Z$ is $(180 - 24 - 98)^\circ$, or 58° .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures of $\angle X$ and $\angle Y$, not the measure of $\angle Z$.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The table gives the perimeters of similar triangles TUV and XYZ , where $\overline{TU}$ corresponds to $\overline{XY}$. The length of $\overline{TU}$ is 6.

	Perimeter
Triangle TUV	50
Triangle XYZ	150

What is the length of $\overline{XY}$?

Answer

- A. 2
- B. 6
- C. 18
- D. 56

Correct Answer: C

Rationale

Choice C is correct. It’s given that triangle TUV is similar to triangle XYZ , and $\overline{TU}$ corresponds to $\overline{XY}$. If two triangles are similar, then the ratio of their perimeters is equal to the ratio of their corresponding sides. It’s given that the perimeter of triangle TUV is 50, the perimeter of triangle XYZ is 150, and the length of $\overline{TU}$ is 6. Let n represent the length of $\overline{XY}$. It follows that $\frac{50}{150} = \frac{6}{n}$, or $\frac{1}{3} = \frac{6}{n}$. Multiplying each side of this equation by n yields $\frac{n}{3} = 6$. Multiplying each side of this equation by 3 yields $n = 18$. Therefore, the length of $\overline{XY}$ is 18.

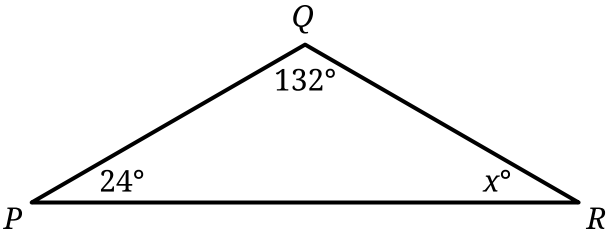
Choice A is incorrect. This is the solution to $\frac{3}{1} = \frac{6}{n}$, not $\frac{1}{3} = \frac{6}{n}$.

Choice B is incorrect. This is the length of $\overline{TU}$, not $\overline{XY}$.

Choice D is incorrect. This is the sum of the length of $\overline{TU}$ and the perimeter of triangle TUV , not the length of $\overline{XY}$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the triangle shown, $PQ = QR$. What is the value of x ?

Answer

- A. 156
- B. 66
- C. 48
- D. 24

Correct Answer: D

Rationale

Choice D is correct. The sum of the angle measures of a triangle is 180° . It's given that the measure of angle Q is 132° , the measure of angle P is 24° , and the measure of angle R is x° . It follows that $x + 132 + 24 = 180$, or $x + 152 = 180$. Subtracting 152 from both sides of this equation yields $x = 24$.

Alternate approach: It's given that in the triangle shown, $PQ = QR$. Thus, triangle PQR is an isosceles triangle. Since base angles of an isosceles triangle are congruent, the measure of angle P is equal to the measure of angle R . Therefore, the measure of angle R is 24° , and the value of x is 24.

Choice A is incorrect. This is the sum of the measures, in degrees, of angles Q and P , not the measure, in degrees, of angle R .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. This is the sum of the measures, in degrees, of angles P and R , not the measure, in degrees, of angle R .

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a volume of **377** cubic centimeters. The area of the base of the cylinder is **13** square centimeters. What is the height, in centimeters, of the cylinder?

Correct Answer: 29

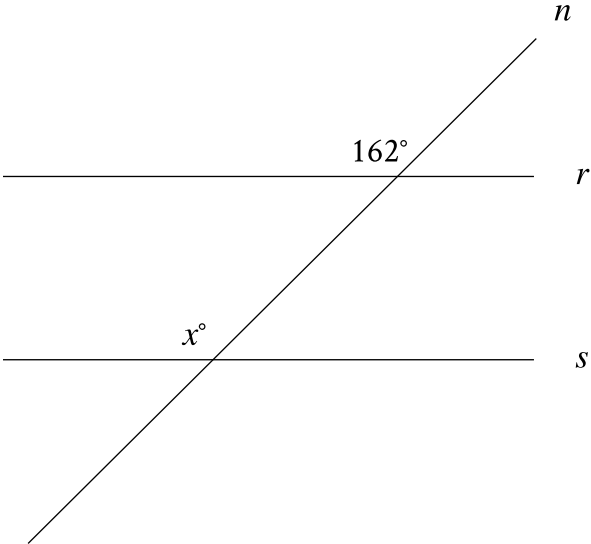
Rationale

The correct answer is **29**. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. Since the base of the cylinder is a circle with radius r , the area of the base of the cylinder is πr^2 . It's given that a right circular cylinder has a volume of **377** cubic centimeters; therefore, $V = 377$. It's also given that the area of the base of the cylinder is **13** square centimeters; therefore, $\pi r^2 = 13$. Substituting **377** for V and **13** for πr^2 in the formula $V = \pi r^2 h$ yields $377 = 13h$. Dividing both sides of this equation by **13** yields $29 = h$. Therefore, the height of the cylinder, in centimeters, is **29**.

Question ID: c349b88b

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Easy

Question



Note: Figure not drawn to scale.

In the figure, line n intersects lines r and s . Line r is parallel to line s . What is the value of x ?

Correct Answer: 162

Rationale

The correct answer is **162**. It's given that line r is parallel to line s . Since line n intersects both lines r and s , it's a transversal. The angles in the figure marked as 162° and x° are on the same side of the transversal, where one is an interior angle with line s as a side, and the other is an exterior angle with line r as a side. Thus, the marked angles are corresponding angles. When two parallel lines are intersected by a transversal, corresponding angles are congruent and, therefore, have equal measure. It follows that the value of x is **162**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Circle N has a radius of **7** millimeters (**mm**). Circle M has an area of **64π mm²**. What is the total area, in **mm²**, of circles N and M?

- Answer**
- A. **113π**
- B. **92π**
- C. **78π**
- D. **15π**

Correct Answer: A

Rationale

Choice A is correct. It’s given that circle N has a radius of **7 mm**. The area of a circle is given by the expression **πr^2** , where **r** is the radius of the circle. Substituting **7** for **r** in the expression yields **$\pi(7)^2$** , or **49π** , for the area, in **mm²**, of circle N. It’s also given that the area of circle M is **64π mm²**. Adding the two areas yields **$49\pi + 64\pi$** , or **113π mm²**. Therefore, the total area, in **mm²**, of circles N and M is **113π** .

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Question ID: c3f4d8c1

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

A right circular cylinder has a base diameter of **22** centimeters and a height of **6** centimeters. What is the volume, in cubic centimeters, of the cylinder?

Answer

- A. **132** π
- B. **264** π
- C. **726** π
- D. **2,904** π

Correct Answer: C

Rationale

Choice C is correct. The volume, V , of a right circular cylinder is given by the formula $V = \pi r^2 h$, where r is the radius of the base of the cylinder and h is the height of the cylinder. It's given that a right circular cylinder has a height of **6** centimeters. Therefore, $h = 6$. It's also given that the cylinder has a base diameter of **22** centimeters. The radius of a circle is half the diameter of the circle. Since the base of a right circular cylinder is a circle, it follows that the radius of the base of the right circular cylinder is $\frac{22}{2}$, or **11**, centimeters. Therefore, $r = 11$. Substituting **11** for r and **6** for h in the formula $V = \pi r^2 h$ yields $V = \pi(11)^2(6)$, which is equivalent to $V = \pi(121)(6)$, or $V = 726\pi$. Therefore, the volume, in cubic centimeters, of the cylinder is **726** π .

Choice A is incorrect. This is the volume of a right circular cylinder that has a base diameter of $2\sqrt{22}$, not **22**, centimeters and a height of **6** centimeters.

Choice B is incorrect. This is the volume of a right circular cylinder that has a base diameter of $4\sqrt{11}$, not **22**, centimeters and a height of **6** centimeters.

Choice D is incorrect. This is the volume of a right circular cylinder that has a base diameter of **44**, not **22**, centimeters and a height of **6** centimeters.

Question ID: 6c1320cc

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

Which expression represents the area, in square inches, of a square with a side length of **35** inches?

Answer

- A. $\sqrt{35}$
- B. $35 + 35$
- C. $(4)(35)$
- D. 35^2

Correct Answer: D

Rationale

Choice D is correct. The area of a square is given by s^2 , where s is the length of each side of the square. It's given that the length of each side of a square is **35** inches. Substituting **35** for s in the expression s^2 yields 35^2 . Therefore, the expression that represents the area, in square inches, of the square is 35^2 .

Choice A is incorrect and may result from conceptual errors.

Choice B is incorrect and may result from conceptual errors.

Choice C is incorrect. This expression represents the perimeter, in inches, of the square rather than the area, in square inches.

Question ID: d1610d1a

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The area of a triangle is **36** square units. If the height of this triangle is **2** units, what is the length, in units, of the base of this triangle?

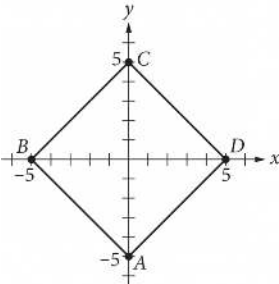
Correct Answer: 36

Rationale

The correct answer is **36**. The area, ***A***, of a triangle is given by the formula **$A = \frac{1}{2}bh$** , where ***b*** represents the length of the base of the triangle and ***h*** represents its height. It's given that the triangle has an area of **36** square units and a height of **2** units. Substituting **36** for ***A*** and **2** for ***h*** in the formula **$A = \frac{1}{2}bh$** yields **$36 = \frac{1}{2}b(2)$** , or **$36 = b$** . Thus, the length, in units, of the base of this triangle is **36**.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question



In the xy-plane shown, square ABCD has its diagonals on the x- and y-axes. What is the area, in square units, of the square?

Answer

- A. 20
- B. 25
- C. 50
- D. 100

Correct Answer: C

Rationale

Choice C is correct. The two diagonals of square ABCD divide the square into 4 congruent right triangles, where each triangle has a vertex at the origin of the graph shown. The formula for the area of a triangle is $A = \frac{1}{2}bh$, where b is the base length of the triangle and h is the height of the triangle. Each of the 4 congruent right triangles has a height of 5 units and a base length of 5 units. Therefore, the area of each triangle is $A = \frac{1}{2}(5)(5)$, or 12.5 square units. Since the 4 right triangles are congruent, the area of each is $\frac{1}{4}$ of the area of square ABCD. It follows that the area of the square ABCD is equal to 4×12.5 , or 50 square units.

Choices A and D are incorrect and may result from using 5 or 25, respectively, as the area of one of the 4 congruent right triangles formed by diagonals of square ABCD. However, the area of these triangles is 12.5. Choice B is incorrect and may result from using 5 as the length of one side of square ABCD. However, the length of a side of square ABCD is $5\sqrt{2}$.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question

In triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. The length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. What is the value of $\frac{BC}{BD}$?

Answer

- A. $\frac{15}{38}$
- B. $\frac{15}{23}$
- C. $\frac{23}{15}$
- D. $\frac{38}{15}$

Correct Answer: D

Rationale

Choice D is correct. It's given that in triangle ABC , the measure of angle B is 90° and $\overline{BD}$ is an altitude of the triangle. Therefore, the measure of angle BDC is 90° . It follows that angle B is congruent to angle D and angle C is congruent to angle C . By the angle-angle similarity postulate, triangle ABC is similar to triangle BDC . Since triangles ABC and BDC are similar, it follows that $\frac{AC}{AB} = \frac{BC}{BD}$. It's also given that the length of $\overline{AB}$ is 15 and the length of $\overline{AC}$ is 23 greater than the length of $\overline{AB}$. Therefore, the length of $\overline{AC}$ is $15 + 23$, or 38 . Substituting 15 for AB and 38 for AC in the equation $\frac{AC}{AB} = \frac{BC}{BD}$ yields $\frac{38}{15} = \frac{BC}{BD}$. Therefore, the value of $\frac{BC}{BD}$ is $\frac{38}{15}$.

Choice A is incorrect. This is the value of $\frac{BD}{BC}$.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Question ID: 017166ab

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Medium

Question

The side length of a square is **55 centimeters (cm)**. What is the area, **in cm²**, of the square?

Answer

- A. 110
- B. 220
- C. 3,025
- D. 12,100

Correct Answer: C

Rationale

Choice C is correct. The area **A , in square centimeters (cm²)**, of a square with side length **s , in cm**, is given by the formula **$A = s^2$** . It's given that the square has a side length of **55 cm**. Substituting **55** for **s** in the formula **$A = s^2$** yields **$A = 55^2$** , or **$A = 3,025$** . Therefore, the area, **in cm²**, of the square is **3,025**.

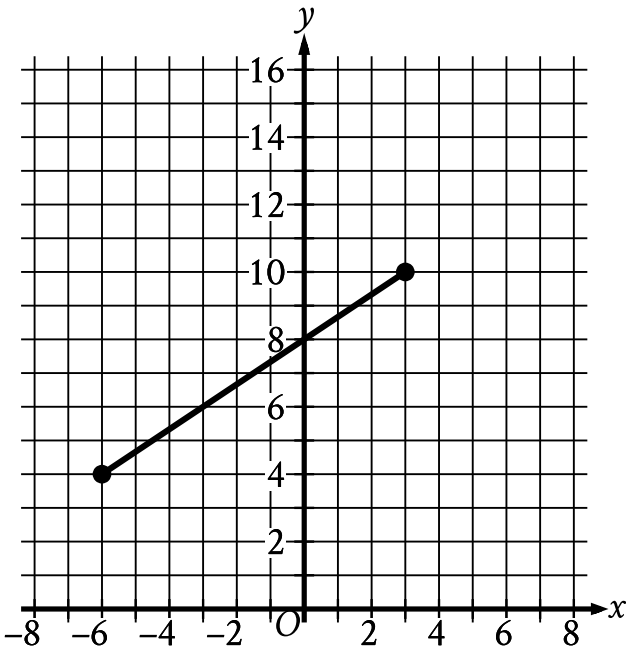
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect. This is the perimeter, **in cm**, of the square, not its area, **in cm²**.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question



The line segment shown in the xy -plane represents one of the legs of a right triangle. The area of this triangle is $36\sqrt{13}$ square units. What is the length, in units, of the other leg of this triangle?

Answer

- A. 12
- B. 24
- C. $3\sqrt{13}$
- D. $18\sqrt{13}$

Correct Answer: B

Rationale

Choice B is correct. The length of a segment in the xy -plane can be found using the distance formula, $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$, where (x_1, y_1) and (x_2, y_2) are the endpoints of the segment. The segment shown has endpoints at $(-6, 4)$ and $(3, 10)$. Substituting $(-6, 4)$ and $(3, 10)$ for (x_1, y_1) and (x_2, y_2) , respectively, in the distance formula yields $\sqrt{(3 - (-6))^2 + (10 - 4)^2}$, or $\sqrt{9^2 + 6^2}$, which is equivalent to $\sqrt{81 + 36}$, or $\sqrt{117}$. Let x represent the length, in units, of the other leg of this triangle. The area, A , of a right triangle can be calculated using the formula $A = \frac{1}{2}bh$, where b and h are the lengths of the legs of the triangle. It's given that the area of the triangle is $36\sqrt{13}$ square units. Substituting $36\sqrt{13}$ for A , $\sqrt{117}$ for b , and x for h in the formula $A = \frac{1}{2}bh$ yields $36\sqrt{13} = \frac{1}{2}(\sqrt{117})(x)$. Multiplying both sides of this equation by 2 yields $72\sqrt{13} = x\sqrt{117}$. Dividing both sides of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{13}}{\sqrt{117}} = x$. Multiplying the numerator and denominator on the left-hand side of this equation by $\sqrt{117}$ yields $\frac{72\sqrt{1,521}}{117} = x$, or $\frac{72(39)}{117} = x$, which is equivalent to $\frac{2,808}{117} = x$, or $24 = x$. Therefore, the length, in units, of the other leg of this triangle is 24.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect. $3\sqrt{13}$ is equivalent to $\sqrt{117}$, which is the length, in units, of the line segment shown in the xy -plane, not the length, in units, of the other leg of the triangle.

Choice D is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Medium

Question

Triangles ABC and DEF are congruent, where A corresponds to D , and B and E are right angles. The measure of angle A is 69° . What is the measure, in degrees, of angle F ?

Correct Answer: 21

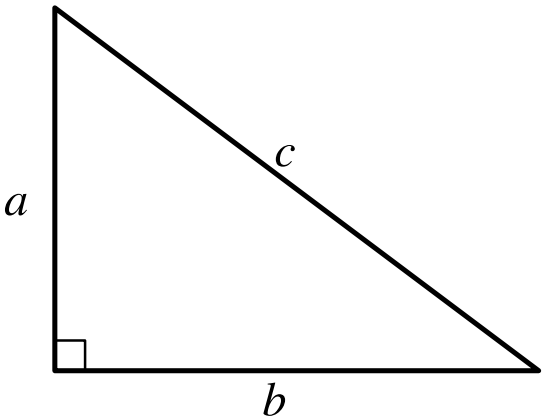
Rationale

The correct answer is **21**. It's given that triangles ABC and DEF are congruent with angle A corresponding to angle D . Corresponding angles of congruent triangles are congruent and, therefore, have equal measure. It's given that the measure of angle A is 69° . It follows that the measure of angle D is also 69° . It's given that angle E is a right angle. Therefore, the measure of angle E is 90° . Let x represent the measure, in degrees, of angle F . Since the measures of the angles in a triangle sum to 180° , it follows that $69 + 90 + x = 180$, or $159 + x = 180$. Subtracting **159** from both sides of this equation yields $x = 21$. Therefore, the measure, in degrees, of angle F is **21**.

Question ID: f6ea61ff

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Right triangles and trigonometry	Easy

Question



Note: Figure not drawn to scale.

For the right triangle shown, $a = 4$ and $b = 5$. Which expression represents the value of c ?

- Answer
- A. $4 + 5$
 - B. $\sqrt{(4)(5)}$
 - C. $\sqrt{4 + 5}$
 - D. $\sqrt{4^2 + 5^2}$

Correct Answer: D

Rationale

Choice D is correct. By the Pythagorean theorem, if a right triangle has a hypotenuse with length c and legs with lengths a and b , then $c^2 = a^2 + b^2$. In the right triangle shown, the hypotenuse has length c and the legs have lengths a and b . It's given that $a = 4$ and $b = 5$. Substituting 4 for a and 5 for b in the Pythagorean theorem yields $c^2 = 4^2 + 5^2$. Taking the square root of both sides of this equation yields $c = \pm\sqrt{4^2 + 5^2}$. Since the length of a side of a triangle must be positive, the value of c is $\sqrt{4^2 + 5^2}$.

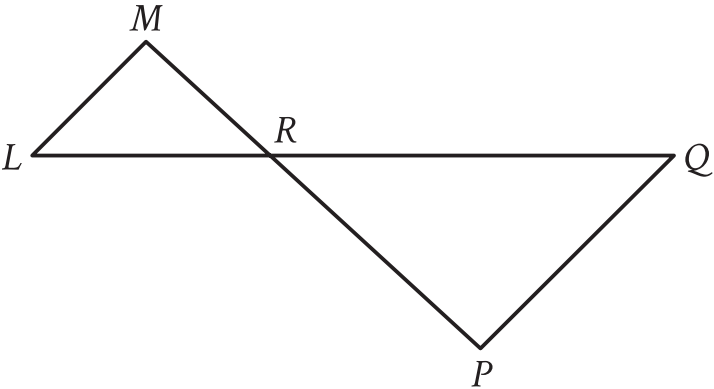
Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice C is incorrect and may result from conceptual or calculation errors.

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Lines, angles, and triangles	Hard

Question



Note: Figure not drawn to scale.

In the figure, $\overline{LQ}$ intersects $\overline{MP}$ at point R , and $\overline{LM}$ is parallel to $\overline{PQ}$. The lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively. The area of $\triangle LMR$ is 36 square units. What is the area of $\triangle PQR$, in square units?

Answer

- A. $\frac{576}{49}$
- B. $\frac{144}{7}$
- C. 63
- D. $\frac{441}{4}$

Correct Answer: D

Rationale

Choice D is correct. The figure shows that $\angle MRL$ and $\angle PRQ$ are vertical angles. Since vertical angles are congruent, $\angle MRL$ and $\angle PRQ$ are congruent. It's given that $\overline{LM}$ is parallel to $\overline{PQ}$. The figure also shows that $\overline{LQ}$ intersects $\overline{LM}$ and $\overline{PQ}$. If two parallel segments are intersected by a third segment, alternate interior angles are congruent. Thus, $\angle MLR$ and $\angle PQR$ are congruent. Since $\triangle LMR$ and $\triangle PQR$ have two pairs of congruent angles, the triangles are similar, where $\overline{MR}$ in $\triangle LMR$ corresponds to $\overline{PR}$ in $\triangle PQR$. Since the areas of similar triangles are proportional to the squares of the lengths of corresponding sides, it follows that $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$. It's given that the lengths of $\overline{MR}$ and $\overline{RP}$ are 8 and 14 units, respectively, and the area of $\triangle LMR$ is 36 square units. Substituting 14 for RP , 8 for MR , and 36 for $\text{area of } \triangle LMR$ in the equation $\frac{\text{area of } \triangle PQR}{\text{area of } \triangle LMR} = \left(\frac{RP}{MR}\right)^2$ yields $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{14}{8}\right)^2$, which is equivalent to $\frac{\text{area of } \triangle PQR}{36} = \left(\frac{7}{4}\right)^2$, or $\frac{\text{area of } \triangle PQR}{36} = \frac{49}{16}$. Multiplying each side of this equation by 36 yields $\text{area of } \triangle PQR = 36\left(\frac{49}{16}\right)$, or $\text{area of } \triangle PQR = \frac{441}{4}$. Therefore, the area of $\triangle PQR$ is $\frac{441}{4}$ square units.

Choice A is incorrect. This is the result of multiplying 36 by $\left(\frac{8}{14}\right)^2$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice B is incorrect. This is the result of multiplying 36 by $\frac{8}{14}$ instead of by $\left(\frac{14}{8}\right)^2$.

Choice C is incorrect. This is the result of multiplying **36** by $\frac{14}{8}$ instead of by $\left(\frac{14}{8}\right)^2$.

Question ID: 72e7f927

Assessment	Test	Domain	Skill	Difficulty
PSAT/NMSQT and PSAT 10	Math	Geometry and Trigonometry	Area and volume	Hard

Question

A cube has a volume of **474,552** cubic units. What is the surface area, in square units, of the cube?

Correct Answer: 36504

Rationale

The correct answer is **36,504**. The volume of a cube can be found using the formula $V = s^3$, where s represents the edge length of a cube. It's given that this cube has a volume of **474,552** cubic units. Substituting **474,552** for V in $V = s^3$ yields $474,552 = s^3$. Taking the cube root of both sides of this equation yields $78 = s$. Thus, the edge length of the cube is **78** units. Since each face of a cube is a square, it follows that each face has an edge length of **78** units. The area of a square can be found using the formula $A = s^2$. Substituting **78** for s in this formula yields $A = 78^2$, or $A = 6,084$. Therefore, the area of one face of this cube is **6,084** square units. Since a cube has **6** faces, the surface area, in square units, of this cube is **6(6,084)**, or **36,504**.